INCH-POUND

MIL-DTL-32742(AR)

9 MARCH 2023

DEPARTMENT OF DEFENSE
DETAIL SPECIFICATION
HANDLING QUALITIES
FOR
MILITARY ROTORCRAFT

Comments, suggestions, or questions on this document should be addressed to (U.S. Army DEVCOM, Aviation and Missile Center, 5400 Fowler Rd, FCDD-AMR-MD/Tod Jordan) Redstone Arsenal, AL, 35898-5000 or emailed to tod.n.jordan.civ@mail.mil). Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 1520

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

CONTENTS

CONTENTS ..... **II**

1. SCOPE **1**

1.1. SCOPE. 1

2. APPLICABLE DOCUMENTS **1**

2.1. GENERAL. 1
2.2. GOVERNMENT DOCUMENTS. 1
2.2.1. Specifications, standards, and handbooks. *1*
2.2.2. Other Government documents, drawings, and publications. *1*
2.3. ORDER OF PRECEDENCE. 2

3. REQUIREMENTS **2**

3.1. GENERAL. 2
3.1.1. Required agility. *2*
3.1.2. Mission-Task-Elements. *2*
3.1.3. Multi-crew rotorcraft. Unless otherwise stated, all requirements shall apply for the primary pilot station. 2
3.1.4. Levels of handling qualities. *2*
3.1.4.1. Predicted levels of handling qualities. 2
3.1.4.2. Assigned levels of handling qualities. 2
3.1.4.3. Interpretation of predicted vs. assigned levels of handling qualities 2
3.1.5. Levels for normal states. *3*
3.1.6. Flight beyond the service flight envelopes. *3*
3.1.7. Rotorcraft failures. *3*
3.1.7.1. Rotorcraft transients following failures. 3
3.1.7.2. Indication of failures. 3
3.1.8. Rotorcraft limits. *3*
3.1.9. Devices for indication, warning, prevention, and recovery. *3*
3.1.10. Residual oscillations. *3*
3.1.11. Rotor RPM governing. *4*
3.1.12. Rotorcraft status. *4*
3.2. RESPONSE-TYPES. 4
3.2.1. Required response-types. *4*
3.2.1.1. Character of rate response-types. 4
3.2.1.2. Character of rate command response-types. 4
3.2.1.3. Rate command with heading (direction) hold. 4
3.2.1.4. Character of attitude hold and heading hold. 4
3.2.1.5. Heading hold in hover. 4
3.2.1.6. Character of attitude command response-types. 5
3.2.1.6.1. Attitude command response-type in hover and low speed 5
3.2.1.6.2. Attitude command response-type in forward flight. 5
3.2.1.7. Character of translational rate response-types. 5
3.2.1.8. Character of vertical rate response-types. 5
3.2.1.8.1. Altitude (height) hold relaxation. 5
3.2.1.8.2. Character of vertical rate command with altitude (height) hold. 5
3.2.1.9. Character of yaw response to lateral controller. 5
3.2.1.9.1. Turn coordination. 6
3.2.1.9.2. Turn coordination additional requirement. 6

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.2.2. Non-specified response-types. *6*
3.2.3. Inputs to control actuator. *6*
3.2.4. Transition between airborne and ground operations. *6*
3.2.5. Transfer between response-types. *6*
3.2.5.1. Annunciation of response-type to the pilot. 6
3.2.5.2. Control forces during transfer 6
3.2.5.3. Control system blending. 6
3.3. HOVER AND LOW SPEED REQUIREMENTS. 6
3.3.1. Equilibrium characteristics. *6*
3.3.2. Small-amplitude pitch (roll) attitude change. *7*
3.3.2.1. Short-term response to control inputs (bandwidth). 7
3.3.2.2. Short-term pitch (roll) response to disturbance inputs. 7
3.3.2.3. Mid-term response to control longitudinal (lateral) inputs. 7
3.3.2.3.1. Fully attended operations. 7
3.3.2.3.2. Divided attention operations. 7
3.3.3. Moderate-amplitude pitch (roll) attitude changes (attitude quickness). *7*
3.3.4. Large-amplitude pitch (roll) attitude changes. *8*
3.3.5. Small-amplitude yaw attitude changes. *8*
3.3.5.1. Short-term response to yaw control inputs (bandwidth). 8
3.3.5.2. Short-term yaw response to disturbance inputs. 8
3.3.5.3. Yaw rate response to lateral gusts. 8
3.3.5.4. Mid-term response to Yaw control inputs. 8
3.3.5.4.1. Fully attended operations. 8
3.3.5.4.2. Divided attention operations. 8
3.3.6. Moderate-amplitude heading attitude changes (attitude quickness). *9*
3.3.7. Large-amplitude heading changes. *9*
3.3.8. Hover and low speed interaxis coupling. *9*
3.3.8.1. Yaw due to collective for aggressive agility. 9
3.3.8.2. Pitch due to roll and roll due to pitch coupling for aggressive agility. 9
3.3.8.3. Pitch due to roll and roll due to pitch coupling for target acquisition and tracking. 9
3.3.9. Response to collective controller. *9*
3.3.9.1. Height response characteristics. 10
3.3.9.2. Torque response. 10
3.3.9.3. Vertical axis control power. 11
3.3.9.4. Short-term height response to disturbance inputs. 11
3.3.10. Position hold. *11*
3.3.10.1. Short-term position response to disturbance. 11
3.3.10.2. Position hold annunciation. 11
3.3.11. Translational rate response type. *11*
3.3.11.1. Short-term translational rate response to disturbance inputs. 12
3.3.12. Pitch (roll) response to externally slung loads. *12*
3.4. FORWARD FLIGHT REQUIREMENTS. 13
3.4.1. Pitch attitude response to longitudinal controller. *13*
3.4.1.1. Short-term response (bandwidth). 13
3.4.1.2. Mid-term response to control inputs. 13
3.4.1.2.1. Fully attended operations. 13
3.4.1.2.2. Divided attention operations. 13
3.4.1.3. Mid-term response – maneuvering stability. 13
3.4.1.3.1. Control feel and stability in maneuvering flight at constant speed. 13
3.4.1.3.2. Control forces in maneuvering flight. 14
3.4.2. Pitch control power. *14*
3.4.3. Flight path control. *14*

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.4.3.1. Flight path response to pitch attitude (frontside). 14
3.4.3.2. Flight path response to collective controller (backside). 14
3.4.4. Longitudinal static stability. *15*
3.4.5. Roll attitude response to lateral controller. *15*
3.4.5.1. Small-amplitude roll attitude response to control inputs (bandwidth). 15
3.4.5.2. Moderate amplitude attitude changes (attitude quickness). 15
3.4.5.3. Large-amplitude roll attitude changes. 15
3.4.6. Roll-sideslip coupling. *15*
3.4.6.1. Bank angle oscillations. 15
3.4.6.2. Turn coordination. 16
3.4.7. Yaw response to yaw controller. *16*
3.4.7.1. Small-amplitude yaw response for target acquisition and tracking (bandwidth). 16
3.4.7.2. Large-amplitude heading changes for aggressive agility. 16
3.4.7.3. Yaw control with speed change. 16
3.4.8. Lateral-directional stability. *16*
3.4.8.1. Lateral-directional oscillations. 16
3.4.8.2. Spiral stability. 17
3.4.9. Lateral-directional response characteristics in steady sideslips. *17*
3.4.9.1. Yaw control in steady sideslips (directional stability). 17
3.4.9.2. Bank angle in steady sideslips. 17
3.4.9.3. Lateral control in steady sideslips. 17
3.4.9.4. Positive effective dihedral limit. 17
3.4.10. Forward flight interaxis coupling. *18*
3.4.10.1. Pitch attitude due to collective control. 18
3.4.10.1.1. Small collective inputs. 18
3.4.10.1.2. Large collective inputs. 18
3.4.10.1.3. Pitch control in autorotation. 18
3.4.10.2. Roll due to pitch coupling for aggressive agility. 18
3.4.10.3. Pitch due to roll coupling for aggressive agility. 18
3.4.10.4. Pitch due to roll and roll due to pitch for target acquisition and tracking. 18
3.5. CONTROLLER CHARACTERISTICS. 18
3.5.1. Conventional controllers. *18*
3.5.1.1. Centering and breakout forces. 18
3.5.1.2. Force gradients. 19
3.5.1.3. Limit control force. 19
3.5.2. Sensitivity and gradients. *19*
3.5.3. Cockpit control free play. *19*
3.5.4. Control harmony. *19*
3.5.5. Trimming characteristics. *19*
3.5.6. Dynamic coupling. *19*
4. VERIFICATION 19
4.1. GENERAL. 19
4.1.1. Analysis. *20*
4.1.2. Simulation. *20*
4.1.3. Flight. *21*
4.2. DETERMINATION OF USABLE CUE ENVIRONMENT. 21
4.2.1. Characteristic of test rotorcraft. *22*
4.2.2. Applicable Mission-Task-Elements. *22*
4.2.3. Dispersions among visual cue rating. *22*
4.3. LEVELS OF HANDLING QUALITIES. 22

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

4.4. MISSION-TASK-ELEMENTS. 22
4.4.1. Conduct of test. *23*
4.4.2. Performance standards. *23*
4.4.3. Visual cueing. *23*
4.5. TESTING WITH EXTERNALLY SLUNG LOADS. 24
4.6. INTERPRETATION OF SUBJECTIVE REQUIREMENTS. 24

5. PACKAGING 24

5.1. PACKAGING. 24

6. NOTES 24

6.1. INTENDED USE. 24
6.2. DEFINITIONS. 25
6.2.1. Acronyms. *25*
6.2.2. Flight envelopes. *26*
6.2.2.1. Operational flight envelopes (OFE). 26
6.2.2.2. Service flight envelopes (SFE). 26
6.2.3. Configuration. *26*
6.2.4. Loadings. *26*
6.2.5. Flight conditions. *26*
6.2.6. Settings. *26*
6.2.7. States. *27*
6.2.8. Rotorcraft status. *27*
6.2.9. Degree of pilot attention. *27*
6.2.9.1. Fully attended operation. 27
6.2.9.2. Divided attention operation. 27
6.2.10. IMC operations. *27*
6.2.11. Levels of handling qualities. *27*
6.2.12. Landing gear. *27*
6.2.13. Load mass ratio. *27*
6.2.14. Mission-task-element (MTE). *27*
6.2.15. Response-Type. *27*
6.2.16. Sideslips. *28*
6.2.17. Control free play. *28*
6.2.18. Useable cueing environment. *28*
6.2.19. Winds. *28*
6.2.19.1. Calm winds. 28
6.2.19.2. Light winds. 28
6.2.19.3. Moderate winds. 28
6.2.20. Near-earth operations. *28*
6.2.21. Stabilized hover. *28*
6.2.22. Step input. *28*
6.2.23. Speed ranges. *28*
6.2.23.1. Hover. 28
6.2.23.2. Low Speed. 28
6.2.23.3. Forward flight. 28
6.2.24. Objectionable. *28*
6.3. SUBJECT TERM (KEY WORD) LISTING. 29

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. MISSION-TASK-ELEMENTS (MTE) 30
TABLE II. LEVELS FOR ROTORCRAFT FAILURE STATES 31
TABLE III. TRANSIENTS FOLLOWING FAILURES 31
TABLE IV. MINIMUM RESPONSE-TYPES FOR HOVER AND LOW SPEED – NEAR EARTH 32
TABLE V. MINIMUM RESPONSE-TYPES FOR FORWARD FLIGHT (PITCH AND ROLL) 32
TABLE VI. LIMIT COCKPIT CONTROL FORCES, POUNDS 33
TABLE VII. REQUIREMENTS FOR LARGE-AMPLITUDE ATTITUDE CHANGES – HOVER AND LOW SPEED 33
TABLE VIII. MAXIMUM VALUES FOR HEIGHT RESPONSE PARAMETERS – HOVER AND LOW SPEED 34
TABLE IX. MAXIMUM VALUES FOR FLIGHT PATH RESPONSE PARAMETERS - FORWARD FLIGHT 34
TABLE X. REQUIREMENTS FOR LARGE-AMPLITUDE ROLL ATTITUDE CHANGES – FORWARD FLIGHT 34
TABLE XI. LIMIT COCKPIT CONTROL FORCES, POUNDS 35
TABLE XII. ALLOWABLE BREAKOUT FORCES, POUNDS - HOVER AND LOW SPEED 35
TABLE XIII. ALLOWABLE BREAKOUT FORCES, POUNDS - FORWARD FLIGHT 35
TABLE XIV. ALLOWABLE CONTROL FORCE GRADIENTS, POUNDS/INCH 36
TABLE XV. REQUIREMENTS/VERIFICATION MATRIX 37
TABLE XVI. ROTORCRAFT STATUS AND FLIGHT CONDITIONS FOR VERIFICATION 38
FIGURE 1. HOVER 39
FIGURE 2. LANDING 40
FIGURE 3. SLOPE LANDING AND LIFTOFF 41
FIGURE 4. HOVERING TURN 42
FIGURE 5. HOVER HEADING CAPTURE 43
FIGURE 6. PRECISION LOAD PLACEMENT 44
FIGURE 7. PIROUETTE 45
FIGURE 8. VERTICAL MANEUVER 46
FIGURE 9. DEPART/ABORT 47
FIGURE 10. LATERAL REPOSITION 48
FIGURE 11. SLALOM 49
FIGURE 12. VERTICAL REMASK 50
FIGURE 13. ACCELERATION AND DECELERATION 51
FIGURE 13. ACCELERATION AND DECELERATION - CONTINUED 52
FIGURE 14. SIDESLIP 53
FIGURE 14. SIDESLIP - CONTINUED 54
FIGURE 15. DECELERATION TO DASH 55
FIGURE 16. TRANSIENT TURN 56

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

FIGURE 17. PUSHUP/PUSHOVER 57
FIGURE 18. ROLL REVERSAL 58
FIGURE 19. TURN TO TARGET 59
FIGURE 20. HIGH YO-YO 60
FIGURE 21. LOW YO-YO 61
FIGURE 22. DECELERATING APPROACH 62
FIGURE 23. ILS APPROACH 62
FIGURE 24. MISSED APPROACH 63
FIGURE 25. SPEED CONTROL 63
FIGURE 26. DEFINITION OF HANDLING QUALITIES LEVELS 64
FIGURE 27. RESPONSE FOR ATTITUDE HOLD AND HEADING HOLD RESPONSE TYPES 65
FIGURE 28. VISUAL CUE RATING SCALE 65
FIGURE 29. REQUIREMENTS FOR SMALL-AMPLITUDE PITCH (ROLL) ATTITUDE CHANGES – HOVER AND LOW SPEED 66
FIGURE 30. DEFINITIONS OF BANDWIDTH AND PHASE DELAY 67
FIGURE 31. DISTURBANCE REJECTION BLOCK DIAGRAMS 68
FIGURE 32. DEFINITION OF DISTURBANCE REJECTION BANDWIDTH (DRB) AND PEAK (DRP) 69
FIGURE 33. LIMITS ON PITCH (ROLL) OSCILLATIONS - HOVER AND LOW SPEED 69
FIGURE 34. REQUIREMENTS FOR MODERATE-AMPLITUDE PITCH (ROLL) ATTITUDE CHANGES – HOVER AND LOW SPEED 70
FIGURE 35. REQUIREMENTS FOR SMALL-AMPLITUDE HEADING CHANGES – HOVER AND LOW SPEED 71
FIGURE 36. LIMITS ON YAW OSCILLATIONS - HOVER AND LOW SPEED 71
FIGURE 37. REQUIREMENTS FOR MODERATE-AMPLITUDE HEADING CHANGES – HOVER AND LOW SPEED 72
FIGURE 38. YAW-DUE-TO-COLLECTIVE COUPLING REQUIREMENTS 72
FIGURE 39. REQUIREMENTS FOR PITCH DUE TO ROLL AND ROLL DUE TO PITCH COUPLING FOR TARGET ACQUISITION AND TRACKING 73
FIGURE 40. PROCEDURE FOR OBTAINING EQUIVALENT TIME DOMAIN PARAMETERS FOR HEIGHT RESPONSE TO COLLECTIVE CONTROLLER 73
FIGURE 41. DISPLAYED TORQUE RESPONSE REQUIREMENT 74
FIGURE 42. REQUIREMENTS FOR LONGITUDINAL (LATERAL) TRANSLATIONAL RATE RESPONSE-TYPES – HOVER AND LOW SPEED 75
FIGURE 43. ΔHQR FROM BASELINE FOR EXTERNALLY SLUNG LOADS 76
FIGURE 44. DEFINITIONS OF SLUNG LOAD PARAMETERS 76
FIGURE 45. REQUIREMENTS FOR SMALL-AMPLITUDE PITCH ATTITUDE CHANGES – FORWARD FLIGHT 77
FIGURE 46. REQUIREMENTS FOR SMALL-AMPLITUDE ROLL ATTITUDE CHANGES - FORWARD FLIGHT 77

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

FIGURE 47. REQUIREMENTS FOR MODERATE-AMPLITUDE ROLL ATTITUDE CHANGES – FORWARD FLIGHT 77

FIGURE 48. ROLL-SIDESLIP COUPLING PARAMETERS 78

FIGURE 49. BANK ANGLE OSCILLATION LIMITATIONS.....79

FIGURE 50. SIDESLIP EXCURSION LIMITATIONS 79

FIGURE 51. REQUIREMENT FOR SMALL-AMPLITUDE YAW RESPONSE - FORWARD FLIGHT 80

FIGURE 52. LATERAL-DIRECTIONAL OSCILLATORY REQUIREMENTS.....80

FIGURE 53. USABLE CUE ENVIRONMENTS FOR VISUAL CUE RATINGS 81

FIGURE 54. SUGGESTED COURSE FOR HOVER MANEUVER.....82

FIGURE 55. SUGGESTED COURSE FOR PIROUETTE MANEUVER 83

FIGURE 56. LOAD PLACEMENT MISSION TASK ELEMENT (MTE) 84

FIGURE 57. SUGGESTED COURSE FOR ACCELERATION-DECELERATION MANEUVER 84

FIGURE 58. SUGGESTED COURSE FOR SIDESTEP, LATERAL REPOSITION, AND VERTICAL REMASK MANEUVERS ..85

FIGURE 59. SUGGESTED COURSE FOR SLALOM MANEUVER.....85

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1. Scope. This specification establishes the military-unique handling qualities requirements essential for rotorcraft design, development, production, and acceptance. The specification covers land-based rotorcraft which have primary missions ranging from scout and attack to utility and cargo. Additional requirements or modified standards may be required for rotorcraft that operate from ships in varying sea states.

2. APPLICABLE DOCUMENTS

2.1. General. The following specifications, standards, and handbooks listed in this section are the background and source material for this specification. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of this document.

2.2. Government documents.

2.2.1. Specifications, standards, and handbooks. The following specifications, standards, and handbooks from a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

2.2.2. Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

UNITED STATES GOVERNMENT PRINTING OFFICE

ADS-33E-PRF	-	Aeronautical Design Standard Performance Specification Handling Qualities Requirements for Military Rotorcraft
Special Report AMR-AF-08-07	-	Test Guide for ADS-33E-PRF

(Copies of this document are available from https://everyspec.com)

Special Report FCDD-AMV-19-01	-	Proposed Revisions to Aeronautical Design Standard -33E (ADS-33E-PRF) Toward ADS-33F-PRF
Technical Report 89-A-008	-	Background Information and User's Guide for Handling Qualities Requirements for Military Rotorcraft

(Copies of this document are available from https://discover.dtic.mil/)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

2.3. Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence unless otherwise noted. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1. General. This section provides the general military-unique handling qualities technical requirements that shall be used in the evaluation of rotorcrafts.

3.1.1. Required agility. The quantitative criteria of this specification have multiple boundaries that discriminate between rotorcraft that have to maneuver precisely and aggressively and those that can accomplish their mission tasks with limited agility and maneuverability. TABLE I indicates the limits that shall be met by associating a required agility with the intended Mission-Task-Elements (MTEs). If no criterion is provided for the required agility, the next available lower value shall apply.

3.1.2. Mission-Task-Elements. The MTEs specified in TABLE I shall be accomplished in accordance with the associated flight test maneuver in FIGURE 1 through FIGURE 25

3.1.3. Multi-crew rotorcraft. Unless otherwise stated, all requirements shall apply for the primary pilot station.

3.1.4. Levels of handling qualities. The overall rotorcraft Level of handling qualities shall be a combination of the two distinct methods of assessment, Predicted Levels and Assigned Levels.

3.1.4.1. Predicted levels of handling qualities. To obtain the Predicted Levels of handling qualities, the rotorcraft's flying qualities parameters shall be determined and compared with the criteria limits appropriate to the rotorcraft's operational requirements. For the predicted Level of handling qualities to be Level 1, the rotorcraft shall meet the Level 1 standards for all of the criteria.

3.1.4.2. Assigned levels of handling qualities. To determine the Assigned Level of handling qualities, test pilots shall use the Cooper-Harper Handling Qualities Rating (HQR) Scale in FIGURE 26. Definition of handling qualities levels**Error! Reference source not found.**in 6.2.11 to assess the workload and task performance required to perform the designated MTEs in 3.1.2. For the assigned Level of handling qualities to be Level 1, the rotorcraft shall be rated Level 1 for all of the MTEs designated as appropriate to the rotorcraft's operational requirements. The procuring activity shall judge the acceptability of any degradations when performing an MTE in light winds, and with load mass ratios greater than 0.33.

3.1.4.3. Interpretation of predicted vs. assigned levels of handling qualities. Failing only one or two of the predicted criteria by a small margin shall not necessarily determine the overall outcome. To resolve this, assessment of appropriate MTEs in 3.1.2 shall be the deciding factor. The MTEs to be used for such testing must be mutually agreed upon by the developer and procuring activity. The MTE assessment shall be conducted in flight or on a validated simulator, obtaining the HQRs per the methodology defined in 4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.1.5. Levels for normal states. The minimum Levels of flying qualities shall be Level 1 in the Operational Flight Envelopes and Level 2 in the Service Flight Envelopes.

3.1.6. Flight beyond the service flight envelopes. Flight beyond the Service Flight Envelope that does not involve structural failure, or unrecoverable loss of rotor RPM, shall be recoverable to the SFE without undue pilot skill.

3.1.7. Rotorcraft failures. The allowable degradation in handling qualities due to one or more system failures is specified in TABLE II as a function of the probability per flight hour that the failed state will be encountered. The degradation in handling qualities is quantified in terms of Levels, by application of the criteria in 3.2 through 3.5, and piloted evaluations of designated MTEs in 3.1.2. The methodology for calculating the handling qualities Level for each failed state is given below:

a. Review the Failure Modes and Effects Analysis (FMEA) and identify and tabulate all failures that are expected to adversely affect handling qualities.

b. Calculate the handling qualities performance criterion parameters (e.g., Bandwidth and Phase Delay, Attitude Quickness, etc.) for each of the tabulated failures.

c. Plot the parameters on the criterion boundaries (e.g., Bandwidth vs. Phase Delay), and note the Level that results. The criterion that results in the highest Level (most degraded handling qualities) defines the Level associated with that failure. Accomplish for all failure identified as adversely affecting handling qualities.

d. Enter TABLE II with the probability of encountering each Level to determine compliance. The probability of encountering a given Level is obtained from the System Safety Analysis (sum of probabilities of all independent failures that result in that Level).

3.1.7.1. Rotorcraft transients following failures. The rotorcraft transient following a flight control system failure or combination of flight control system failures shall not exceed the limits of TABLE III.

3.1.7.2. Indication of failures. Immediate and easily interpreted indications of failures shall be provided if such failures require a change of strategy or crew action.

3.1.8. Rotorcraft limits. The rotorcraft shall provide the pilot with clear and unambiguous warnings and indications of 10% control margins remaining. In near-earth operations, the warnings and indications shall be interpretable by the pilot with eyes out of the cockpit.

3.1.9. Devices for indication, warning, prevention, and recovery. Special devices used for indication, warning, prevention, and recovery during normal or inadvertent operation, shall not create a hazard to the rotorcraft, or prohibit flight within the Operational Flight Envelope.

3.1.10. Residual oscillations. Any sustained oscillations in any axis in calm air shall not interfere with the pilot's ability to perform the designated MTEs in 3.1.2. For Level 1, oscillations in attitude and in acceleration at the pilot's station greater than 0.5 degrees and 0.05g shall be considered excessive for any Response-Type and MTE. These requirements shall apply with the cockpit controls fixed and free. Residual motions that are classified as a vibration shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

be excluded from this requirement. Residual motions that are to be classified as vibrations shall be subject to procuring activity approval.

3.1.11. Rotor RPM governing. The rotor RPM shall remain within the limits established by the Service Flight Envelope during the execution of the designated MTEs in 3.1.2 conducted within in the Operational Flight Envelope.

3.1.12. Rotorcraft status. For each of the Settings and States defined in 6.2.6 and 6.2.7 a set of Loadings, Configurations, and Flight Conditions (6.2.8) shall be selected by the contractor for demonstrating compliance with this specification. This selection shall include combinations that are critical from the point of handling qualities and shall be submitted to the Government for approval.

3.2. Response-types.

3.2.1. Required response-types. The minimum required response-types shall be in accordance with TABLE IV for hover and low speed, and TABLE V for forward flight.

3.2.1.1. Character of rate response-types. A response that exhibits a monotonic or oscillatory divergence in attitude or airspeed following a pulse control input of the cockpit controller is defined as a Rate Response-Type. For pulse input to the longitudinal or lateral controller, feet may remain on the pedals. For a pulse input to the pedals, hands may remain on the pitch/roll controller.

3.2.1.2. Character of rate command response-types. A response that produces a linear time response in pitch, roll, or yaw attitude following a step input of the cockpit controller force.

3.2.1.3. Rate command with heading (direction) hold. For hover, the yaw controller inputs required to maintain constant heading during rolling maneuvers shall not be objectionably large or complex.

3.2.1.4. Character of attitude hold and heading hold. If Attitude Hold or Heading (Direction) Hold is specified as a required Response-Type in 3.2.1, the response to a pulse input shall be as illustrated in FIGURE 27. Response for attitude hold and heading hold response types. The pulse shall be inserted directly into the control actuator, unless it can be demonstrated that the pulse cockpit controller input will produce the same response. The maximum required attitude/heading displacement from trim is ±10 degrees for pitch and heading, and ±20 degrees for roll.

Attitude and heading shall return to within one degree for all attitude/heading changes up to 10 degrees, and to within 10 percent of peak for attitude/heading changes above 10 degrees, following a pulse input, in less than 10 seconds for Useable Cueing Environment (UCE)=1, and in less than 7 seconds for UCE>1 or flight in IMC.

The attitude or heading shall remain within the specified limit for at least 30 seconds for Level 1.

3.2.1.5. Heading hold in hover. Following a release of the directional controller the rotorcraft shall capture the reference heading (in degrees) with transient excursions limited to one overshoot, which shall not exceed 2.5 degrees. In no case shall a divergence result from activation of the Heading Hold mode.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.2.1.6. Character of attitude command response-types. If Attitude Command is specified as a required Response-Type in 3.2.1, a step cockpit pitch (roll) controller force input shall produce a proportional pitch (roll) attitude change within 6 seconds. The attitude shall remain essentially constant between 6 and 12 seconds following the step input. However, the pitch (roll) attitude may vary between 6 and 12 seconds following the input, if the resulting ground-referenced translational longitudinal (lateral) acceleration is constant, or its absolute value is asymptotically decreasing towards a constant.

3.2.1.6.1. Attitude command response-type in hover and low speed. Regardless of attitude response, this requirement is satisfied if the resulting inertial translational longitudinal (lateral) acceleration is constant, or its absolute value is asymptotically decreasing towards a constant.

3.2.1.6.2. Attitude command response-type in forward flight. The requirement shall apply for cyclic force inputs equal to at least three pounds or a deflection of one inch, whichever is less. This requirement may also be satisfied in the longitudinal axis if:

a. The gradient of longitudinal cyclic position vs. airspeed is less than or equal to 0.04 in/kt about the trim airspeed, and

b. The attitude hold verification criteria of 3.2.1.4 is met.

3.2.1.7. Character of translational rate response-types. A response that produces a proportional steady translational rate, with respect to the earth, in the appropriate direction following a constant pitch and roll controller force and deflection inputs.

3.2.1.8. Character of vertical rate response-types. The rotorcraft shall have a Vertical Rate Response-type if a constant deflection (force if an isometric control is used) of the vertical axis controller from trim produces a constant steady-state vertical velocity.

3.2.1.8.1. Altitude (height) hold relaxation. The rotorcraft system requirement of a Vertical Rate Command with Altitude (Height) Hold may be relaxed if the Vertical Translational Rate Visual Cue Rating is 2 or better as defined in FIGURE 28. Visual cue rating scale and divided-attention operation is not required. The vertical response shall meet the definition of the Vertical Rate Response-Type in 3.2.1.8.

3.2.1.8.2. Character of vertical rate command with altitude (height) hold. A response that returns the rotorcraft to its original altitude without objectionable delays and with no overshoot following an altitude deviation induced by insertion and removal of an input directly into the vertical-axis actuator. For hover and low speed, the rotorcraft shall automatically hold altitude with respect to a flat surface for land-based operations, or with respect to the mean surface/flight deck deviation for a rough sea or sea-based operations, with the altitude controller free. For Level 1, the altitude deviation during the performance of the following MTEs shall not exceed desired Degraded Visual Environment (DVE) performance requirements of the Hover, Hovering Turn, Pirouette, and either Sidestep for Scout-Attack categories, or Lateral Reposition for Utility and Cargo categories.

3.2.1.9. Character of yaw response to lateral controller.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.2.1.9.1. Turn coordination. For low-speed and forward flight, during banked turn with any available Heading Hold modes disengaged, the rotorcraft heading response to lateral controller inputs shall remain aligned to the direction of flight.

3.2.1.9.2. Turn coordination additional requirement. Turn Coordination (TC) defined in 3.2.1.9.1 shall be provided as an available Response-Type for the slalom MTE in low-speed flight and for divided attention operations in IMC in forward flight. TC is not required at airspeeds below 15 knots.

3.2.2. Non-specified response-types. Response-Types not explicitly defined in 3.2.1 shall meet the requirements of this specification. In addition, their operation shall not result in excessive excursions in rotorcraft attitudes, or require objectionably complex or unfamiliar control strategies.

3.2.3. Inputs to control actuator. Control input shaping may be used to achieve the necessary command-response relationship for backup flight control systems. The requirements to check for adequate disturbance rejection via inputs directly into the control actuator (3.2.1.4 and 3.2.1.8.2) shall be waived for Levels 2 and 3.

3.2.4. Transition between airborne and ground operations. There shall be no tendency for uncommanded, divergent motions of any primary flight control surface when the rotorcraft is in contact with any potential landing platform.

3.2.5. Transfer between response-types. The transients and trim changes caused by the intentional transfer between Response-Types shall not be objectionable.

3.2.5.1. Annunciation of response-type to the pilot. If more than one response-type can be selected in a given axis, there shall be a clear and easily interpretable annunciation to the pilot indicating which of the Response-Types are currently engaged or armed.

3.2.5.2. Control forces during transfer. Following transfer between Response-Types, control forces required to suppress transients shall not exceed one-third of the appropriate limit control forces in TABLE VI. The rotorcraft shall initially be trimmed at a fixed operating point, and during the transfer the heading, altitude, rate of climb or descent, and speed shall be maintained without use of the trimmer controls.

3.2.5.3. Control system blending. Blending between response-types shall be essentially linear with time and occur within the time limitations specified below:

Blending during Deceleration: $2 \text{ sec} < t_{blend} < 10 \text{ sec}$

Blending during Acceleration: $2 \text{ sec} < t_{blend} < 5 \text{ sec}$

3.3. Hover and low speed requirements. The hover and low speed requirements shall apply throughout the applicable portions of the Operational and Service Flight Envelopes for operations up to 45 knots ground speed.

3.3.1. Equilibrium characteristics. The equilibrium pitch and roll attitudes required to achieve a no-wind hover, and to achieve equilibrium flight in a 35 knot relative wind from any

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

direction, shall not result in restriction to field-of-view or objectionable characteristics that would interfere with the accomplishment of the designated MTEs in 3.1.2.

3.3.2. Small-amplitude pitch (roll) attitude change.

3.3.2.1. Short-term response to control inputs (bandwidth). The pitch (roll) response to longitudinal (lateral) cockpit control position inputs shall meet the limits specified in FIGURE 29. Requirements for small-amplitude pitch (roll) attitude changes – hover and low speed. The rotorcraft's pitch (roll) bandwidth (ω_{BW}) and phase delay (τ_p) parameters shall be obtained from frequency responses as defined in FIGURE 30. Definitions of bandwidth and phase delay. It is desirable to also meet this criterion for controller force inputs. If the bandwidth for force inputs falls outside the specified limits, flight testing shall be conducted to determine that the force feel system is not excessively sluggish. For Attitude Command Response-Types, if the bandwidth defined by gain margin is less than the bandwidth defined by phase margin, or is undefined, the rotorcraft may be PIO prone. In this case flight testing shall be performed to determine acceptability.

3.3.2.2. Short-term pitch (roll) response to disturbance inputs. If an Attitude Hold response-type is required in 3.2.1 and the system meets the Character of Attitude Hold Response-Type in 3.2.1.4, the attitude response to disturbance input as defined in FIGURE 31. Disturbance rejection block diagrams shall have a Disturbance Rejection bandwidth (DRB) no less than 0.5 rad/sec in pitch attitude and 0.9 rad/sec in roll attitude. In addition, Disturbance Rejection Peak (DRP) shall be no greater than 5.0 dB in pitch attitude or roll attitude. The DRB and DRP parameters shall be obtained from frequency responses as defined in FIGURE 32. Definition of disturbance rejection bandwidth (DRB) and peak (DRP).

3.3.2.3. Mid-term response to control longitudinal (lateral) inputs. The mid-term response characteristics to inputs directly into the longitudinal (lateral) cockpit controller shall apply at all frequencies and for oscillations of all magnitudes that are noticeable to the pilot. Use of an Attitude Hold Response-Type shall constitute compliance with this requirement, as long as any oscillatory modes following a cockpit controller pulse input have an effective damping ratio of at least $\zeta = 0.35$. If Attitude Hold is not available, the applicable criterion shall depend on the required degree of pilot attention according to 3.3.2.3.1 and 3.3.2.3.2.

3.3.2.3.1. Fully attended operations. The fully attended mid-term response shall meet the limits in FIGURE 33a.

3.3.2.3.2. Divided attention operations. The divided attention mid-term response shall meet the limits in FIGURE 33b, except that the Level 1 damping ratio, ζ, shall not be less that $\zeta = 0.35$ at any frequency.

3.3.3. Moderate-amplitude pitch (roll) attitude changes (attitude quickness). The ratio of peak pitch (roll) rate to change in pitch (roll) attitude, $q_{pk}/\Delta\theta_{pk}$ ($p_{pk}/\Delta\phi_{pk}$), shall meet the limits specified in FIGURE 34. The required attitude changes shall be made as rapidly as possible from one steady attitude to another without significant reversal in the sign of the cockpit control input relative to the trim position. The attitude change for compliance with this requirement shall vary from 5 degrees in pitch (10 degrees in roll) to the limits of the Operational Flight

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

Envelope or 30 degrees in pitch (60 degrees in roll), whichever is less. This requirement is not applicable for Response-Types that are designated as applicable only to UCE = 2 or 3.

3.3.4. Large-amplitude pitch (roll) attitude changes. The achievable angular rate (for Rate Response-Types) or attitude change from trim (for Attitude Response-Types) shall be at least those specified in TABLE VII. Angular rates or attitudes are achieved by inputting appropriate cockpit control inputs in each axis (nose up and down or roll right and left) while limiting excursions in other axes. Response-Types that are designated as applicable exclusively to UCE = 2 or 3 are only required to meet the Limited Agility Requirements in 3.1.1.

3.3.5. Small-amplitude yaw attitude changes.

3.3.5.1. Short-term response to yaw control inputs (bandwidth). The yaw response to directional cockpit control position inputs shall meet the limits specified in FIGURE 35. The bandwidth (ω_{BW}) and phase delay (τ_p) parameters are obtained from attitude to frequency responses as defined in FIGURE 30. It is desirable to also meet this criterion for controller force inputs. If the bandwidth for force inputs falls outside the specified limits, flight testing shall be conducted to determine that the force feel system is not excessively sluggish.

3.3.5.2. Short-term yaw response to disturbance inputs. If a Heading (Direction) Hold response-type is required in 3.2.1 and the system meets the Character of Heading Hold Response-Type in 3.2.1.4, the heading response to disturbance inputs as defined in FIGURE 31 shall have a Disturbance Rejection Bandwidth (DRB) no less than 0.7 rad/sec and Disturbance Rejection Peak (DRP) no greater than 5.0 dB in heading. The DRB and DRP parameters shall be obtained from frequency responses as defined in FIGURE 32.

3.3.5.3. Yaw rate response to lateral gusts. The peak yaw rate within the first three seconds following a step lateral gust input shall be such that the ratio, r_{pk}/V_g, shall not exceed 0.30 (deg/sec)/(ft/sec) for Level 1 or 1.0 (deg/sec)/(ft/sec) for Level 2. This requirement shall apply for lateral gust magnitudes from 10 to 25 knots in the presence of a steady wind of up to 25 knots from the most critical direction, except that the total wind velocity need not exceed 35 knots. This requirement shall be demonstrated with analysis and simulation.

3.3.5.4. Mid-term response to Yaw control inputs. The mid-term response characteristics shall apply at all frequencies and for oscillations of all magnitudes that are noticeable to the pilot. Use of a Heading Hold Response-Type shall constitute compliance with this paragraph, as long as any oscillatory modes following a cockpit controller pulse input have an effective damping ratio of at least $\zeta = 0.35$. If Heading Hold is not available, the applicable criterion shall depend on the required degree of pilot attention according to 3.3.5.4.1 and 3.3.5.4.2.

3.3.5.4.1. Fully attended operations. For fully attended operation, the mid-term response shall meet the oscillation limits in FIGURE 36a, except that the Level 1 limit on effective damping ratio, ζ, for oscillations with natural frequencies, ω_n, greater than 0.5 rad/sec is relaxed from 0.35 to 0.19.

3.3.5.4.2. Divided attention operations. For divided attention operations, the mid-term response shall meet the limits in FIGURE 36b, except that the Level 1 damping ratio, ζ, shall not be less than 0.19 at any frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.3.6. Moderate-amplitude heading attitude changes (attitude quickness). The ratio of peak yaw rate to change in heading, $r_{pk}/\Delta\psi_{pk}$, shall meet the limits specified in FIGURE 37. The required heading changes shall be made as rapidly as possible from one steady heading to another and without significant reversals in the sign of the cockpit control input relative to the trim position. This requirement is not applicable for Response-Types that are designated as applicable only to UCE = 2 or 3.

3.3.7. Large-amplitude heading changes. The achievable yaw rate in hover shall be at least the values specified in TABLE VII. The specified angular rates shall be achieved about the yaw axis while limiting excursions in the other axes with the appropriate control inputs, and with main rotor RPM at the lower sustained operating limit. Response-Types that are designated as applicable only to UCE = 2 or 3 shall meet at least the Limited Agility requirements in3.1.1.

3.3.8. Hover and low speed interaxis coupling. This section defines the hover and low speed requirements for off axes coupling response. Control inputs in one axis should not result in objectionable responses in one or more of the other axes.

3.3.8.1. Yaw due to collective for aggressive agility. The yaw rate response to abrupt step collective control inputs with the directional controller fixed shall not exceed the limits specified in FIGURE 38. The directional controller may be free if the rotorcraft is equipped with a heading hold function. Pitch and roll attitudes shall be maintained essentially constant. In addition, there shall be no objectionable yaw oscillations following step or ramp collective changes in the positive and negative directions. Oscillations involving yaw rates greater than 5 deg/sec shall be deemed objectionable.

3.3.8.2. Pitch due to roll and roll due to pitch coupling for aggressive agility. The ratio of peak off-axis attitude response from trim within 4 seconds to the desired (on-axis) attitude response from trim at 4 seconds, $\Delta\theta_{pk}/\Delta\phi_4$ ($\Delta\phi_{pk}/\Delta\theta_4$), following an abrupt lateral (longitudinal) cockpit control step input, shall not exceed ± 0.25 for Level 1 or ± 0.60 for Level 2. Heading shall be maintained essentially constant.

3.3.8.3. Pitch due to roll and roll due to pitch coupling for target acquisition and tracking. The pitch due to roll (q/p) and roll due to pitch (p/q) coupling for Target Acquisition and Tracking shall not exceed the limits specified in FIGURE 39. The average q/p and average p/q are derived from ratios of pitch and roll frequency responses. Specifically, average q/p is defined as the magnitude of pitch-due-to-roll control input (q/δ_{lat}) divided by roll-due-to-roll control input (p/δ_{lat}) averaged between the bandwidth and neutral-stability (phase = -180 deg) frequencies of the pitch-due-to-pitch control inputs (θ/δ_{lon}). Similarly, average p/q is defined as the magnitude (p/δ_{lon}) divided by (q/δ_{lon}) between the roll-axis (ϕ/δ_{lat}) bandwidth and neutral stability frequencies. Off-axis inputs shall be minimized while generating the frequency response data. Multi-input/multi-output frequency response identification techniques shall be used to analyze the frequency responses to account for the effect of inadvertent multiple-axis control inputs which may have been present during excitation.

3.3.9. Response to collective controller.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.3.9.1. Height response characteristics. The vertical rate response shall have a qualitative first-order appearance for at least 5 seconds following a step collective input. If the most rapid input achievable is not a clear step, the time zero shall be defined as shown in FIGURE 40. Pitch, roll, and heading excursions shall be minimized. The limits on the parameters defined by the following equivalent first-order vertical-rate-to-collective transfer function shall be in accordance with TABLE VIII.

$$\frac{\dot{h}}{\delta_c} = \frac{Ke^{-(\tau_{\dot{h}_{eq}} s)}}{T_{\dot{h}_{eq}} s + 1}$$

The equivalent system parameters shall be obtained using the time domain fitting method defined below.

The coefficient of determination, r^2, shall be greater than 0.97 and less than 1.03 for compliance with the requirement.

Obtain readings ft/sec from response to step collective input at intervals of no greater than t = 0.05 sec for a time span of 5 seconds – a total of $n = 5/\Delta t + 1$ data points (minimum n = 101).

Use a three variable nonlinear least squares algorithm to obtain a best fit curve to this data in the time domain using the following form for the estimated $\dot{h}(\dot{h}_{est})$

$$\dot{h}_{est}(t) = K\left[1 - \exp\left[-(t - \tau_{\dot{h}_{eq}})/T_{\dot{h}_{eq}}\right]\right] \quad \text{for } t > 0$$

where t is time (sec) and K, $1/T_{\dot{h}_{eq}}$ and $\tau_{\dot{h}_{eq}}$ are the variables. (Note $\tau_{\dot{h}_{eq}}$ may be less than zero.)

The function to be minimized is the sum of squares of the error (e), defined as,

$$e^2 = \sum_{i=1}^{n}\left[\dot{h}(t = t_i) - \dot{h}_{est}(t = t_i)\right]^2$$

where t_i is the time (sec) at the ith observed data point

The goodness of fit of the estimated curve shall be determined by the coefficient of determination (r^2) which is defined as

$$r^2 = \frac{\sum_{i=1}^{n}\left[\dot{h}_{est}(t = t_i) - \dot{h}_m\right]^2}{\sum_{i=1}^{n}\left[\dot{h}(t = t_i) - \dot{h}_m\right]^2}$$

where $\dot{h}_m$ is the mean of the observed $\dot{h}$,

$$\dot{h}_m = \sum_{i=1}^{n}\frac{\dot{h}(t = t_i)}{n}$$

3.3.9.2. Torque response. Torque, or any other parameter displayed to the pilot as a measure of the maximum allowable power that can be commanded without exceeding engine or transmission limits, shall have dynamic response characteristics that fall within the limits

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

specified in FIGURE 41. This requirement shall apply if the displayed parameter must be manually controlled by the pilot to avoid exceeding displayed limits.

3.3.9.3. Vertical axis control power. From a spot OGE hover with the wind vector from the most critical direction at a velocity of up to 35 knots, and with the most critical loading and density altitude, the rotorcraft shall attain a minimum vertical rate in not more than 1.5 seconds.

Level 1: 160 ft/min

Level 2: 55 ft/min

Level 3: 40 ft/min

3.3.9.4. Short-term height response to disturbance inputs. If a Vertical Rate with Altitude (Height) Hold response-type is required in 3.2.1 and the system meets the Character of Vertical Rate Command with Altitude (Height) Hold Response-Type in 3.2.1.8.2, the height response to disturbance inputs as defined in FIGURE 31 shall have a Disturbance Rejection Bandwidth (DRB) no less than 0.17 rad/sec and Disturbance Rejection Peak (DRP) no greater than 4.0 dB in height (z). The DRB and DRP parameters shall be obtained from frequency responses as defined in FIGURE 32.

3.3.10. Position hold. This requirement is applicable when Position Hold is a required Response-Type in 3.2.1. When Position Hold is engaged, the rotorcraft shall automatically hold its position with respect to a ground fixed or shipboard hover reference. The rotorcraft shall maintain its position within a 10 ft diameter circle during a 360 degree turn in a steady wind of up to 15 knots. The 360-degree turn shall be accomplished by the use of the directional controller with the pitch and roll controllers free and collective control as required to maintain constant altitude. The maneuver shall be completed in less than 10 seconds if Aggressive agility is required, 30 seconds if only Moderate agility is required and 45 seconds for Limited agility. The pitch and roll attitudes shall not exceed ±18 degrees from trim at any point in the 360-degree turn.

3.3.10.1. Short-term position response to disturbance. If Position Hold is required response-type in 3.2.1 and meets the requirement in 3.3.10, the position response to disturbance inputs as defined in FIGURE 31 shall have a Disturbance Rejection Bandwidth (DRB) no less than 0.17 rad/sec in both axial (x) and lateral (y) displacements. In addition, Disturbance Rejection Peak (DRP) shall be no greater than 4.0 dB in both axial (x) and lateral (y) displacements. The DRB and DRP parameters shall be obtained from frequency responses as defined in FIGURE 32.

3.3.10.2. Position hold annunciation. While Position Hold is engaged, there shall be a clear annunciation to the pilot indicating status of the Position Hold function.

3.3.11. Translational rate response type. For Response-Types designated as Translational Rate Command, the translational rate response to step cockpit pitch (roll) control position or force inputs shall have a qualitative first order appearance, and shall have an equivalent rise time, $T_{\dot{x}_{eq}}$ ($T_{\dot{y}_{eq}}$), no less than 2.5 seconds and no greater than 5 seconds. The parameter $T_{\dot{x}_{eq}}$ ($T_{\dot{y}_{eq}}$) is

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

defined by FIGURE 42a. For rotorcraft to obtain the Handling Qualities Rating of Level 1, the following shall be verified:

a. The pitch and roll attitudes shall not exhibit objectionable overshoots in response to a step cockpit controller input.

b. Zero cockpit control force and deflection shall correspond to zero translational rate with respect to fixed objects.

c. There shall be no objectionable overshoots in the response of translational rate to control inputs. The gradient of translational rate with control inputs shall be smooth and continuous.

In addition, for centerstick controllers, the variation in translational rate with control deflection should lie within the limits of FIGURE 42b. For sidestick controllers, the variation in translational rate with control force should lie within the limits of FIGURE 42c.

3.3.11.1. Short-term translational rate response to disturbance inputs. If a Translational Rate response-type is required in 3.2.1 and the system meets the Character of Translational Rate Response-Type in 3.2.1.7, the transitional rate response to disturbance inputs are defined in FIGURE 31 shall have a Disturbance Rejection Bandwidth (DRB) no less than 0.34 rad/sec in surge (u) and 0.54 rad/sec in sway (v). In addition, Disturbance Rejection Peak (DRP) shall be no greater than 5.0 dB in surge or sway. The DRB and DRP parameters shall be obtained from frequency responses as defined in FIGURE 32.

3.3.12. Pitch (roll) response to externally slung loads. With an externally slung load of less than 0.25 Load-Mass Ratio (LMR), the load bandwidth ($\omega_{BW\text{-}LD}$) and delta dB (Δ_{dB}) parameters shall fall within the delta HQR < 1 region of FIGURE 43. Configurations with LMRs greater than 0.25 may fall within the delta HQR > 1 region of FIGURE 43 if the baseline pitch (roll) attitude frequency response is Level 1 in accordance with the boundaries in FIGURE 29c, FIGURE 29d, and FIGURE 29e as well as with concurrence of the procuring activity.

The baseline pitch (roll) attitude frequency response corresponds to the aircraft internally loaded to the lesser of either the gross weight as the aircraft plus the externally slung load, or the aircraft maximum internally loaded gross weight. This criteria may be used as an alternative to Hover MTE evaluations of externally slung load configurations with approval of the procuring activity.

The load bandwidth ($\omega_{BW\text{-}LD}$) and delta dB (Δ_{dB}) parameters shall be obtained from the baseline and externally loaded pitch (roll) attitude frequency responses as defined in FIGURE 44.

$\omega_{BW\text{-}LD}$ is the lesser of -135 deg crossing or frequency of minimum phase due to the load mode.

Δ_{dB} is the deformation of magnitude curve due to externally slung load, estimated from centroid as follows:

a. Define areas between frequency responses:

- Determine the minimum and maximum frequency for the area by finding the intersections of the two frequency responses closest to the load mode frequency (or the load mode frequency ± 0.2 rad/s if crossings exceed load mode freq. ± 0.2 rad/s)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

- Calculate the area strictly under the curve between these frequencies

b. Calculate centroid from: centroid = $\frac{\sum_{i=1}^{n} dA_i * y_i}{A}$

c. Rate command Δ_{dB}: pitch Δ_{dB} = 4*centroid, roll Δ_{dB} = 3*centroid
Attitude command Δ_{dB}: pitch Δ_{dB} = 3*centroid, roll Δ_{dB} = 3*centroid

3.4. Forward flight requirements. The forward flight requirements shall apply throughout the applicable portions of the Operational and Service Flight envelopes for operations at greater than 45 knots groundspeed.

3.4.1. Pitch attitude response to longitudinal controller.

3.4.1.1. Short-term response (bandwidth). The pitch attitude response to longitudinal cockpit control position inputs shall meet the limits specified in FIGURE 45. The bandwidth (ω_{BW_θ}) and phase delay (τ_{p_θ}) parameters shall be obtained from frequency responses as defined in FIGURE 30. It is desirable to also meet this criterion for controller force inputs. If the bandwidth for force inputs falls outside the specified limits, flight testing shall be conducted to determine that the force feel system is not excessively sluggish. For Attitude Command Response-Types, if the bandwidth defined by gain margin is less than the bandwidth defined by phase margin, or is undefined, the rotorcraft may be prone to Pilot Induced Oscillations (PIO). In this case flight testing shall determine acceptability.

3.4.1.2. Mid-term response to control inputs. The mid-term response characteristics shall apply at all frequencies and for oscillations of all magnitudes that are noticeable to the pilot. Use of an Attitude Hold Response-Type shall constitute compliance with this requirement, as long as any oscillatory modes following a pulse controller input have an effective damping ratio of at least $\zeta = 0.35$. If Attitude Hold is not available, the applicable criterion shall depend on the required degree of pilot attention according to 3.4.1.2.1 and 3.4.1.2.2.

3.4.1.2.1. Fully attended operations. Mid-term response for fully attended operations shall meet the limits in FIGURE 33a.

3.4.1.2.2. Divided attention operations. Mid-term response for divided attention operations shall meet the limits in FIGURE 33b, except that the Level 1 damping ratio, ζ, shall not be less than 0.35 at any frequency.

3.4.1.3. Mid-term response – maneuvering stability. The following maneuvering stability requirements are applicable at all airspeeds greater than 45 knots.

3.4.1.3.1. Control feel and stability in maneuvering flight at constant speed. The incremental control force required to produce a change in normal load factor (n_z) and pitch rate (q) shall be in the same sense (aft force – increase load factor, forward force – decrease load factor) as required to initiate the change. In steady turning flight at constant airspeed, and in pullups and pushovers, for Levels 1 and 2 there shall be no tendency for the rotorcraft pitch attitude or angle of attack to diverge aperiodically. The requirement shall apply for all local gradients.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.4.1.3.2. Control forces in maneuvering flight. The variations in longitudinal cockpit control force with steady-state normal acceleration shall have no objectionable nonlinearities throughout the Operational Flight Envelope. At no time shall a negative local gradient be permitted. In addition, deflection of the longitudinal cockpit controller shall not lead the control force at any frequency below 5 rad/sec. For Level 1 the following requirements shall apply:

a. For centerstick controllers, the local force gradient, Fs/n_z shall be not less than 3 lb/g and no greater than 15 lb/g.

b. For sidestick controllers, the local force gradient shall be not less than 3 lb/g and no greater than 6 lb/g.

c. The local slope of Fs vs. n_z should be constant over the range of normal accelerations within the Operational Flight Envelopes. A variation of more than 50 percent shall be considered as excessive.

3.4.2. Pitch control power. The rotorcraft shall meet the following load factor generation criteria. For Level 1, from trimmed, unaccelerated flight the rotorcraft shall achieve the load factor limits specified in the Operational Flight Envelopes during turns or pull-up/push-over maneuvers. For Level 2, the following criteria shall apply:

a. There must be sufficient pitch control authority to accelerate from 45 knots to the maximum level flight airspeed, and to decelerate back to 45 knots at constant altitude

b. There must be sufficient pitch authority to maintain altitude if full power is applied at 45 knots and if minimum power is applied at maximum level flight airspeed.

3.4.3. Flight path control. When operating at airspeeds on the front side of the power-required curve, the limits in 3.4.3.1 shall apply. For operation at airspeeds on the backside of the power-required curve, or when the limits in 3.4.3.1 cannot be met, the limits in 3.4.3.2 shall apply. For the purposes of this requirement, frontside operation shall be defined when the slope of the steady-state response of flight path angle vs. airspeed, $\Delta\gamma_{ss}/\Delta V_{ss}$, resulting from a step change in pitch attitude, with collective held fixed, is negative. Backside operation shall be defined when this slope is positive or zero.

3.4.3.1. Flight path response to pitch attitude (frontside). The vertical rate response shall not lag the pitch attitude response by more than 45 degrees at all frequencies below 0.40 rad/sec for Level 1 and 0.25 rad/sec for Level 2.

3.4.3.2. Flight path response to collective controller (backside). The vertical rate response shall have a qualitative first-order appearance for at least 5 seconds following a step collective input. Pitch attitude excursions shall be limited so that they have a negligible effect on the vertical rate response. The limits on the parameters defined by the following equivalent first-order vertical-rate-to-collective transfer function shall be in accordance with the values specified in TABLE IX.

$$\frac{\dot{h}}{\delta_c} = \frac{Ke^{-(\tau_{\dot{h}_{eq}} s)}}{T_{\dot{h}_{eq}} s+1}$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

The equivalent system parameters are to be obtained using the time domain fitting method defined in 3.3.9.1. The coefficient of determination, r^2, shall be greater than 0.97 and less than 1.03 for compliance with this requirement.

3.4.4. Longitudinal static stability. Push (pull) force on the longitudinal controller shall always be required to increase (decrease) speed. Without retrimming, steady state changes in controller force for increased (decreased) speed shall be push (pull) or zero. Meeting the requirements for pitch Attitude Command in 3.2.1.6 constitutes compliance with longitudinal static stability in this specification.

3.4.5. Roll attitude response to lateral controller.

3.4.5.1. Small-amplitude roll attitude response to control inputs (bandwidth). The roll attitude response to lateral cockpit control position inputs shall meet the limits specified in FIGURE 46. The bandwidth (ω_{BW_ϕ}) and phase delay (τ_{p_ϕ}) parameters shall be obtained from attitude to cockpit controller inputs frequency response as defined in FIGURE 30. It is desirable to also meet this criterion for controller force inputs. If the bandwidth for force inputs falls outside the specified limits, flight testing shall be conducted to determine that the force feel system is not excessively sluggish. For Attitude Command Response-Types, if the bandwidth defined by gain margin is less than the bandwidth defined by phase margin, or is undefined, the rotorcraft may be PIO prone. In this case flight testing shall be performed to determine acceptability.

3.4.5.2. Moderate amplitude attitude changes (attitude quickness). The ratio of peak roll rate to change in bank angle, $p_{pk}/\Delta\phi_{pk}$, shall meet the limits specified in FIGURE 47. The required attitude changes must be made as rapidly as possible from one steady attitude to another without significant reversals in the sign of the cockpit control input relative to the trim position. The attitude changes required for compliance with this requirement shall vary from 10 degrees to the limits of the Operational Flight Envelope or 60 degrees, whichever is less. The parameters in FIGURE 47 are defined in FIGURE 34e.

3.4.5.3. Large-amplitude roll attitude changes. The achievable roll rate (for Rate Response-Types) or attitude change from trim (for Attitude Response-Types) shall be not less than those specified in TABLE X. Yaw control may be used to reduce sideslip that retards roll rate (not to produce sideslip that augments roll rate).

3.4.6. Roll-sideslip coupling. The requirements on roll-sideslip coupling shall apply for both right and left lateral control commands of all magnitudes up to the magnitude required to meet the roll performance requirements of 3.4.5.2. The cockpit yaw controller shall be free. The parameters defined in FIGURE 48 shall be used.

3.4.6.1. Bank angle oscillations. The value of the parameter ϕ_{OSC}/ϕ_{AV} following a pulse lateral control command for Rate Response-Types or step command for Attitude Response-Types shall be within the limits specified in FIGURE 49 for Levels 1 and 2. The input shall be as abrupt as practical. For Levels 1 and 2, ϕ_{AV} shall always be in the direction of the lateral control command.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.4.6.2. Turn coordination. The amount of sideslip resulting from abrupt lateral control commands shall not be excessive or require complicated or objectionable directional control coordination. The ratio of the maximum change in sideslip angle to the initial peak magnitude in roll response, $|\Delta\beta/\phi_1|$, for an abrupt lateral control pulse command for Rate Response-Types or step command for Attitude Response-Types, shall not exceed the limit specified in FIGURE 50. In addition, if the ratio $|\phi/\beta|_d$ exceeds 0.20, then the product of $0.20\times\left|\Delta\beta/\phi_1\right|\times|\phi/\beta|_d$ shall not exceed the limits in FIGURE 50.

3.4.7. Yaw response to yaw controller.

3.4.7.1. Small-amplitude yaw response for target acquisition and tracking (bandwidth). The heading response to cockpit yaw control position inputs shall meet the limits specified in FIGURE 51. The bandwidth (ω_{BW_θ}) and phase delay (τ_{p_θ}) parameters shall be obtained from frequency responses as defined in FIGURE 30. It is desirable to also meet this criterion for controller force inputs. If the bandwidth for force inputs falls outside the specified limits, flight testing shall be conducted to determine that the force feel system is not excessively sluggish.

3.4.7.2. Large-amplitude heading changes for aggressive agility. The heading changes in 1 second following an abrupt step displacement of the yaw control shall not be less than:

Level 1: The lesser of 16 degrees or β_L

Level 2: The lesser of 8 degrees or ½ β_L

Level 3: The lesser of 4 degrees or ¼ β_L

Where β_L is the sideslip limit of the Operational Flight envelope in degrees. Other controls shall be fixed but may be used to reduce pitch and roll attitude excursions.

3.4.7.3. Yaw control with speed change. With the rotorcraft initially trimmed directionally, yaw control shall be sufficient to maintain heading constant with the yaw controller, at constant bank angle, when speed is rapidly increased or decreased 30 percent from the trim speed or 20 knots, whichever is less (except where limited by the boundaries of the Service Flight Envelope). For pedal controllers, the yaw control forces shall not be greater than one-half those of TABLE XI. For other yaw control types, the forces required shall not be objectionable to the pilot. These requirements shall be satisfied without retrimming and accomplished at constant power (altitude varies), and at constant altitude (power varies).

3.4.8. Lateral-directional stability.

3.4.8.1. Lateral-directional oscillations. The frequency, ω_n, and damping ratio, ζ, of the lateral-directional oscillations following a yaw doublet control input, shall meet the minimums specified in FIGURE 52. This requirement shall also be met for a roll control pulse input. The requirements shall be met with controls fixed and with them free for inputs of any magnitude that might be experienced in operational use. If the oscillation is nonlinear with amplitude, the requirements shall apply to each cycle of the oscillation. The lack of oscillations implies compliance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.4.8.2. Spiral stability. Following a lateral pulse control input, the time for the bank angle to double amplitude shall be greater than the following:

Level 1: 20.0 seconds

Level 2: 12.0 seconds

Level 3: 4.0 seconds

These requirements shall be met with the cockpit controls free and the rotorcraft trimmed for straight and level flight. The values specified apply to an exponential divergence and should not depend on the size of the control input. If the variation of roll angle with time is linear following the pulse control input, this requirement is satisfied.

3.4.9. Lateral-directional response characteristics in steady sideslips. The requirements of 3.4.9.1 through 3.3.9.4 shall be met while performing yaw-control-induced, steady zero-yaw-rate sideslips in the worst flight condition with the rotorcraft trimmed for straight flight.

3.4.9.1. Yaw control in steady sideslips (directional stability). For the sideslips specified in 3.4.9, right yaw control deflection and force shall be required in left sideslips and left yaw control deflection and force shall be required in right sideslips. For Levels 1 and 2, the following requirements shall apply. Between sideslip angles of ±15 degrees, or the sideslip limit of the Operational Flight Envelopes, whichever is less, the variation of yaw controller deflection and force shall be essentially linear with sideslip. For larger sideslip angles, an increase in yaw control deflection shall always be required for an increase in sideslip, and the following requirements shall apply:

Level 1: The gradient of sideslip angle with yaw control force shall not reverse slope.

Level 2: The gradient of sideslip angle with yaw control is permitted to reverse slope provided the sign of the yaw control force does not reverse.

The term gradient does not include that portion of the yaw control force versus sideslip-angle curve within the preloaded breakout force or friction band.

3.4.9.2. Bank angle in steady sideslips. For the sideslips specified in 3.4.9, an increase in right bank angle shall accompany an increase in right sideslip, and an increase in left bank angle shall accompany an increase in left sideslip.

3.4.9.3. Lateral control in steady sideslips. For the sideslips specified in 3.4.9, right (left) sideslips shall not require left (right) lateral control deflection or force. For Levels 1 and 2, the variation of lateral control deflection and force with sideslip angle shall be essentially linear.

3.4.9.4. Positive effective dihedral limit. For Level 1, positive effective dihedral (right roll control for right sideslip and left roll control for left sideslip) shall never be so great that more than 75 percent of the roll control power available to the pilot and no more than 11 pounds of roll control force (for centerstick controllers) are required for sideslip angles that might be experienced in performing the designated MTEs in 3.1.2. The corresponding limits for Level 2 shall be 90 percent and 13.5 pounds.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

3.4.10. Forward flight interaxis coupling. Control inputs in one axis should not result in objectionable responses in one or more of the other axes while performing any of the designated MTEs in 3.1.2.

3.4.10.1. Pitch attitude due to collective control.

3.4.10.1.1. Small collective inputs. The peak change in pitch attitude from trim, $\Delta\theta_{peak}$, occurring within the first 3 seconds following a step change in collective causing less than 20% torque change, shall be such that the ratio $\left|\Delta\theta_{peak}/\Delta n_{z_{peak}}\right|$ is no greater than 1.0 deg/ft/sec^2, where $\Delta n_{z_{peak}}$ is the peak incremental normal acceleration from 1g flight.

3.4.10.1.2. Large collective inputs. The peak change in pitch attitude from trim, $\Delta\theta_{peak}$, occurring within the first 3 seconds following a step change in collective causing greater than or equal to 20% torque change, shall be such that the ratio $\left|\Delta\theta_{peak}/\Delta n_{z_{peak}}\right|$ is no greater than 0.5 deg/ft/sec^2 in the up direction and 0.25 deg/ft/sec^2 in the down direction.

3.4.10.1.3. Pitch control in autorotation. During an autorotation to touchdown from any point in the Operational Flight Envelopes, there shall be at least 10 percent of the total (stop-to-stop) pitch control margin remaining throughout the maneuver.

3.4.10.2. Roll due to pitch coupling for aggressive agility. The ratio of peak roll attitude response from trim within 4 seconds to the desired pitch attitude response from trim at 4 seconds, $\Delta\theta_{pk}/\Delta\theta_4$, shall not exceed ± 0.25 for Level 1 or ± 0.60 for Level 2.

3.4.10.3. Pitch due to roll coupling for aggressive agility. The pitch response resulting from bank to bank maneuvering shall not be objectionable to the pilot.

3.4.10.4. Pitch due to roll and roll due to pitch for target acquisition and tracking. The pitch due to roll (q/p) and roll due to pitch (p/q) coupling for Target Acquisition and Tracking shall not exceed the limits specified in FIGURE 39. The average q/p and average p/q are derived from ratios of pitch and roll frequency responses. Specifically, average q/p is defined as the magnitude of pitch-due-to-roll control input (q/δ_{lat}) divided by roll-due-to-roll control input (p/δ_{lat}) averaged between the bandwidth and neutral-stability (phase = -180 deg) frequencies of the pitch-due-to-pitch control inputs (θ/δ_{lon}). Similarly, average p/q is defined as the magnitude (p/δ_{lon}) divided by (q/δ_{lon}) between the roll-axis (ϕ/δ_{lat}) bandwidth and neutral stability frequencies. Off-axis inputs shall be minimized while generating the frequency response data. Multi-input/multi-output frequency response identification techniques shall be used to analyze the frequency responses to account for the effect of inadvertent multiple-axis control inputs which may have been present during the excitation.

3.5. Controller characteristics.

3.5.1. Conventional controllers.

3.5.1.1. Centering and breakout forces. Pitch, roll, and yaw controls shall exhibit positive centering in flight at any normal trim setting. The combined effects of centering, breakout force, stability, and force gradient shall not produce objectionable flight characteristics or permit noticeable departures from trim conditions with controls free. Breakout forces, including

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

friction, preload, etc., refer to the cockpit control force required to start movement of the control surface in flight. The breakout forces shall be within the limits specified in TABLE XII for hover and low speed and TABLE XIII for forward flight. The change in breakout force with speed shall not be objectionable. Measurement of breakout forces on the ground will ordinarily suffice in lieu of actual flight measurement, provided qualitative agreement between ground measurement and flight observation can be established.

3.5.1.2. Force gradients. The pitch, roll, and yaw control force gradients shall be within the range specified in TABLE XIV throughout the range of control deflections. In addition, the force produced by a one-inch travel from trim by the gradient chosen shall not be less than the breakout force. For the remaining control travel, the local gradients shall not change by more than 50 percent in one inch of travel.

3.5.1.3. Limit control force. Unless otherwise specified in particular requirements, the maximum control forces required, without retrimming, for any maneuver consistent with the specified Mission-Task-Elements in 3.1.2 shall not exceed the values stated in TABLE VI.

3.5.2. Sensitivity and gradients. The pitch, roll, yaw, and collective controller sensitivities and gradients shall be consistent with the rotorcraft dynamic response characteristics in each axis at all flight conditions. In no case shall the controller sensitivity or gradient produce responses that are objectionably abrupt or sluggish.

3.5.3. Cockpit control free play. The free play in each control, that is, any motion of the cockpit control that does not move the appropriate moment- or force-producing device in flight, shall not be objectionable.

3.5.4. Control harmony. The control forces, displacements, and sensitivities of the pitch, roll, yaw, and collective controls shall be compatible, and their responses shall be harmonious.

3.5.5. Trimming characteristics. It shall be possible to trim controller forces to zero for all unaccelerated flight conditions. Actuation of the trim device to null controller force shall not produce an objectionable change in rotorcraft attitude or translational rate. Operation of the trim controls shall not require a force that is objectionably low or high. The trimmer shall not produce objectionable stick jump, trim rates, or slippage in the trim control position.

3.5.6. Dynamic coupling. There shall be no tendency for dynamic coupling between the rotorcraft and the controller with or without the pilot in the loop. In particular, there shall be no lightly damped, high frequency oscillations that cease when the pilot releases the controller.

4. VERIFICATION

4.1. General. Compliance with the requirements of this specification shall be demonstrated using analysis, simulation, and flight test at appropriate milestones during the rotorcraft design and development. In absence of other guidance from the system specification, TABLE XV and TABLE XVI shall be observed. TABLE XV indicates the appropriate milestones for analysis, simulation, and flight test. TABLE XVI indicates the range of Rotorcraft Status and Flight Conditions from which specific combinations shall be selected to

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

demonstrate compliance with this specification. Complexity and scope of these demonstrations are defined in 4.1.1, 4.1.2, and 4.1.3.

4.1.1. Analysis. Initially the quantitative flying qualities criteria shall be assessed analytically (non-piloted) using available math models that represent the aerodynamic, flight control system, and engine/fuel control characteristics of the rotorcraft. By PDR, and thereafter, analytical checks shall be accomplished using full nonlinear math models including best estimates of engine/fuel controls, rotor system, and SCAS elements. The Configurations, Loadings, and Flight Control modes should be well defined. HQ shall be evaluated at selected conditions throughout the OFE, at least for Normal States. By CDR the Failure States shall be defined and included in the analyses. Analysis shall be continued throughout the development program to supplement the simulation and flight testing.

4.1.2. Simulation. Piloted simulation with representative cockpit controls and displays including the feel system shall be used by the CDR milestone. The simulator shall include a visual attachment but does not require a motion system unless such a system is specified by the system specification. The simulation shall include a validated representation of all controls and displays that will be available to the pilot for control of the rotorcraft. Rotorcraft Status to be tested shall include all mission-required Configurations at nominal and critical Loadings. All Settings shall be available and evaluated. In addition, failure transients and steady state Failure States shall be investigated. Although the piloted simulation environment is primarily to investigate the MTEs, many of the quantitative tests are intended to include the pilot vehicle interface and so a sample shall be repeated in the piloted simulator environment. Any discrepancies between the piloted simulator results and the off-line math model results shall be investigated and resolved.

It has been found that tests should be performed to determine the Simulated Day Usable Cue Environment of the simulator (SIMDUCE), using the test methodology for determination of the UCE. If the SIMDUCE is greater than one, it is likely that the pilot ratings (HQRs) will be in the Level 2 range (HQR between 4 and 6), even though the actual rotorcraft would be Level 1. The key indicator of acceptable Level 1 HQ is that the pilot comments clearly associate the Level 2 deficiencies with deficient visual cueing.

Simulated vision aids shall be calibrated to ensure the visual acuity is consistent with the estimated performance of that vision aid in the DVE specified as part of the operating environment. It should be possible to achieve Level 1 pilot ratings in the simulated DVE, when employing the flight control modes designed for operations in DVE.

Validation of the simulator shall include comparisons of the response of the math model and the visual scene with responses obtained from the validated non-real-time simulation to identical cockpit controller inputs. The lag between the cockpit control input and the out-the-window visual scene response shall not cause a change from the rotorcrafts true Bandwidth and Phase Delay that is equivalent to more than ½ a point in estimated HQR. The increment of HQR is obtained by interpolation on the appropriate Bandwidth criteria figure. If the out-the-window scene is generated by a remotely located sensor, the head tracking dynamic response and the image source location shall be represented.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

4.1.3. Flight. For SVR verification, flight tests shall be performed to demonstrate both the quantitative and subjective requirements of this specification and to show that the predicted Levels of HQ and the assigned Levels of HQ are as required.

Based on the analysis and simulation results the contractor shall develop a test plan to adequately demonstrate the full-mission capability and define limiting conditions. The test plan shall be subject to approval by the procuring activity.

All of the quantitative criteria shall be tested. The tests shall be performed for a sample of Configurations and Loadings, all Normal States and selected Failure States, at a range of Flight Conditions. The sample shall be sufficient to assess validity of the analysis and simulation results, and if adequate fidelity is demonstrated further testing may be limited in scope. Emphasis shall be on data points that are critical from the standpoint of handling qualities and safety, but shall also demonstrate performance at important nominal mission conditions.

Qualitative flight test evaluations shall be performed to provide an overall check of the HQ. All applicable MTEs shall be demonstrated in primary mission configuration, with primary or augmented flight control mode in normal state(s) with Loadings that are most critical for HQ. Selected MTEs shall be performed to evaluate the HQ with secondary flight control mode (or configurations corresponding to $Pf > 10^{-5}$). All ground-referenced MTEs shall be performed in calm winds. In addition, the Hover, Hovering Turn, Pirouette, and Vertical Maneuver shall be performed in the Good Visual Environment (GVE) in light winds. The procuring activity will decide if the extent of HQR degradation due to light winds is acceptable.

All of the quantitative testing, and the flight tests of the GVE MTEs, shall be conducted in day GVE. The DVE MTEs shall be tested in real DVE when evaluating the primary or augmented SCAS in normal state. The Secondary flight control mode may be tested in simulated DVE.

It is not necessary to perform the MTE evaluations in a high density altitude environment unless the rotorcraft has been shown to not meet one or more of the quantitative boundaries, and satisfactory results in the MTE tests are being used to support a deviation.

4.2. Determination of usable cue environment. The Usable Cue Environment (UCE) shall be obtained from FIGURE 53 using visual cue ratings (VCRs) obtained from the flight assessments specified below. The VCRs shall be made by at least three pilots using the scale shown in FIGURE 28. The mean VCRs for each task shall be obtained by separately taking the average of all the pilot ratings in each axis. This will result in five average VCRs for each task: pitch, roll, and yaw attitude, and vertical and horizontal translational rate. This shall be reduced to two VCRs by taking the worst (numerically highest) average VCR among pitch, roll and yaw attitude, and between vertical and horizontal translational rate, for each task. These VCRs for attitude and translational rate shall be plotted on FIGURE 53 to obtain a UCE for each task. Points falling on a boundary in FIGURE 53 shall be considered to lie in the region of numerically higher UCE. The largest UCE value obtained in this process shall be used in TABLE IV to determine the required Response-Type. The visual cue ratings shall be determined using all displays and vision aids that are expected to be operationally available to the pilot, in the Degraded Visual Environments.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

4.2.1. Characteristic of test rotorcraft. The test rotorcraft shall have response characteristics that rank no higher than a Rate Response-Type as defined in 3.2.1.1 and 3.2.1.2. It shall be shown to be a Level 1 rotorcraft by demonstrating compliance with the applicable MTEs specified in 4.2.2, performed to DVE standards but in the GVE (UCE = 1).

4.2.2. Applicable Mission-Task-Elements. The following Mission-Task-Elements shall be flown when making the UCE assessments: Hover, Landing, Pirouette, Acceleration and Deceleration (or Depart/Abort), Sidestep (or Lateral Reposition), and Vertical Maneuver. The standard task description and DVE performance limits for each of these maneuvers shall apply when making the VCR determinations.

4.2.3. Dispersions among visual cue rating. Subject to the procuring activity, a level of UCE shall be assigned or additional pilots shall be used if the standard deviation of the individual visual cue ratings among the pilots is greater than 0.75.

4.3. Levels of handling qualities. Handling qualities depend on the rotorcraft's flying characteristics, the piloting task being performed, the Usable Cue Environment, wind and turbulence, and any additional demands on the pilot. Levels of handling qualities are defined by the 1 to 3, 4 to 6, and 7 to 8 ranges of the Cooper-Harper Handling Qualities Rating (HQR) Scale defined in FIGURE 26. This specification uses two distinct methods of establishing Levels of handling qualities, Predicted Levels and Assigned Levels. Predicted Levels are obtained from quantitative criteria that are based on HQR data from engineering in-flight and ground-based simulation. This knowledge is not complete; data are not available to fully define the required limit for all flying qualities parameters which taken together will ensure good handling qualities for all mission tasks. Assigned Levels are obtained from test pilots using the HQR Scale defined in FIGURE 26 to assess the workload and task performance required to perform designated MTEs in 3.1.2. The MTE flight test maneuvers are not sufficiently comprehensive to represent all mission maneuvers in all environments that a particular rotorcraft may be called upon to perform, nor is it practical to perform the designated MTEs throughout the OFE for all loadings. Since neither the Predicted Levels nor the Assigned Levels provide a complete comprehensive assessment of handling qualities, compliance with both methods is necessary to maximize the likelihood of an accurate assessment of the Level of handling qualities. If there are conflicts between the Predicted Levels and Assigned Levels, the quantitative criteria and the MTE results shall be scrutinized to determine the cause of the discrepancy. On the basis of this investigation, the procuring activity will determine whether or not compliance has been achieved or if further testing is required.

4.4. Mission-Task-Elements. The designated MTEs provide a basis for an overall assessment of the rotorcraft's ability to perform certain critical tasks, and result in an assigned level of Handling Qualities. To allow for different standards of precision and aggressiveness, the performance standards for each task are listed separately for different rotorcraft categories and for both Good Visual Conditions (GVE) and Degraded Visual Conditions (DVE). Generally, GVE means clear daylight with good cueing and unaided vision (UCE=1). DVE means specifically the operational environment defined in rotorcraft performance specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

Typically, DVE will be night with some level of illumination (moon and overcast) while using the actual mission equipment vision aid (UCE>1).

4.4.1. Conduct of test. The designated MTEs shall be performed with all combinations of manual flight control modes and displays available to the pilot and used as they would normally be used in the conduct of the maneuver. Altitude and position requirements refer to a selected reference point on the rotorcraft that is to be determined by the testing activity. Generally, this should be close to the pilot's eye point, but for small rotorcraft the Hovering Turn and Turn to Target may be performed about the center of gravity. All altitudes given are height above ground level. A description of a suggested test course is provided for each maneuver. However, the test course markings and detail are left to the discretion of the testing activity. The terms "calm winds," "moderate winds," "stabilized hover," and "landing gear" are defined in 6.2.

Each MTE shall be assessed by at least three test pilots. These test pilots shall each assign a subjective rating using the Cooper-Harper Handling Qualities Rating scale defined in FIGURE 26. The arithmetic average across all pilots of the Cooper-Harper Handling Qualities Ratings forms the overall rating for the MTE. All individual ratings and associated evaluation commentary shall be documented and supplied to the Government. The maneuvers shall be performed to assess the rotorcraft in configurations and states that are most critical for handling qualities in accordance with associated verification criteria.

4.4.2. Performance standards. The use of Cooper-Harper Handling Qualities Ratings requires the definition of numerical values for desired and adequate performance. These performance limits are set primarily to drive the level of aggressiveness and precision to which the maneuver is to be performed. Compliance with the performance standards may be measured subjectively from the cockpit or by the use of ground observers. It is not necessary to utilize complex instrumentation for these measurements. Experience has shown that lines painted on the rotorcraft and markers on the ground are adequate to provide sufficient cues for ground or onboard observers to perceive whether the rotorcraft is within desired or adequate performance parameters. In any event, the contractor shall develop a scheme for demonstrating compliance that uses at least outside observers and in-cockpit observations. This plan will be subject to approval by the Government. The evaluation pilot is to be advised any time his performance fails to meet the desired limits, immediately following the completion of the maneuver, and before the pilot rating is assigned. In cases where the performance does not meet the specified limits, it is acceptable for the evaluation pilot to make as many repeat runs as necessary to ensure that this is a consistent result. Repeat runs to improve performance may expose handling qualities deficiencies. Such deficiencies should be an important factor in the assigned pilot rating.

4.4.3. Visual cueing. If the inability to meet a performance standard in GVE is due to a lack of visual cueing, the test course should be modified to provide the required pilot cues. This is allowed in the context that the purpose of these maneuvers is to check rotorcraft handling, not problems associated with a lack of objects on the test course. To accommodate different performance standards the test courses may be modified to conduct the DVE maneuvers. If

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

additional modifications in visual cueing are required to enhance the visual cues in the DVE, the changes shall be defined and submitted to the Government for approval prior to testing.

4.5. Testing with externally slung loads. Testing of applicable MTEs with externally slung loads shall be accomplished with a load mass ratio of 0.30 or the maximum load that will be used for operational missions, whichever is less. If load mass ratios of greater than 0.30 will be used operationally, a configuration with the maximum load mass shall also be tested. The procuring activity will decide if HQR degradations at high load mass ratios are acceptable. Testing shall be accomplished in both GVE and DVE if required by the rotorcraft system specification.

4.6. Interpretation of subjective requirements. In several instances throughout the specification, subjective terms such as essentially constant, objectionable delays, excessively sluggish, and complex coordination, have been employed where insufficient information exists to establish quantitative criteria. Such requirements shall be interpreted with due regard to the intent of the Level definitions of 3.1.4. Final determination of compliance with requirements so worded shall be determined through flight test or other suitable means subject to procuring activity approval.

5. PACKAGING

5.1. Packaging. For acquisitions purposes, the packaging requirements shall be as specified in the contract or order. When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but not mandatory.)

6.1. Intended use. This performance specification establishes requirements for the flying qualities and handling qualities of rotorcraft or rotorcraft systems. It is intended that the specification should cover rotorcraft that have primary missions ranging from scout and attack to utility and cargo. Additional requirements or modified standards may be required for rotorcraft that operate from small ships in sea states resulting in more than small ship motion.

The requirements are intended for application in development of a new system, or for a major system upgrade. If all requirements are met, they should assure that no limitation on flight safety, or on the capability to perform intended missions, will result from deficiencies in flying qualities. The criteria and MTEs can also be used as a development tool since they provide quantitative benchmarks and calibrated maneuver standards which may be valuable for assessing the need for upgrades, or for diagnosing the cause of deficient performance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

Application of this specification to a specific system requires minimal tailoring. This tailoring basically consists of selecting the appropriate MTEs corresponding to the operational mission and defining several aspects of the operational environment. Guidance on tailoring these topics will be provided by the procuring activity.

6.2. Definitions.

6.2.1. Acronyms. The acronyms used in this specification are defined as follows:

A	-Analysis methods for verification
ACAH	-Attitude Command/Attitude Hold Response-Type
AH	-Attitude Hold
CDR	-Critical Design Review – Midterm project milestone
DH	-Direction [Heading] Hold
DVE	-Degraded Visual Environment to be defined by system specification, but typically night with some form of visual aid such as night vision goggles
FFR	-First Flight Readiness Review – Project milestone
F_s/n	-Control force per g
GVE	-Good Visual Environment, typically clear day light with adequate visual cues
HH	-Height [Altitude] Hold
HQ	-Handling Qualities
HQR	-Cooper-Harper Handling Qualities Rating
IMC	-Instrument Meteorological Conditions
MTE	-Mission-Task-Element, specifically refers to maneuvers in TABLE I.
NA	-Not Applicable
n_L (+)	-Positive load factor limit of the Operational Flight Envelope
n_L (-)	-Negative load factor limit of the Operational Flight Envelope
OFE	-Operational Flight Envelope
OGE	-Out of Ground Effect
PDR	-Preliminary Design Review – Early project milestone
Pf	-Probability of encountering a failure, typically expressed as failure per flight hour
PH	-Position Hold
PIO	-Pilot-Induced Oscillation
RATE	-Rate Response-Type
RCAH	-Rate Command/Attitude Hold Response-Type
RCDH	-[Yaw] Rate Command/Direction [Heading] Hold Response-Type
RCHH	-[Vertical] Rate Command/Height [Altitude] Hold Response-Type
RPM	-Revolutions per minute
SCAS	-Stability and Control Augmentation System
SFE	-Service Flight Envelope
SFR	-System Functional Review – Early milestone in project evolution
S	-Simulation method for verification

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

SVR	-System Verification Review – Late milestone in project evolution
T	-Test method for verification
TC	-Turn Coordination
TRC	-Translational Rate Command
UCE	-Usable Cue Environment
VCR	-Visual Cue Rating
V_H	-Maximum level flight airspeed at maximum continuous power
✓	-Specification require is applicable

6.2.2. Flight envelopes. The Flight Envelopes should define and clearly indicate the effects of rotorcraft configuration, loadings, settings and states.

6.2.2.1. Operational flight envelopes (OFE). The Operational Flight Envelopes define the boundaries within which the rotorcraft must be capable of operating in order to accomplish the rotorcraft's system specification operational missions. These envelopes are defined in terms of combinations of airspeed, altitude, load factor, rate-of-climb, side-velocity, and any other parameters specified by the system specification, as necessary to accomplish the operational missions. Any warnings or indications of limiting or dangerous flight conditions, required by 3.1.9, should occur outside the OFEs.

6.2.2.2. Service flight envelopes (SFE). The Service Flight Envelopes is derived from rotorcraft limits as distinguished from mission requirements. These envelopes are expressed in terms of the parameters used to define the OFEs, plus any additional parameters deemed necessary to define the appropriate limits. The inner boundaries of the SFEs are defined as coincident with the outer boundaries of the OFEs. The outer boundaries of the SFEs are defined by one or more of the following: uncommanded rotorcraft motions, or structural, engine/power-train, or rotor system limits. The magnitude of the differences between the inner and outer boundaries of the SFEs should be based on the guarantee of adequate margins as required by 3.1.8 and 3.1.9.

6.2.3. Configuration. A configuration is defined by the external geometry. This includes the positions of variable systems such as landing gear or flaps, location of external stores and carriage of sling loads.

6.2.4. Loadings. Loadings refer to the mass properties of a configuration and will be reflected in the total mass or weight, the center of gravity location, and the various moments of inertia.

6.2.5. Flight conditions. Flight Condition is defined by a unique combination of the parameters which define the Operational Flight Envelope.

6.2.6. Settings. Settings refers to the selected functionality of rotorcraft components or systems that affect rotorcraft response, or UCE, which can be activated or deactivated by the pilot.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

6.2.7. States. Rotorcraft States are Normal when the various systems are functioning as selected. Failure States exist when the functionality is modified by one or more malfunctions in rotorcraft components or systems that affect rotorcraft response or UCE.

6.2.8. Rotorcraft status. The Rotorcraft Status is defined by a unique combination of Configuration, Loading, Setting and State.

6.2.9. Degree of pilot attention. Some requirements depend on whether or not the pilot must attend to tasks other than flying the rotorcraft.

6.2.9.1. Fully attended operation. The pilot flying the rotorcraft can devote full attention to attitude and flight path control. Requirements for divided attention are minimal.

6.2.9.2. Divided attention operation. The pilot flying the rotorcraft is required to perform non-control-related side tasks for a moderate period of time.

6.2.10. IMC operations. Instrument Meteorological Conditions (IMC) operations imply control of the rotorcraft solely with reference to the flight instruments. Occurs when the rotorcraft is clear of all obstacles.

6.2.11. Levels of handling qualities. Levels of handling qualities are defined by the 1 to 3, 4 to 6, and 7 to 8 ranges of the Cooper-Harper Handling Qualities Rating (HQR) Scale in FIGURE 26. This specification uses two distinct methods of establishing Levels of handling qualities, Predicted Levels and Assigned Levels. Predicted Levels are obtained by comparing the rotorcraft's flying qualities parameters with the boundaries appropriate to the rotorcraft's operational requirements. Assigned Levels are obtained from test pilots performing the designated MTEs. The results of these two methods are combined to determine the overall Level of handling qualities.

6.2.12. Landing gear. Wheels or skids used to support the rotorcraft while on the ground.

6.2.13. Load mass ratio. For externally slung loads, the load mass ratio is the ratio of external load mass to mass of rotorcraft plus external load.

6.2.14. Mission-task-element (MTE). An element of a mission that can be treated as a handling qualities task.

6.2.15. Response-Type. A characterization of the rotorcraft response to a control input in terms of well recognized stability augmentation systems (i.e., Rate, Rate Command/Attitude Hold, etc.). However, it is not necessary to use a stability augmentation system to achieve the specified characteristics. Response-type ranking from least to most stabilization is defined as follows:

1. RATE
2. RATE = RCDH
3. ACAH + RCDH
4. ACAH + RCDH + RCHH
5. ACAH + RCDH + RCHH + PH
6. TRC + RCDH + RCHH + PH

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

6.2.16. Sideslips. For sideslips in 3.4.9, right yaw control deflections are force are required in left sideslips and left yaw control deflection and force are required in right sideslips.

6.2.17. Control free play. Any motion of the cockpit control that does not move the appropriate moment- or force-producing device in flight.

6.2.18. Useable cueing environment. The measure of the quality of vision aid to provide the cues needed for rotorcraft control (e.g., fine grained texture in degraded visual environment). The Usable Cue Environment is obtained from FIGURE 53. Usable cue environments for visual cue ratings using the visual cue ratings (VCRs) scale in FIGURE 28. Visual cue rating scale during flight assessments.

6.2.19. Winds.

6.2.19.1. Calm winds. Winds with a steady component of less than 5 knots.

6.2.19.2. Light winds. Winds with a steady component of between 10 and 15 knots.

6.2.19.3. Moderate winds. Winds with a steady component of between 20 and 35 knots.

6.2.20. Near-earth operations. Operations sufficiently close to the ground or fixed objects on the ground, or near water and in the vicinity of ships, oil derricks, etc., that flying is primarily accomplished with reference to outside objects

6.2.21. Stabilized hover. A stabilized hover is one in which there are no objectionable oscillations in rotorcraft attitude or position.

6.2.22. Step input. A step input is defined as a rapid change in the controller force or position from one constant value to another. The input should be made as rapidly as possible without exciting undesirable structural or rotor modes, or approaching any rotorcraft safety limits. This differs from the classical definition, where the change occurs in zero time.

6.2.23. Speed ranges. Ground speed is defined to be the speed with respect to a hover reference which may itself be moving, such as for shipboard operations.

6.2.23.1. Hover. Hovering flight is defined as all operations occurring at ground speeds less than 15 knots.

6.2.23.2. Low Speed. Low-speed flight is defined as all operations occurring at ground speeds between 15 and 45 knots.

6.2.23.3. Forward flight. Forward flight is defined as all operations with a ground speed greater than 45 knots.

6.2.24. Objectionable. Aircraft characteristics that, while not necessarily unsafe, negatively impact the pilot's ability to perform any of the aircraft's missions or result in a measurable degradation in predicted or assigned levels of handling qualities.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

6.3. Subject term (key word) listing.

Agility
Disturbance
Dynamic
Evaluation
Equilibrium
Flying qualities
Flight test
Hover
Mission task element
Oscillations
Sensitivity
Testing

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE I. Mission-Task-Elements (MTE)

MTE	REQUIRED AGILITY	ROTORCRAFT CATEGORY				EXTERNALLY SLUNG LOAD
		ATTACK	SCOUT	UTILITY	CARGO	
Task in GVE						
Hover	L	✓	✓	✓	✓	✓
Landing	L	✓	✓	✓	✓	
Slope Landing	L	✓	✓	✓	✓	
Hovering Turn	M	✓	✓	✓	✓	
Hover Heading Capture	M	✓	✓	✓	✓	
Pirouette	M	✓	✓	✓	✓	
Vertical Maneuver	M	✓	✓	✓	✓	✓
Depart/Abort	M			✓	✓	✓
Lateral Reposition	M			✓	✓	✓
Slalom	M	✓	✓	✓	✓	
Vertical remask	A	✓	✓			
Acceleration and Deceleration	A	✓	✓			
Sidestep	A	✓	✓			
Deceleration to Dash	A	✓	✓	✓		
Transient Turn	A	✓	✓	✓		
Pullup/Pushover	A	✓	✓	✓		
Roll Reversal	A	✓	✓	✓		
Turn to Target	T	✓	✓			
High Yo-Yo	T	✓	✓			
Low Yo-Yo	T	✓	✓			
Task in DVE						
Hover	L	✓	✓	✓	✓	✓
Landing	L	✓	✓	✓	✓	
Hovering Turn	L	✓	✓	✓	✓	
Pirouette	L	✓	✓	✓	✓	
Vertical Maneuver	L	✓	✓	✓	✓	✓
Depart/Abort	L			✓	✓	✓
Lateral Reposition	L			✓	✓	✓
Slalom	L	✓	✓	✓		
Acceleration and Deceleration	L	✓	✓			
Sidestep	L	✓	✓			
Task in IMC						
Decelerating Approach	L	✓	✓	✓	✓	✓
ILS Approach	L	✓	✓	✓	✓	
Missed Approach	L	✓	✓	✓	✓	
Speed Control	L	✓	✓	✓	✓	

Notes: ✓ = Suggested maneuver to apply with appropriate performance standards
L = Limited Agility
M = Moderate Agility
A = Aggressive Agility
T = Target Acquisition and Tracking

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE II. Levels for rotorcraft failure states

PROBABILITY OF ENCOUNTERING	WITHIN OPERATIONAL FLIGHT ENVELOPE	WITHIN SERVICE FLIGHT ENVELOPE
Level 2 after failure	$< 2.5 \times 10^{-3}$ per flight hr	
Level 3 after failure	$< 2.5 \times 10^{-5}$ per flight hr	$< 2.5 \times 10^{-3}$ per flight hr
Loss of control	$< 2.5 \times 10^{-7}$ per flight hr	

TABLE III. Transients following failures

LEVEL	FLIGHT CONDITION		
	HOVER AND LOW SPEED	FORWARD FLIGHT	
		NEAR-EARTH	UP-AND-AWAY
1	3° roll, pitch, yaw 0.05g n_x, n_y, n_z No recovery action for 3.0 sec	Both Hover and Low Speed and Forward Flight Up-and-Away requirements apply	Stay within OFE. No recovery action for 10 sec
2	10° attitude change or 0.2g acceleration. No recovery action for 3.0 sec	Both Hover and Low Speed and Forward Flight Up-and-Away requirements apply	Stay within OFE. No recovery action for 5.0 sec
3	24° attitude change or 0.4g acceleration. No recovery action for 3.0 sec	Both Hover and Low Speed and Forward Flight Up-and-Away requirements apply	Stay within OFE. No recovery action for 3.0 sec

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE IV. Minimum response-types for hover and low speed – near earth

MTE	UCE = 1		UCE = 2		UCE = 3	
	Level 1	Level 2	Level 1	Level 2	Level 1	Level 2
Required Response-Type for all MTEs. Additional requirements for specific MTEs are given below.	RATE	RATE	ACAH	RATE + RCDH	TRC+RCDH + RCHH+PH	ACAH
Hover			RCDH+RCHH			RCDH+RCHH
Landing			RCDH			RCDH
Slope landing			RCDH			RCDH
Hovering turn			RCDH			RCDH
Hover heading capture			RCDH			RCDH
Pirouette			RCDH			RCDH
Vertical Maneuver			RCDH			RCDH
Depart/Abort			RCDH+RCHH			RCDH+RCHH
Lateral Reposition			RCDH+RCHH			RCDH+RCHH
Slalom	NA	NA	RCDH			RCDH
Vertical remask			RCDH			RCDH
Acceleration and deceleration			RCDH+RCHH			RCDH+RCHH
Sidestep			RCDH+RCHH			RCDH+RCHH
Turn to Target			RCDH+RCHH			RCDH+RCHH
Divided attention required	RCDH+RCHH +PH		RCDH+RCHH			RCDH+RCHH

TABLE V. Minimum response-types for forward flight (pitch and roll)

MTE	VMC		IMC and UCE = 2,3	
	Level 1	Level 2	Level 1	Level 2
Decelerating approach in IMC conditions	NA	NA	AH + 3-cue flight director	Rate
ILS Approach	NA	NA	Rate	Rate
Missed Approach	NA	NA	Rate	Rate
Speed Control	NA	NA	Rate	Rate
Divided attention required	AH	Rate	Autopilot	Autopilot
Slalom	Rate	Rate	NA	NA
Deceleration to Dash	Rate	Rate	NA	NA
Transient Turn	Rate	Rate	NA	NA
Pullup/Pushover	Rate	Rate	NA	NA
Roll Reversal	Rate	Rate	NA	NA
High Yo-yo	Rate	Rate	NA	NA
Low Yo-yo	Rate	Rate	NA	NA

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE VI. Limit cockpit control forces, pounds

COCKPIT CONTROL	HOVER AND LOW SPEED			FORWARD FLIGHT		
	LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 1	LEVEL 2	LEVEL 3
Pitch (Centerstick)	15.0	20.0	40.0	30.0	35.0	40.0
Roll (Centerstick)	10.0	15.0	20.0	15.0	20.0	25.0
Yaw (Pedals)	30.0	40.0	80.0	75.0	100.0	125.0
Collective	10.0	10.0	10.0	10.0	10.0	10.0

TABLE VII. Requirements for large-amplitude attitude changes – hover and low speed

AGILITY CATEGORY MTE	RATE RESPONSE-TYPES						ATTITUDE COMMNAD RESPONSE-TYPES			
	ACHIEVABLE ANGULAR RATE (deg/sec)						ACHIEVABLE ANGLE (deg)			
	LEVEL 1			LEVELS 2 AND 3			LEVEL 1		LEVELS 2 AND 3	
	Pitch	Roll	Yaw	Pitch	Roll	Yaw	Pitch	Roll	Pitch	Roll
Limited Agility Hover Landing Slope Landing	±6	±21	±9.5	±3	±15	±5	±15	±15	±7	±10
Moderate Agility Hovering Turn Hover Heading Capture Pirouette Vertical Maneuver Depart/Abort Lateral Reposition Slalom	±13	±50	±22	±6	±21	±9.5	+20 -30	±60	±13	±30
Aggressive Agility Vertical Remask Acceleration Deceleration Sidestep Target Acquisition and Tracking Turn to Target	±30	±50	±40	±13	±50	±22	±30	±60	+20 -30	±30

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE VIII. Maximum values for height response parameters – hover and low speed

LEVEL	$T_{i_{heq}}$ (sec)	$\tau_{i_{heq}}$ (sec)
1	5.0	0.20
2	∞	0.30

TABLE IX. Maximum values for flight path response parameters - forward flight

LEVEL	$T_{i_{heq}}$ (sec)	$\tau_{i_{heq}}$ (sec)
1	5.0	0.20
2	10.0	0.30

TABLE X. Requirements for large-amplitude roll attitude changes – forward flight

AGILITY CATEGORY MTE	RATE RESPONSE-TYPES		ATTITUDE COMMNAD RESPONSE-TYPES	
	ACHIEVABLE ANGULAR RATE (deg/sec)		ACHIEVABLE BANK ANGLE (deg)	
	LEVEL 1	LEVEL 2	LEVEL 1	LEVEL 2
Limited Agility Decel Approach in IMC ILS Approach Missed Approach Speed Control	±15	±12	±25	±15
Moderate Agility Slalom	±50	±21	±45	±30
Aggressive Agility Deceleration to Dash Transient Turn Pullup/Pushover Roll Reversal	±50	±21	±90	30
Target Acquisition and Tracking High Yo-Yo Low Yo-Yo	±90	±50	Unlimited	±60

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE XI. Limit cockpit control forces, pounds

COCKPIT CONTROL	HOVER AND LOW SPEED			FORWARD FLIGHT		
	LEVEL 1	LEVEL 2	LEVEL 3	LEVEL 1	LEVEL 2	LEVEL 3
Pitch (Centerstick)	15.0	20.0	40.0	30.0	35.0	40.0
Roll (Centerstick)	10.0	15.0	20.0	15.0	20.0	25.0
Yaw (Pedals)	30.0	40.0	80.0	75.0	100.0	125.0
Collective	10.0	10.0	10.0	10.0	10.0	10.0

TABLE XII. Allowable breakout forces, pounds - hover and low speed

COCKPIT CONTROL	LEVEL 1	LEVEL 2	LEVEL 3
	min/max	min/max	max
Pitch (Centerstick)	0.5/1.5	0.5/3.0	6.0
Roll (Centerstick)	0.5/1.5	0.5/2.0	4.0
Yaw (Pedals)	2.0/7.0	1.0/7.0	14.0
Collective	1.0/3.0	0.75/4.5	6.0

TABLE XIII. Allowable breakout forces, pounds - forward flight

COCKPIT CONTROL	CARGO	
	min	max
Pitch (Centerstick)	0.5	5.0
Roll (Centerstick)	0.5	4.0
Yaw (Pedals)	2.0	14.0

NOTE: The values of this Table are for Levels 1 and 2. For Level 3 the minimum values may be doubled.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE XIV. Allowable control force gradients, pounds/inch

COCKPIT CONTROL	LEVEL 1		LEVEL 2	
	min	max	min	max
Pitch	0.5	3.0	0.5	5.0
Roll	0.5	2.5	0.5	5.0
Yaw	3.0	10.0	2.0	20.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE XV. Requirements/verification matrix

PARAGRAPH NO.	REQUIREMENT	VERIFICATION METHOD/EVENT				
		SFR	PDR	CDR	FFR	SVR
3.1.2	Mission-Task-Elements			S	S	F
3.1.7	Rotorcraft Failures	A	A	A	S	T
3.1.8	Rotorcraft Limits			A	S	F
3.1.10	Residual Oscillations					F
3.2.1	Required Response-Type	A	T	T	T	F
3.2.5	Transfer Between Response-Types		A	A	S	F
3.3	Hover and Low Speed					
3.3.1	Equilibrium Characteristics		A	A	A	F
3.3.2	Small-Amplitude Pitch (Roll) Attitude	A	A	A	A	F
3.3.2.2	Short-Term Pitch (Roll) Response to Disturbance Inputs		A	A	A	AV
3.3.3	Moderate-Amplitude Pitch (Roll) Attitude Changes	A	A	A	A	F
3.3.4	Large-amplitude Pitch (Roll) Attitude Changes	A	A	A	A	F
3.3.5	Small-Amplitude Yaw Attitude Changes	A	A	A	A	F
3.3.5.2	Short-Term Yaw Response to Disturbance Inputs		A	A	A	AV
3.3.6	Moderate-Amplitude Heading Changes	A	A	A	A	F
3.3.7	Large-Amplitude Heading Changes	A	A	A	A	F
3.3.8	Hover and Low Speed Interaxis Coupling		A	A	A	F
3.3.9	Response to Collective Controller	A	A	A	A	F
3.3.9.4	Short-Term Height Response to Disturbance Inputs		A	A	A	AV
3.3.10	Position Hold		A	A	A	F
3.3.10.1	Short-Term Position Response to Disturbance Inputs		A	A	A	AV
3.3.11	Translational Rate Response-Type		A	A	A	F
3.3.11.1	Short-Term Translational Rate Response to Disturbance Inputs		A	A	A	AV
3.3.12	Pitch (Roll) Response to Externally Slung Loads		A	A	A	F
3.4	Forward Flight					
3.4.1	Pitch Attitude Response to Longitudinal Controller	A	A	A	A	F
3.4.2	Pitch Control Power	A	A	A	A	F
3.4.3	Flight Path Control				A	F
3.4.4	Longitudinal Static Stability	A	A	A	A	F
3.4.5	Roll Attitude Response to Lateral Controller	A	A	A	A	F
3.4.6	Roll-Sideslip Coupling		A	A	A	F
3.4.7	Yaw Response to Yaw Controller	A	A	A	A	F
3.4.8	Lateral-Directional Stability	A	A	A	A	F
3.4.9	Lateral-Directional Characteristics in Sideslips		A	A	A	F
3.4.10	Forward Flight Interaxis Coupling		A	A	A	F
3.5	Controller Characteristics	A	A	S	T	F

Methods of Verification
A – Analysis
S – Piloted Simulation
F – Flight Test
T – Testing, miscellaneous
AV – Analysis with flight validated model

Events:
SFR - System Functional Review
PDR - Preliminary Design Review
CDR - Critical Design Review
FFR - First Flight Readiness Review
SVR - System Verification Review

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

TABLE XVI. Rotorcraft status and flight conditions for verification

<table>
<tr><th rowspan="2">HQ LEVEL REQD</th><th colspan="4">ROTORCRAFT STATUS</th><th rowspan="2">FLIGHT CONDITIONS</th><th colspan="2">REQTS TO BE SATISFIED</th></tr>
<tr><th>CONFIG-URATIONS</th><th>LOADINGS</th><th>SETTINGS</th><th>STATES</th><th>FQ CRITERIA</th><th>APPLIC-ABLE MTE</th></tr>
<tr><td rowspan="2">1</td><td rowspan="8">As required by the operational missions</td><td rowspan="8">Possible range for applicable configurations</td><td>Primary SCAS</td><td rowspan="3">Normal and Pf > 2.5x10^{-3}</td><td>GVE OFE</td><td rowspan="8">All for rotorcraft category</td><td rowspan="4">All for mission</td></tr>
<tr><td>Primary or Augmented SCAS</td><td>GVE OFE</td></tr>
<tr><td rowspan="2">2</td><td>Primary SCAS</td><td>DVE OFE</td></tr>
<tr><td rowspan="2">Secondary SCAS</td><td rowspan="2">Normal and Pf > 2.5x10^{-5}</td><td>GVE OFE</td></tr>
<tr><td>3</td><td>DVE OFE</td></tr>
<tr><td rowspan="2">2</td><td>Primary SCAS</td><td>Normal</td><td>GVE SFE</td><td rowspan="2">N/A</td></tr>
<tr><td>Primary or Augmented SCAS</td><td>Normal</td><td>DVE SFE</td></tr>
<tr><td>3</td><td rowspan="2">Backup</td><td rowspan="2">PF>2.5x10^{-7}</td><td>GVE</td><td>NOE Egress</td></tr>
<tr><td>Con-trol-lable</td><td>DVE</td><td>N/A</td><td>UP-&-away egress</td></tr>
</table>

Pf = Probability of encountering a failure per flight hour.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives. Check ability to precisely control the rotorcraft position during the final descent to a precision landing point.Check pilot-vehicle dynamics when pilot is forced into tight compensatory tracking behavior.

b. Description of maneuver. Starting from an altitude of greater than 10 ft, maintain an essentially steady descent to a prescribed landing point. It is acceptable to arrest sink rate momentarily to make last-minute corrections before touchdown.

c. Description of test course. This task may be performed using the hover course (FIGURE 54) with the designated landing point being directly under the reference point on the rotorcraft when the pilot's eye is at the hover point.

d. Performance standard - Landing.

	Scout/Attack		**Cargo/Utility**	
	GVE	DVE	GVE	DVE
DESIRED PERFORMANCE				
• Accomplish a gentle landing with a smooth continuous decent, with no objectionable oscillations	✓	N/A	✓	N/A
• Once altitude is below 10 ft, complete the landing within X seconds	10 sec	10 sec	10 sec	10 sec
• Touch down within ±X ft longitudinally of the designated reference point	1 ft	1 ft	3 ft	3 ft
• Touch down within ±X ft laterally of the designated reference point	0.5 ft	0.5 ft	2 ft	2 ft
• Attain a rotorcraft heading at touchdown that is aligned with the reference heading within ±X deg:	5 deg	5 deg	5 deg	5 deg
• Final position shall be the position that existed at touchdown. It is not acceptable to adjust the rotorcraft positon and heading after all elements of the landing gear have made contact with the pad.	✓	✓	✓	✓
ADEQUATE PERFORMANCE				
• Touch down and remain within ±X ft of the designated landing point	3 ft	3 ft	3 ft	3 ft
• Attain a rotorcraft heading at touchdown that is aligned with the reference heading within ±X deg:	10 deg	10 deg	10 deg	10 deg

FIGURE 1. Hover

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check adequacy of any stability and control augmentation system changes that respond to partial or full landings.
- Check ability to precisely coordinate control of the heave axis and lateral axis with either the left or right part of the landing gear in contact with the ground.
- Check ability to precisely coordinate control of the heave axis and longitudinal axis with either the aft or forward part of the landing gear on the ground.

b. Description of maneuver. Perform a vertical landing to a sloped surface with the rotorcraft longitudinal axis oriented perpendicular to the fall line. Also perform vertical landings to a sloped surface with the rotorcraft longitudinal axis oriented parallel to the fall line. The landings shall be made with the nose pointed uphill and downhill, and with the up-slope to the left and right. For all of the slope landings, follow the following procedure. Once the upslope landing gear is in contact with the ground, maintain a level rotorcraft attitude for a short period of time, and then gently lower the downslope landing gear to the ground. Raise the downslope landing gear, keeping the upslope landing gear in contact with the ground, and maintain a level rotorcraft attitude for a short time before liftoff.

c. Description of test course. The test area shall consist of sloped terrain that is at least 75% of the rotorcraft slope landing performance limits. The landing area shall be clearly marked on the ground.

d. Performance standards – Slope Landing and Liftoff.

	GVE	DVE
DESIRED PERFORMANCE		
• Touch down and maintain a final position within an area that is X ft longer than the rotorcraft landing gear	6 ft	6 ft
• Touch down and maintain a final position within an area that is X ft wider than the rotorcraft landing gear	4 ft	4 ft
• Maintain heading within ±X deg:	5 deg	5 deg
• Maintain a level rotorcraft attitude with one part of the landing gear in contact with the ground and the rest in the air for at least X seconds before lowering and raising the downhill part of the landing gear	5 sec	5 sec
• No perceptible horizontal drift at touchdown or liftoff	✓	✓
• Any load limits shall remain within the OFE	✓	✓
ADEQUATE PERFORMANCE		
• Touch down and maintain a final position within an area that is X ft longer than the rotorcraft landing gear	12 ft	12 ft
• Touch down and maintain a final position within an area that is X ft wider than the rotorcraft landing gear	8 ft	8 ft
• Maintain heading within ±X deg:	10 deg	10 deg
• Maintain a level rotorcraft attitude with one part of the landing gear in contact with the ground and the rest in the air for at least X seconds before lowering and raising the downhill part of the landing gear	5 sec	5 sec
• No perceptible horizontal drift at touchdown or liftoff	✓	✓
• Any load limits shall remain within the OFE	✓	✓

FIGURE 2. Landing

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

e. Objectives.

- Check adequacy of any stability and control augmentation system changes that respond to partial or full landings.
- Check ability to precisely coordinate control of the heave axis and lateral axis with either the left or right part of the landing gear in contact with the ground.
- Check ability to precisely coordinate control of the heave axis and longitudinal axis with either the aft or forward part of the landing gear on the ground.

f. Description of maneuver. Perform a vertical landing to a sloped surface with the rotorcraft longitudinal axis oriented perpendicular to the fall line. Also perform vertical landings to a sloped surface with the rotorcraft longitudinal axis oriented parallel to the fall line. The landings shall be made with the nose pointed uphill and downhill, and with the up-slope to the left and right. For all of the slope landings, follow the following procedure. Once the upslope landing gear is in contact with the ground, maintain a level rotorcraft attitude for a short period of time, and then gently lower the downslope landing gear to the ground. Raise the downslope landing gear, keeping the upslope landing gear in contact with the ground, and maintain a level rotorcraft attitude for a short time before liftoff.

g. Description of test course. The test area shall consist of sloped terrain that is at least 75% of the rotorcraft slope landing performance limits. The landing area shall be clearly marked on the ground.

h. Performance standards – Slope Landing and Liftoff.

	GVE	DVE
DESIRED PERFORMANCE		
• Touch down and maintain a final position within an area that is X ft longer than the rotorcraft landing gear	6 ft	6 ft
• Touch down and maintain a final position within an area that is X ft wider than the rotorcraft landing gear	4 ft	4 ft
• Maintain heading within ±X deg:	5 deg	5 deg
• Maintain a level rotorcraft attitude with one part of the landing gear in contact with the ground and the rest in the air for at least X seconds before lowering and raising the downhill part of the landing gear	5 sec	5 sec
• No perceptible horizontal drift at touchdown or liftoff	✓	✓
• Any load limits shall remain within the OFE	✓	✓
ADEQUATE PERFORMANCE		
• Touch down and maintain a final position within an area that is X ft longer than the rotorcraft landing gear	12 ft	12 ft
• Touch down and maintain a final position within an area that is X ft wider than the rotorcraft landing gear	8 ft	8 ft
• Maintain heading within ±X deg:	10 deg	10 deg
• Maintain a level rotorcraft attitude with one part of the landing gear in contact with the ground and the rest in the air for at least X seconds before lowering and raising the downhill part of the landing gear	5 sec	5 sec
• No perceptible horizontal drift at touchdown or liftoff	✓	✓
• Any load limits shall remain within the OFE	✓	✓

FIGURE 3. Slope landing and liftoff

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check for undesirable handling qualities in a moderately aggressive hovering turn.
- Check ability to recover from a moderate rate hovering turn with reasonable precision.
- Check for undesirable interaxis coupling.
- In the DVE, check for undesirable display symbology and dynamics for hover.

b. Description of maneuver. From a stabilized hover at an altitude of less than 20 ft, complete a 180 degree turn. Perform the maneuver in both directions. For a cockpit with side-by-side seating, the evaluation pilot should be allowed to sit in either seat to assess potential detrimental cueing effects on the handling qualities rating. In the GVE, the maneuver shall be accomplished in calm winds and in light winds from the most critical direction. If a critical direction has not been defined, the turn shall be terminated with the wind blowing directly from the rear of the rotorcraft.

c. Description of test course. It is suggested that this maneuver use the test course described for the pirouette (FIGURE 55) with the rotorcraft located at the center of the pirouette circle. An alternate suggestion is to use the hover course with two extra markers placed in the 6 o'clock position relative to the rotorcraft. The maneuver begins with the rotorcraft lined up on these extra markers and the hover target and board located at the rotorcraft's 6 o'clock position.

d. Performance standards – Hovering Turn.

	Scout/Attack		**Cargo/Utility**	
	GVE	DVE	GVE	DVE
DESIRED PERFORMANCE				
• Maintain the longitudinal and lateral position within ±X ft of a point on the ground	3 ft	6 ft	3 ft	6 ft
• Maintain altitude within ±X ft:	3 ft	3 ft	3 ft	3 ft
• Stabilize the final rotorcraft heading at 180 deg from the initial heading within ±X deg:	3 deg	5 deg	5 deg	5 deg
• Complete turn to a stabilized hover (within the desired window) within X seconds from initiation of the maneuver	10 sec	15 sec	15 sec	15 sec
ADEQUATE PERFORMANCE				
• Maintain the longitudinal and lateral position within ±X ft of a point on the ground	6 ft	12 ft	6 ft	12 ft
• Maintain altitude within ±X ft:	6 ft	6 ft	6 ft	6 ft
• Stabilize the final rotorcraft heading at 180 deg from the initial heading within ±X deg:	6 deg	10 deg	10 deg	10 deg
• Complete turn to a stabilized hover (within the adequate window) within X seconds from initiation of the maneuver	15 sec	15 sec	20 sec	20 sec

FIGURE 4. Hovering Turn

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check for undesirable handling qualities in a moderately aggressive hover heading capture.
- Check for the ability to capture a heading and stabilize on that heading with reasonable precision.
- Check for undesirable interaxis coupling

b. Description of maneuver. From a stabilized hover at a wheel height of less than 20 ft, complete a 45 degree turn, stabilize in a hover on the new heading for 3 seconds, complete a 45 degree turn back to the initial heading, and stabilize in a hover. The initial turn shall be performed to both the left and right direction. In GVE, the maneuver shall be accomplished in calm winds and in moderate winds from the most critical direction. If a critical direction has not been defined, the wind shall be blowing directly from the rear of the rotorcraft at the initial/final heading.

c. Description of test course. It is suggested that this maneuver use the hover course (FIGURE 54) with the initial heading aligned with the hover boards and the 45 degree stabilization heading marked by the 45 degree line-up lines.

d. Performance standards – Hover Heading Capture.

	Scout/Attack		**Cargo/Utility**	
	GVE	DVE	GVE	DVE
DESIRED PERFORMANCE				
• Maintain the longitudinal and lateral position within ±X ft of a point on the ground while in a stabilized hover	3 ft	6 ft	3 ft	6 ft
• Maintain altitude within ±X ft:	3 ft	3 ft	3 ft	3 ft
• Stabilize the intermediate/final rotorcraft heading at 45/0 deg from the initial heading within ±X deg	3 deg	5 deg	5 deg	5 deg
• Complete turn to a stabilized hover (within the desired window) within X seconds from initiation of the maneuver	10 sec	15 sec	15 sec	15 sec
ADEQUATE PERFORMANCE				
• Maintain the longitudinal and lateral position within ±X ft of a point on the ground	6 ft	12 ft	6 ft	12 ft
• Maintain altitude within ±X ft:	6 ft	6 ft	6 ft	6 ft
• Stabilize the final rotorcraft heading at 180 deg from the initial heading within ±X deg:	6 deg	10 deg	10 deg	10 deg
• Complete turn to a stabilized hover (within the adequate window) within X seconds from initiation of the maneuver	15 sec	15 sec	20 sec	20 sec

FIGURE 5. Hover heading capture

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check the ability to translate with, stabilize, and set down an external load at a specific location, within a reasonable time limit.
- Check ability to set load down without any residual motion of the load on the ground, such as dragging or swinging.

b. Description of maneuver. Initiate the maneuver at a ground speed between 6 and 10 knots, with a load clearance of 20 feet above ground level. The load placement target shall be oriented approximately 45 degrees relative to the heading of the rotorcraft. The load placement target is a ground referenced point, from which the deviation in the set-down point is measured. The ground track should be such that the rotorcraft will arrive over the target point (FIGURE 56). Once the aircraft is stabilized in a hover over the load placement target, the crew chief will provide verbal instructions to assist the pilot in placing the load. These instructions should follow the form of direction-count-hold as in "Right, 3-2-1, hold" or "Down, 3-2-1, hold" to position the load and set it down.

c. Description of test course. The suggested test course for this maneuver is shown in FIGURE 56. Note that the desired and adequate boxes refer to the load set-down point, not the helicopter position during maneuver.

d. Performance standards - Precision Load Placement. Accomplish the transition to hover in one smooth maneuver. It is not acceptable to accomplish most of the deceleration well before the load target point and then creep up to the final position. The load swing should be contained within the desired boundaries (or adequate if trying for adequate performance) before placing the load on the ground. The load should not perceptibly drift, swing, or drag after initial ground contact. All other performance standards are given below.

	Externally Slung Load	
	GVE	DVE
DESIRED PERFORMANCE		N/A
• Attain a controlled hover within X seconds of initiation of deceleration:	10 sec	
• Maintain altitude during translation and hover within ±X ft:	4 ft	
• Controlled set-down of external load within X seconds of hover:	50 sec	
• Load set-down position should be within a box ±X ft larger than the footprint of the external load on all sides:	3 ft	
• The load should have no perceptible drift at touchdown	✓	
ADEQUATE PERFORMANCE		N/A
• Attain a controlled hover within X seconds of initiation of deceleration:	15 ft	
• Maintain altitude during translation and hover within ±X ft:	6 ft	
• Controlled set-down of external load within X seconds of hover:	120 sec	
• Load set-down position should be within a box ±X ft larger than the footprint of the external load on all sides:	6 ft	

FIGURE 6. Precision load placement

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check ability to accomplish precision control of the rotorcraft simultaneously in the pitch, roll, yaw, and heave axes.
- In the GVE, check ability to control the rotorcraft precisely in a light wind that is continuously varying in direction relative to the rotorcraft heading.
- In the DVE, check for degraded display symbology/dynamics during multiple axis maneuvering.

b. Description of maneuver. Initiate the maneuver from a stabilized hover over a point on the circumference of a 100 ft radius circle with the nose of the rotorcraft pointed at a reference point at the center of the circle, and at a hover altitude of approximately 15 ft. Accomplish a lateral translation around the circle, keeping the nose of the rotorcraft pointed at the center of the circle, and the circumference of the circle under a selected point on the rotorcraft. Maintain essentially constant lateral groundspeed throughout the lateral translation (note: nominal lateral velocity will be approximately 8 knots for the 45-sec and 6 knots for the 60-sec time around the circle). Terminate the maneuver with a stabilized hover over the starting point. Perform the maneuver in both directions. For a cockpit with side-by-side seating, the evaluation pilot should be allowed to sit in either seat to assess potential detrimental cueing effects on the handling qualities rating. In the GVE, the maneuver shall be accomplished in calm winds and in light winds from the most critical direction at the starting point.

c. Description of test course. The test course shall consist of markings on the ground that clearly denote the circular pathways that define desired and adequate performance. The suggested course shown in FIGURE 55 is adequate for the evaluation. It may also be useful to add objects to assist the pilot with vertical cueing, such as a post at the center of the circle.

d. Performance standards - Pirouette.

	GVE	DVE
DESIRED PERFORMANCE		
• Maintain a selected reference point on the rotorcraft within ±X ft of the circumference of the circle	10 ft	10 ft
• Maintain altitude within ±X ft:	3 ft	4 ft
• Maintain heading so that the nose of the rotorcraft points at the center of the circle within ±X deg:	10 deg	10 deg
• Complete the circle and arrive back over the starting point within:	45 sec	60 sec
• Achieve a stabilized hover (within desired hover reference point) within X seconds after returning to the starting point.	5 sec	10 sec
• Maintain the stabilized hover for X sec	5 sec	5 sec
ADEQUATE PERFORMANCE		
• Maintain a selected reference point on the rotorcraft within ±X ft of the circumference of the circle	15 ft	15 ft
• Maintain altitude within ±X ft:	10 ft	10 ft
• Maintain heading so that the nose of the rotorcraft points at the center of the circle within ±X deg:	15 deg	15 deg
• Complete the circle and arrive back over the starting point within:	60 sec	75 sec
• Achieve a stabilized hover (within adequate hover reference point) within X seconds after returning to the starting point.	10 sec	20 sec
• Maintain the stabilized hover for X sec	5 sec	5 sec

FIGURE 7. Pirouette

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives. For a scout/attack rotorcraft this maneuver is to simulate a rapid unmask/remask maneuver, with an aiming task at the unmask. For a utility or cargo rotorcraft, the maneuver is to assess the heave axis controllability with precision station keeping.

- Check for adequate heave damping, i.e., the ability to precisely start and stop a vertical rate.
- Check for adequate vertical control power.
- Check for undesirable coupling between collective and the pitch, roll, and yaw axes.
- Check the characteristics of the heave axis controller, especially if a non-conventional controller is used, e.g., a four-axis sidestick.
- With an external load, check for undesirable effects between the heave controller and the other axes of the rotorcraft and complications caused by the external load dynamics.

b. Description of maneuver. From a stabilized hover at an altitude of 15 ft, initiate a vertical ascent of 25 ft, stabilize for 2 seconds, and then descend back to the initial hover position. With an external load, the maneuver is initiated from a higher altitude to assure a 10 ft load clearance. In the GVE, the maneuver shall be accomplished in calm winds and in light winds from the most critical direction. If a critical direction has not been defined, the hover shall be accomplished with the wind blowing directly from the rear of the rotorcraft.

c. Description of test course. The test course shall consist of markings on the ground that clearly define desired and adequate performance. It is suggested that this maneuver use the hover course (FIGURE 54) with a second reference symbol or hover board set to align at the upper reference.

d. Performance standards - Vertical Maneuver.

	Scout/Attack		**Cargo/Utility**		**Externally Slung Load**	
	GVE	DVE	GVE	DVE	GVE	DVE
DESIRED PERFORMANCE						
• Maintain the longitudinal and lateral position within ±X ft of a point on the ground	6 ft	10 ft	3 ft	3 ft	3 ft	3 ft
• Maintain start/finish altitude within ±X ft:	3 ft	3 ft	3 ft	3 ft	4 ft	4 ft
• Maintain heading within ±X deg:	3 deg	3 deg	5 deg	5 deg	5 deg	5 deg
• Complete the maneuver within:	10 sec	13 sec	13 sec	15 sec	13 sec	15 sec
ADEQUATE PERFORMANCE						
• Maintain the longitudinal and lateral position within ±X ft of a point on the ground	10 ft	20 ft	6 ft	6 ft	6 ft	6 ft
• Maintain start/finish altitude within ±X ft:	6 ft	6 ft	6 ft	6 ft	6 ft	6 ft
• Maintain heading within ±X deg:	6 deg	6 deg	10 deg	10 deg	10 deg	10 deg
• Complete the maneuver within:	15 sec	18 sec	18 sec	18 sec	20 sec	20 sec

FIGURE 8. Vertical maneuver

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check pitch axis and heave axis handling qualities during moderately aggressive maneuvering.
- Check for undesirable coupling between the longitudinal and lateral-directional axes.
- Check for harmony between the pitch axis and heave axis controllers.
- Check for overly complex power management requirements.
- Check for ability to re-establish hover after changing trim.
- With an external load, check for dynamic problems resulting from the external load configuration.

b. Description of maneuver. From a stabilized hover at 35 ft wheel height (or no greater than 35 ft external load height) and 800 ft from the intended endpoint, initiate a longitudinal acceleration to perform a normal departure. Abort the departure and decelerate to a hover such that at the termination of the maneuver, the cockpit shall be within 20 ft of the intended endpoint. It is not permissible to overshoot the intended endpoint and move back. If the rotorcraft stopped short, the maneuver is not complete until it is within 20 ft of the intended endpoint. The acceleration and deceleration phases shall be accomplished in a single smooth maneuver. For rotorcraft that use changes in pitch attitude for airspeed control, a target of approximately 20 degrees of pitch attitude should be used for the acceleration and deceleration. The maneuver is complete when control motions have subsided to those necessary to maintain a stable hover.

c. Description of test course. The test course shall consist of at least a reference line on the ground indicating the desired track during the acceleration and deceleration, and markers to denote the starting and endpoint of the maneuver. The course should also include reference lines or markers parallel to the course reference line to allow the pilot and observers to perceive the desired and adequate longitudinal tracking performance, such as the example shown in FIGURE 57.

d. Performance standards. Depart/Abort.

	Cargo/Utility		**Externally Slung Load**	
	GVE	DVE	GVE	DVE
DESIRED PERFORMANCE				
• Maintain lateral track within ±X ft:	10 ft	10 ft	10 ft	10 ft
• Maintain radar altitude below X ft:	75 ft	75 ft	75 ft*	75 ft*
• Maintain heading within ±X deg:	10 deg	10 deg	10 deg	10 deg
• Time to complete maneuver:	25 sec	25 sec	30 sec	30 sec
• Maintain rotor speed within:	OFE	OFE	OFE	OFE
ADEQUATE PERFORMANCE				
• Maintain lateral track within ±X ft:	20 ft	20 ft	20 ft	20 ft
• Maintain radar altitude below X ft:	100 ft	100 ft	100 ft*	100 ft*
• Maintain heading within ±X deg:	15 deg	15 deg	15 deg	15 deg
• Time to complete maneuver:	30 sec	30 sec	35 sec	35 sec
• Maintain rotor speed within:	SFE	SFE	SFE	SFE

* Altitudes refer to height of external load, maneuver at hover

FIGURE 9. Depart/Abort

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check roll axis and heave axis handling qualities during moderately aggressive maneuvering.
- Check for undesirable coupling between the roll controller and the other axes.
- With an external load, check for dynamic problem resulting from the external load configuration.

b. Description of maneuver. Start in a stabilized hover at 35 ft wheel height (or no greater than 35 ft external load height) with the longitudinal axis of the rotorcraft oriented 90 degrees to a reference line marked on the ground. Initiate a lateral acceleration to approximately 35 knots groundspeed followed by a deceleration to laterally reposition the rotorcraft in a stabilized hover 400 ft down the course within a specified time. The acceleration and deceleration phases shall be accomplished as single smooth maneuvers. The rotorcraft must be brought to within ±10 ft of the endpoint during the deceleration, terminating in a stable hover within this band. Overshooting is permitted during the deceleration, but will show up as a time penalty when the pilot moves back within ±10 ft of the endpoint. The maneuver is complete when a stabilized hover is achieved.

c. Description of test course. The test course shall consist of any reference lines or markers on the ground indicating the desired track and tolerances for the acceleration and deceleration, and markers to denote the starting and endpoint of the maneuver. The course should also include reference lines or markers parallel to the course reference line to allow the pilot and observers to perceive the desired and adequate longitudinal tracking performance, such as the example shown in FIGURE 58.

d. Performance standards – Lateral Reposition.

	Cargo/Utility		**Externally Slung Load**	
	GVE	DVE	GVE	DVE
DESIRED PERFORMANCE				
• Maintain longitudinal track within ±X ft:	10 ft	10 ft	10 ft	10 ft
• Maintain altitude within X ft:	10 ft	10 ft	10 ft	10 ft
• Maintain heading within ±X deg:	10 deg	10 deg	10 deg	10 deg
• Time to complete maneuver:	18 sec	20 sec	25 sec	25 sec
ADEQUATE PERFORMANCE				
• Maintain longitudinal track within ±X ft:	20 ft	20 ft	20 ft	20 ft
• Maintain altitude within X ft:	15 ft	15 ft	15 ft	15 ft
• Maintain heading within ±X deg:	15 deg	15 deg	15 deg	15 deg
• Time to complete maneuver:	22 sec	25 sec	30 sec	30 sec

FIGURE 10. Lateral Reposition

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check ability to maneuver aggressively in forward flight and with respect to objects on the ground.
- Check turn coordination for moderately aggressive forward flight maneuvering.
- Check for objectionable interaxis coupling during moderately aggressive forward flight maneuvering.
- In the DVE, check ability to maneuver moderately aggressive with respect to objects on the ground.

b. Description of maneuver. Initiate the maneuver in level unaccelerated flight and lined up with the centerline of the test course. Perform a series of smooth turns at 500-ft intervals* (at least twice to each side of the course). The turns shall be at least 50 ft from the centerline, with a maximum lateral error of 50 ft. The maneuver is to be accomplished below the reference altitude. Complete the maneuver on the centerline, in coordinated straight flight.

c. Description of test course. The suggested test course for this maneuver is shown in FIGURE 59. Most runways have touchdown stripes at 500-ft intervals that can be conveniently used instead of the pylons. However, if the runway is not 100 ft wide, it will be necessary to use two cones to define each gate (as opposed to one cone and the runway edge as shown in FIGURE 59).

d. Performance standards – Slalom.

	GVE	DVE
DESIRED PERFORMANCE		
• Maintain an airspeed of a least X knots throughout the course	60	30
• Accomplish maneuver below reference altitude of X ft:	Lesser of twice rotor diameter or 100 ft	100 ft
ADEQUATE PERFORMANCE		
• Maintain an airspeed of a least X knots throughout the course	40	15
• Accomplish maneuver below reference altitude of X ft:	100 ft	100 ft

FIGURE 11. Slalom

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check ability to accomplish an aggressive vertical descent close to the ground.
- Check ability to combine vertical and lateral aggressive maneuvering as required to evade enemy fire if observed during a bob-up.

b. Description of maneuver. From a stabilized hover at 75 ft, remask vertically to an altitude below 25 ft. Then rapidly displace the rotorcraft laterally 300 ft and stabilize at a new hover position. During the vertical remask simulate deploying rotorcraft survivability equipment as appropriate. Accomplish the maneuver to the left and to the right.

c. Description of test course. The test course should include markers to denote the desired and adequate performance related to position during the vertical descent and final stabilized hover. It may also be desirable to include a vertical reference to provide cues related to the 25 ft altitude reference. This maneuver assumes that the pilot is remasking behind some object, and such an altitude reference should therefore be available. A suggested test course for this maneuver is shown in FIGURE 58.

d. Performance standards – Vertical Remask

	DESIRED	ADEQUATE
• Achieve an altitude of X or less within 6 seconds of initiation the maneuver.	25 ft	N/A
• During initial stabilized hover, vertical descent, and final stabilized hover, maintain longitudinal and lateral position within ±X ft of a reference point on the ground.	6 ft	6 ft
• Maintain altitude after remask and during displacement within X ft:	±10 ft	+10 and -15 ft
• Maintain lateral ground track within ±X ft:	10 ft	15 ft
• Maintain heading within ±X deg	10 deg	15 deg
• Achieve a stabilized hover within X sec after reaching the final hover position.	5 sec	10 sec
• Achieve the final stabilized hover within X sec of initiating the maneuver.	15 sec	25 sec

FIGURE 12. Vertical remask

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check pitch axis and heave axis handling qualities:
  - GVE: for aggressive maneuvering near the rotorcraft limits of performance
  - DVE: for reasonable aggressive maneuvering in the DVE.
- Check for undesirable coupling between the longitudinal and lateral-directional axes.
- Check for harmony between the heave axis and pitch axis controllers.
- Check for adequate rotor response to aggressive collective inputs.
- Check for overly complex power management requirements.

b. Description of maneuver. Start from a stabilized hover. In the GVE, rapidly increase power to approximately maximum, maintain altitude constant with pitch attitude, and hold collective constant during the acceleration to an airspeed of 50 knots. Upon reaching the target airspeed, initiate a deceleration by aggressively reducing the power and holding altitude constant with pitch attitude. The peak nose-up attitude should occur just before reaching the final stabilized hover. In the DVE, accelerate to a groundspeed of at least 50 knots, and immediately decelerate to hover over a defined point. The maximum nose-down attitude should occur immediately after initiating the maneuver, and the peak nose-up attitude should occur just before reaching the final stabilized hover. Complete the maneuver in a stabilized hover for 5 seconds over the reference point at the end of the course.

c. Description of test course. The test course shall consist of a reference line on the ground indicating the desired track during the acceleration and deceleration, and markers to denote the starting point and endpoint of the maneuver. The distance from the starting point to the final stabilized hover position is a function of the performance of the rotorcraft, and shall be determined based on trial runs consisting of acceleration to the target airspeed, and decelerations to hover as described above. The course should also include reference lines or markers parallel to the course centerline to allow the pilot and observers to perceive desired and adequate lateral tracking performance. A suggested test course is shown in FIGURE 57.

FIGURE 13. Acceleration and deceleration

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

d. Performance standard – Acceleration and Deceleration

	GVE	DVE
DESIRED PERFORMANCE		
• Within X seconds from initiation of the maneuver, achieve at least the greater of 95% maximum continuous power or 95% maximum transient limit that can be sustained for the required acceleration, whichever is greater. If the 95% power results in objectionable pitch attitudes, use the power corresponding to the maximum nose-down pitch attitude that is felt to be acceptable. This pitch attitude shall be considered as a limit of the Operational Flight Envelope (OFE) for NOE flying.	1.5 sec	N/A
• Achieve a nose-down pitch attitude during the acceleration of at least X deg below the hover attitude:	N/A	12 deg
• Maintain altitude below X ft:	50 ft	50 ft
• Maintain lateral track within ±X ft:	10 ft	10 ft
• Maintain heading within ±X deg:	10 deg	10 deg
• Decrease power to less than 5% within X seconds to initiate deceleration.	3 sec	N/A
• Significant increases in power are not allowed until just before the final stabilized hover.	✓	✓
• Achieve a nose-up pitch attitude during the deceleration of at least X deg above the hover attitude. The maximum pitch attitude should occur shortly before the hover.	30deg	15 deg
• Longitudinal tolerance on the final hover point is plus zero, minus a distance equal to X % of the overall rotorcraft length.	50%	50%
• Rotor RPM shall remain within the limits of X without undue pilot compensation	OFE	OFE
ADEQUATE PERFORMANCE		
• Within X seconds from initiation of the maneuver, achieve at least the greater of 95% maximum continuous power or 95% maximum transient limit that can be sustained for the required acceleration, whichever is greater. If the 95% power results in objectionable pitch attitudes, use the power corresponding to the maximum nose-down pitch attitude that is felt to be acceptable. This pitch attitude shall be considered as a limit of the Operational Flight Envelope (OFE) for NOE flying.	3 sec	N/A
• Achieve a nose-down pitch attitude during the acceleration of at least X deg below the hover attitude:	N/A	7 deg
• Maintain altitude below X ft:	70 ft	70 ft
• Maintain lateral track within ±X ft:	20 ft	20 ft
• Maintain heading within ±X deg:	20 deg	20 deg
• Decrease power to less than 5% within X seconds to initiate deceleration.	5 sec	N/A
• Significant increases in power are not allowed until just before the final stabilized hover.	✓	✓
• Achieve a nose-up pitch attitude during the deceleration of at least X deg above the hover attitude. The maximum pitch attitude should occur shortly before the hover.	10 deg	10 deg
• Longitudinal tolerance on the final hover point is plus zero, minus a distance equal to X % of the overall rotorcraft length.	100%	100%
• Rotor RPM shall remain within the limits of X without undue pilot compensation	SFE	SFE

FIGURE 13. Acceleration and deceleration - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check lateral-directional handling qualities for aggressive maneuvering near the rotorcraft limits of performance (GVE), or reasonably aggressive lateral maneuvering (DVE).
- Check for objectionable interaxis coupling.
- Check ability to coordinate bank angle and collective to hold constant altitude.

b. Description of maneuver. Starting from a stabilized hover with the longitudinal axis of the rotorcraft oriented 90 degrees to a reference line marked on the ground, initiate a rapid and aggressive lateral acceleration, holding altitude constant with power. Hold target velocity for 5 seconds and then initiate an aggressive deceleration to hover at constant altitude. The peak bank angle during deceleration should occur just before the rotorcraft comes to a stop. Establish and maintain a stabilized hover for 5 seconds. Immediately repeat the maneuver in the opposite direction.

c. Description of test course. The test course shall consist of any reference lines or markers on the ground indicating the desired track and tolerances for the acceleration and deceleration, and markers to denote the starting and endpoint of the maneuver. The course should also include reference lines or markers parallel to the course reference line to allow the pilot and observers to perceive the desired and adequate longitudinal tracking performance, such as the example shown in FIGURE 58. Note that the end point for this maneuver is not prescribed.

FIGURE 14. Sideslip

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

d. Performance standards – Sidestep.

	GVE	DVE
DESIRED PERFORMANCE		
• Achieve at least 25 degrees of bank angle change from trim, or target airspeed, within X seconds of initiating the maneuver:	1.5 sec	N/A
• Achieve a target airspeed of X knots:	lesser of 40 or (OFE-5) knots	17 knots
• Achieve at least 30 degrees of bank angle within X seconds of initiating deceleration:	1.5 sec	N/A
• Achieve at least X deg of bank angle change from trim during the acceleration and deceleration:	N/A	20 deg
• Maintain a selected reference point on the rotorcraft within ±X ft of the ground reference line:	10 ft	10 ft
• Maintain altitude within ±X ft at a selected altitude below 30 ft:	10 ft	10 ft
• Maintain heading within ±X deg:	10 deg	10 deg
• Achieve a stabilized hover within X seconds after reaching the hover point:	5 sec	10 sec
ADEQUATE PERFORMANCE		
• Achieve at least 25 degrees of bank angle change from trim, or target airspeed, within X seconds of initiating the maneuver:	3.0 sec	N/A
• Achieve a target airspeed of X knots:	lesser of 40 or (OFE-5) knots	17 knots
• Achieve at least 30 degrees of bank angle within X seconds of initiating deceleration:	3.0 sec	N/A
• Achieve at least X deg of bank angle change from trim during the acceleration and deceleration:	N/A	10 deg
• Maintain a selected reference point on the rotorcraft within ±X ft of the ground reference line:	15 ft	15 ft
• Maintain altitude within ±X ft at a selected altitude below 30 ft:	15 ft	15 ft
• Maintain heading within ±X deg:	15 deg	15 deg
• Achieve a stabilized hover within X seconds after reaching the hover point:	10 sec	20 sec

FIGURE 14. Sideslip - Continued

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. **Objectives.**
- Check for poor engine governing or overly complex power management requirement.
- Check pitch, heave, and yaw axis handling qualities for aggressive maneuvering.
- Check for undesirable coupling between the longitudinal and lateral-directional axes, and between the heave axis and longitudinal and lateral-directional axes, for maneuvers requiring large power changes.
- Check for harmony between the heave, pitch, and directional axis controllers.
- Check for adequate rotor response to aggressive collective inputs.

b. **Description of maneuver.** From level unaccelerated flight at the lesser of V_H or120 knots, perform a level deceleration-acceleration. Adjust the pitch attitude to maintain altitude with a full down collective position. As the airspeed decreases to approximately 50 knots, aggressively assume the attitude for maximum acceleration and rapidly increase power to approximately the maximum, and maintain that power until the initial airspeed is reached.

c. **Description of test course.** Any reference line on the ground will serve as an adequate test course for this maneuver.

d. **Performance standards – Deceleration to Dash.** The entire maneuver shall be conducted below 200 ft.

Performance - Deceleration to Dash

	DESIRED	ADEQUATE
• Achieve X% collective within Y sec from the initiation of the deceleration.	0% (full down) 3 sec	Less than 10% 5 sec
• Achieve either X% of maximum continuous power (or X% of the transient limit) within Y sec of initiating the acceleration.	95% 2 sec	80% 3 sec
• During the acceleration, the power shall not exceed any rotorcraft limitation, and shall not fall below X%.	85%	80%
• Without undue pilot compensation, rotor RPM shall remain within the limits of the:	OFE	SFE
• Maintain heading within ±X deg:	5 deg	10 deg
• Maintain altitude below X ft:	200 ft	200 ft
• Maintain altitude within ±X ft:	50 ft	N/A
• Any oscillations or coupling shall not be:	undesirable	objectionable

FIGURE 15. Deceleration to dash

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Insure that handling qualities do not degrade during aggressive maneuvering in all axes.
- Check for undesirable coupling between pitch, roll, and yaw during aggressive maneuvering.

b. Description of maneuver. Starting at the lesser of V_H or 120 knots and an altitude at or below 200 ft, accomplish a 180-degree change in directional flightpath and achieve wings-level attitude in as little time as possible. Use of pedals to induce a lateral acceleration in the direction of the turn is acceptable. Perform the maneuver both to the right and to the left. It is acceptable to reduce collective to increase the rate of speed bleed-off and thereby maximize the turn rate.

c. Description of test course. This maneuver does not require a test course that is marked out on the ground aside from a reference line such as a road or railroad track.

d. Performance standards – Transient Turn.

	DESIRED	ADEQUATE
• Achieve a peak normal load factor of at least X% of the OFE $n_L(+)$:	100% - 0.2g	80%
• Complete the maneuver within X seconds:	10 sec	15 sec
• Maintain altitude within X ft:	±50 ft	Below 300 ft
• Maintain the rotor RPM within the limit of the:	OFE	SFE
• Wings-level attitude on roll out within X degrees of level trim:	5 deg	10 deg

FIGURE 16. Transient turn

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check handling qualities at elevated and reduced load factors and during transition between elevated and reduced load factors.
- Check for undesirable coupling between pitch, roll, and yaw for aggressive maneuvering in forward flight.
- Check for ability to avoid obstacles during high-speed NOE operations.

b. Description of maneuver. From level unaccelerated flight at the lesser of V_H or 120 knots, attain a sustained positive load factor in a symmetrical pullup. Transition, via a symmetrical pushover, to a sustained negative load factor. Recover to level flight as rapidly as possible.

c. Description of test course. This maneuver may be accomplished up-and-away, and no test course is required.

d. Performance standards – Pullup/Pushover

Performance - Pullup/Pushover

	DESIRED	ADEQUATE
• Attain a normal load factor of at least the positive limit of the OFE ($n_L(+)$) within X seconds from the initial control input.	1 sec	2 sec
• Maintain at least $n_L(+)$ for at least X seconds:	2 sec	1 sec
• Accomplish transition from $n_L(+)$ pullup to a pushover of not greater than the negative normal load factor limit of the OFE ($n_L(-)$) within X seconds.	2 sec	4 sec
• Maintain a load factor of not greater than $n_L(-)$ for at least X seconds.	2 sec	1 sec
• Maintain angular deviations in roll and yaw within ±X degrees from the initial unaccelerated level flight condition to completion of the maneuver.	10 deg	15 deg

FIGURE 17. Pushup/Pushover

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check handling qualities while maneuvering with load factors close to the OFE limits.
- Check the roll damping and roll authority during elevated and reduced load factor.
- Check for undesirable coupling between axes during aggressive maneuvering.
- Check the maneuvering stability of the rotorcraft close to the OFE limits.

b. Description of maneuver. Starting in a dive, conduct a series of pullups and pushovers to achieve normal accelerations within 0.10g of the positive [$n_L(+)$] and negative [$n_L(-)$] boundaries of the Operational Flight Envelope. The target normal acceleration should occur as the rotorcraft passes through the level attitude. At this time execute an aggressive roll to a minimum of 45 degrees of bank, and back to zero while maintaining incremental load factor. The maneuvers should be conducted so that the airspeed at the start of the rolling maneuvers is the lesser of V_H or 120 knots.

c. Description of test course. This maneuver may be accomplished up-and-away, and no test course is required.

d. Performance standards – Roll Reversal.

	DESIRED	ADEQUATE
• Achieve a peak roll rate of at least X percent of the maximum steady state roll rate achievable at one g:	50%	30%
• Maintain target normal acceleration within X % of the incremental load factor:	50%	50%
• No oscillation in any axis that is:	undesirable	uncontrollable or persistent
• Any change in roll or pitch response shall not be:	sudden	objectionable or reversals

FIGURE 18. Roll reversal

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check for undesirable handling qualities during a maximum effort, rapid hovering turn.
- Check ability to recover from a rapid hovering turn with sufficient precision to fire a weapon.
- Check for undesirable interaxis coupling.

b. Description of maneuver. From a stabilized hover at an altitude of less than 20 ft, complete a 180 degree turn. Turns shall be completed in both directions. The final heading tolerance should be based on a sight mounted on the rotorcraft, preferably the same sight to be used for operational missions.

c. Description of test course. The test course can consist of any convenient target.

d. Performance standards – Turn to Targets

	DESIRED	ADEQUATE
• Maintain longitudinal and lateral position of a selected point on the rotorcraft within ±X ft of a reference point on the ground.	6 ft	12 ft
• Maintain altitude within ±X ft	3 ft	6 ft
• Stabilize final rotorcraft heading within a tolerance based on the firing constraints of the weapon system to be deployed on the rotorcraft.	✓	N/A
• Stabilize final rotorcraft heading within ±X deg:	N/A	3 deg
• Complete the turn so that a firing solution has been achieved within X from initiation of the maneuver.	5 sec	10 sec

FIGURE 19. Turn to target

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check handling qualities during reduced and elevated load factors.
- Check the short term-response characteristics of the rotorcraft through aggressive pitch pointing tasks.
- Check for undesirable coupling between pitch, roll, and yaw during aggressive maneuvering.

b. Description of maneuver. Two aircraft are required to perform this air-combat maneuver. The maneuver is initiated from level unaccelerated flight with both aircraft at a constant airspeed equal to the V_H of the test rotorcraft. The test rotorcraft is positioned at least 500 ft in trail behind the target aircraft. The target aircraft then decelerates 20 knots, to (V_H - 20), causing the test rotorcraft to close on the target. When the range between the two aircraft decreases to approximately 300 ft, the target aircraft initiates a 45-degree banked turn at constant altitude, and holds the turn to roll out after 180 deg heading change. The test rotorcraft delays until the line-of-sight reaches 30 degrees, at which time the pilot initiates a climbing turn toward the target, with a nose-up pitch attitude of 15 to 30 degrees. The resulting deceleration causes a decrease in the rate of closure from above. When the closure rate is no longer apparent, and the range to the target is approximately 200 to 500 ft, the test rotorcraft rapidly lowers the nose to achieve a firing solution within missile launch constraints.

c. Description of test course. This maneuver may be accomplished up-and-away, and no test course is required.

d. Performance standards – High Yo-Yo.

	DESIRED	ADEQUATE
• Interaxis coupling shall not be:	undesirable	objectionable
• Maintain the missile launch constraints for X sec:	7 sec	4 sec
• Acquire the target with no tendency for pitch overshoots	✓	✓

FIGURE 20. High Yo-Yo

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check handling qualities during reduced and elevated load factors.
- Check the short-term response characteristics of the rotorcraft through aggressive pitch pointing tasks.
- Check for undesirable coupling between pitch, roll, and yaw during aggressive maneuvering.

b. Description of maneuver. Two aircraft are required to perform this air-combat maneuver, and the target aircraft must be capable of achieving airspeeds of at least V_H of the test rotorcraft. The maneuver is initiated from level unaccelerated flight, with both aircraft at an airspeed equal to (V_H - 20 knots) of the test rotorcraft. The test rotorcraft is positioned approximately 200 ft in trail behind the target aircraft. The target aircraft then accelerates 20 knots to V_H, resulting in a steady increase in range between the two aircraft. When the range between the two aircraft increases to approximately 300 ft, the target aircraft executes a 45-degree banked turn at constant altitude and holds the turn to roll out after 180 deg heading change. The test rotorcraft delays until the line-of-sight reaches 30 degrees, at which time the pilot initiates a diving turn in the direction of the target with a nose-down pitch attitude of 15 to 30 degrees. The resulting acceleration causes the test rotorcraft to begin to close on the target from below. When a rate of closure on the target is apparent, and the range to the target is within 500 ft, the test rotorcraft rapidly raises the nose and tracks the target to achieve a firing solution within missile launch constraints.

c. Description of test course. This maneuver may be accomplished up and away, and no test course is required.

d. Performance standards – Low Yo-Yo

	DESIRED	ADEQUATE
• Interaxis coupling shall not be:	undesirable	objectionable
• Maintain the missile launch constraints for X sec:	7 sec	4 sec
• Acquire the target with no tendency for pitch overshoots	✓	✓
• Maintain power within the transient range for as long as possible, without exceeding the time limit specified for the rotorcraft, or until simulated missile launch. Maintain at least 95% of maximum continuous power when rotorcraft limitations prohibit operation in the transient range.	✓	N/A
• Maintain power within ±10% of maximum continuous. If +10% exceeds a limit, do not exceed that limit.	N/A	✓

FIGURE 21. Low Yo-Yo

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check ability to perform precision glideslope and localizer tracking to very low decision height and groundspeed with a reasonable pilot workload.
- Check ability to precisely control airspeed and to perform a deceleration while descending on the glideslope.

b. Description of maneuver. Starting on a 4-degree glideslope at an airspeed of 100 knots, perform a manual deceleration to an airspeed of 25 knots at an altitude of 50 ft. Guidance commands may be generated using onboard sensors, or from ground-based transmitters.

c. Performance standards – Decelerating Approach.

	DESIRED	ADEQUATE
• Maintain glideslope within ±X ft:	12.5 ft	25 ft
• Maintain localizer within ±X ft:	50 ft	75 ft
• Maintain airspeed within ±X knots of the reference:	5 knots	10 knots

FIGURE 22. Decelerating approach

a. Objectives.

- Check ability to perform precise flight path and speed control.

b. Description of maneuver. Start the maneuver eight miles from the runway with an offset from the localizer of 1 nm, on a 45 degree intercept to the localizer. At an airspeed of 120 knots or V_H and altitude of 1500 ft, intercept the localizer and slow the rotorcraft to 90 knots prior to glideslope intercept. Track the ILS to a 200 ft decision height at an airspeed of 90 knots.

c. Performance standards – ILS Approach.

	DESIRED	ADEQUATE
• Maintain airspeed of 90 knots within ±X knots:	5 knots	10 knots
• Prior to glideslope intercept, maintain altitude within ±X ft:	100 ft	200 ft
• Maintain glideslope and localizer within X dots:	1 dot	2 dots

FIGURE 23. ILS approach

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

a. Objectives.

- Check longitudinal flight control variations in a high-workload, divided-attention task.

b. Description of maneuver. After performing an ILS approach to Decision Height, initiate a climb on runway heading to an altitude of 500 ft at an airspeed of 80 knots. At 500 ft, turn right to a heading 90 degrees from runway heading. Level off at 1,000 ft and accelerate to 100 knots. Once steady at this condition, turn right to a heading of 180 degrees from runway heading and climb to 2,000 ft. Once level at 2000 ft and steady on 100 knots, accelerate to 130 knots or V_H.

c. Performance standards – Missed Approach.

	DESIRED	ADEQUATE
• Maintain target altitudes within ±X ft:	100 ft	200 ft
• Maintain target airspeeds within ±X knots:	5 knots	10 knots

FIGURE 24. Missed approach

a. Objectives.

- Investigate airspeed control to assess adequacy of stick force gradient with airspeed.

b. Description of maneuver. From trimmed level flight at 90 knots, decelerate to 70 knots and retrim for hands-off flight. Then accelerate to 90 knots and retrim for hands off flight. Finally, accelerate to 110 knots and retrim.

c. Performance standards – Speed Control.

	DESIRED	ADEQUATE
• Maintain altitude within ±X ft:	100 ft	200 ft
• Trim hands-off at target airspeed within ±X knots:	3 knots	5 knots
• Change from one trim airspeed to another within X minutes:	1 minute	2 minutes
• Maintain heading within ±X deg:	5 deg	10 deg

FIGURE 25. Speed Control

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

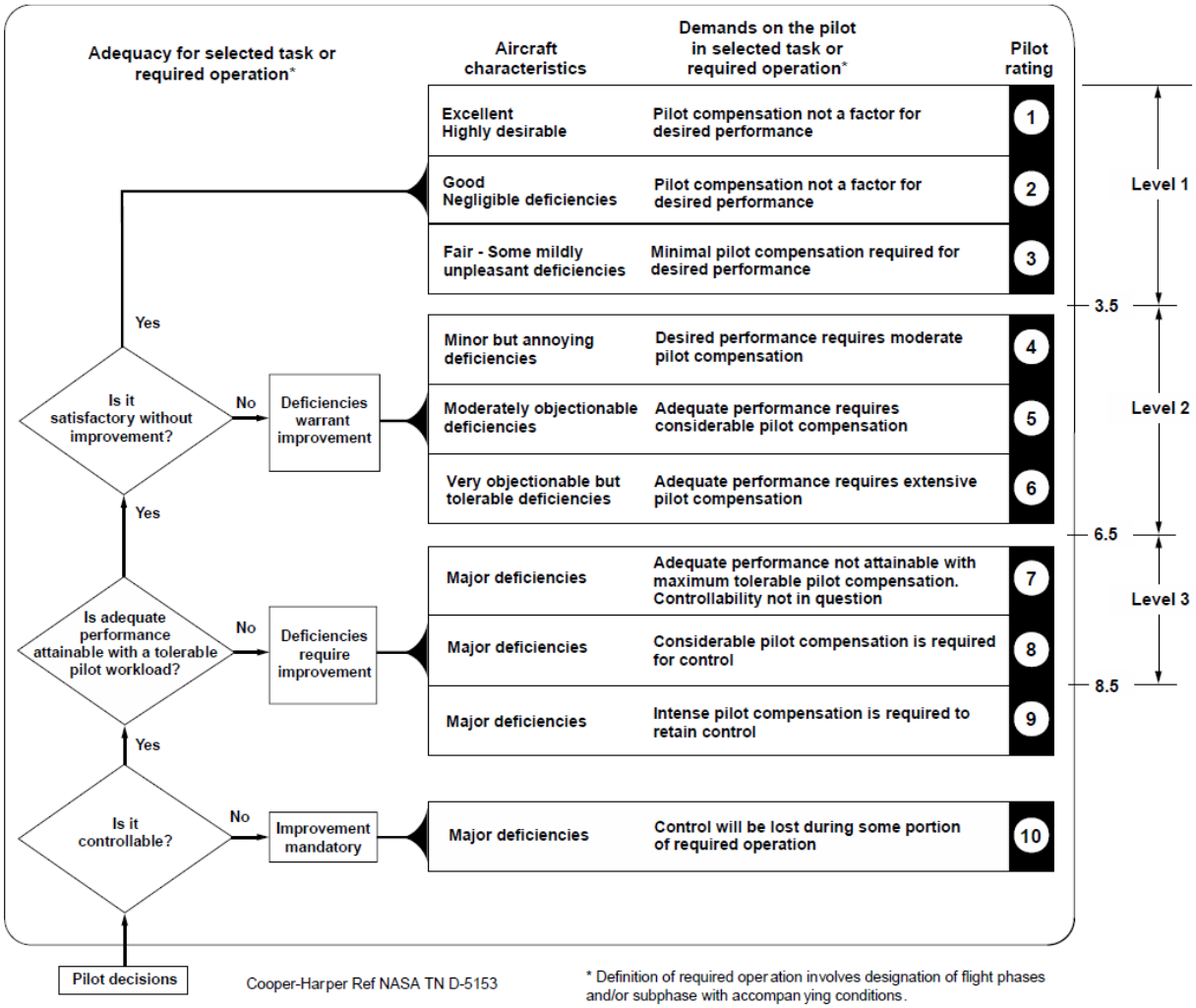
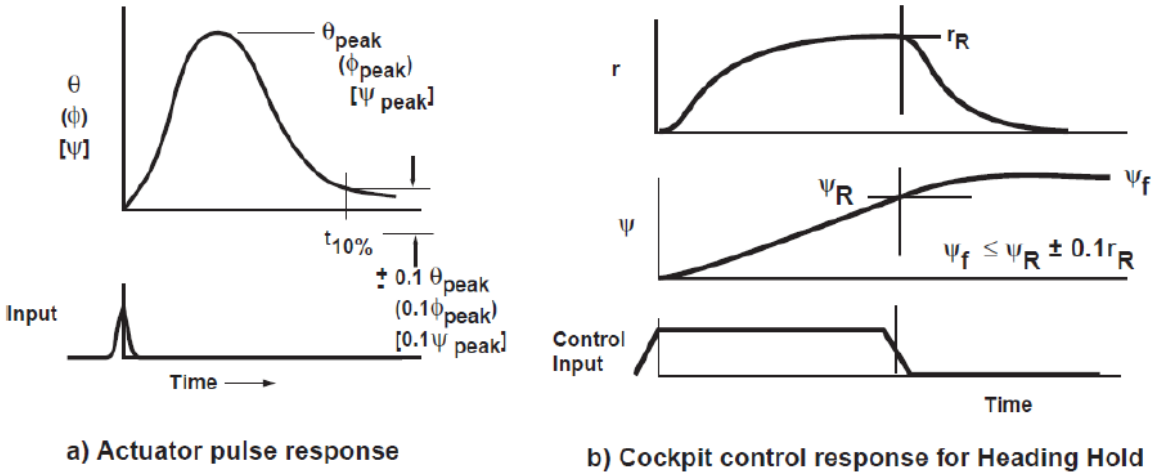


FIGURE 26. Definition of handling qualities levels

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



a) Actuator pulse response

b) Cockpit control response for Heading Hold

FIGURE 27. Response for attitude hold and heading hold response types



Pitch, roll and yaw attitude, and lateral-longitudinal, and vertical translational rates shall be evaluated for stabilizati effectiveness according to the following definitions:

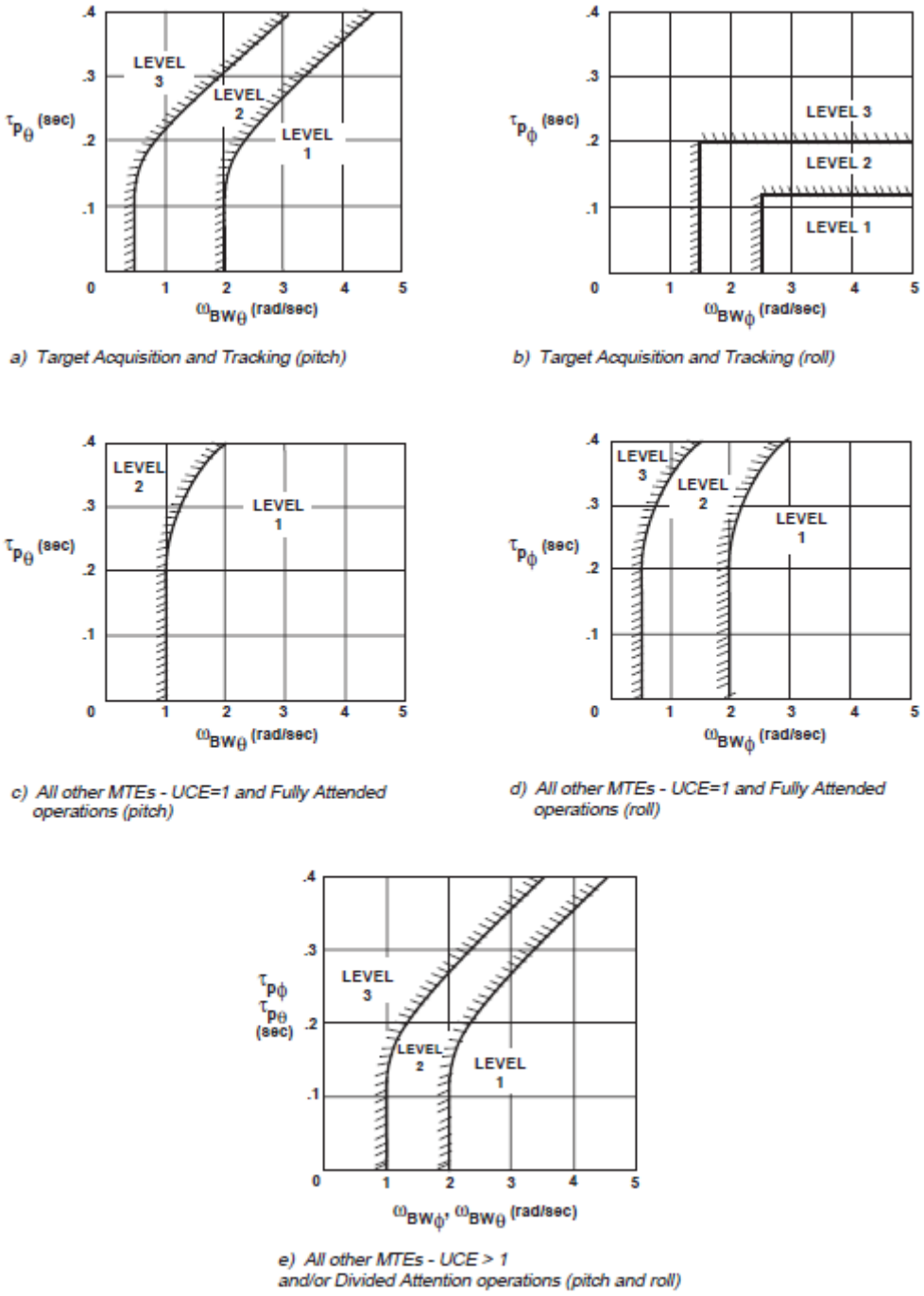
Good : Can make aggressive and precise corrections with confidence and precision is good.

Fair : Can make limited corrections with confidence and precision is only fair.

Poor: Only small and gentle corrections are possible, and consistent precision is not attainable.

FIGURE 28. Visual cue rating scale

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



a) Target Acquisition and Tracking (pitch)

b) Target Acquisition and Tracking (roll)

c) All other MTEs - UCE=1 and Fully Attended operations (pitch)

d) All other MTEs - UCE=1 and Fully Attended operations (roll)

e) All other MTEs - UCE > 1 and/or Divided Attention operations (pitch and roll)

FIGURE 29. Requirements for small-amplitude pitch (roll) attitude changes – hover and low speed

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

Phase delay:

$$\tau_p = \frac{\Delta\Phi 2\omega_{180}}{57.3\,(2\omega_{180})}$$

Note: If phase is nonlinear between ω_{180} and 2 ω_{180}, τ_p shall be determined from a linear least squares fit to phase curve between ω_{180} and 2 ω_{180}

Caution:
For ACAH, if $\omega_{BW_{gain}} < \omega_{BW_{phase}}$, or if $\omega_{BW_{gain}}$ is indeterminate, the rotorcraft may be PIO prone for super-precision tasks or aggressive pilot technique.

Rate response-types:

ω_{BW} is lesser of $\omega_{BW_{gain}}$ and $\omega_{BW_{phase}}$

Attitude Command/Attitude Hold Response-Types (ACAH):

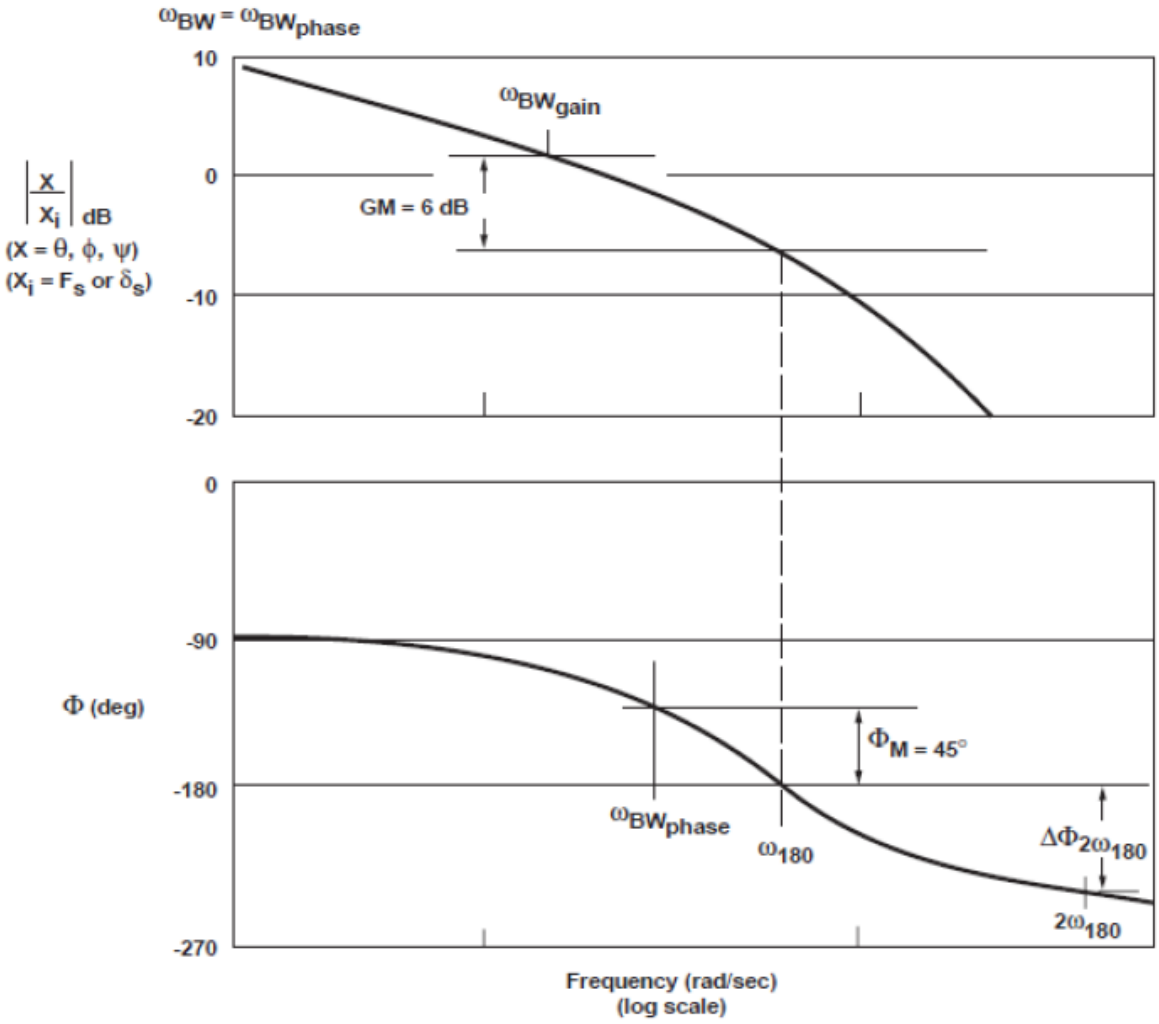
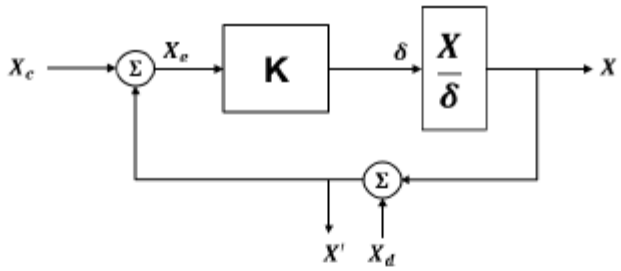


FIGURE 30. Definitions of bandwidth and phase delay

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

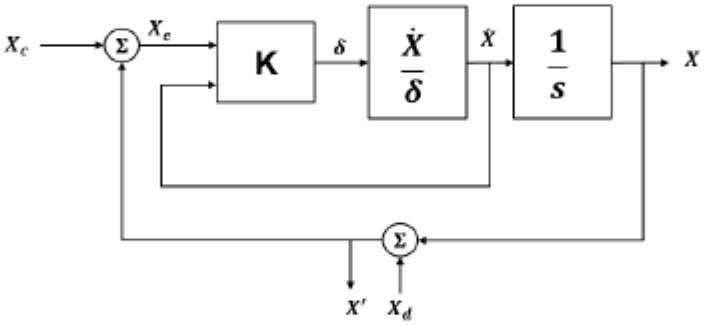


DRB/DRP calculated from magnitude of X'/X_d frequency response

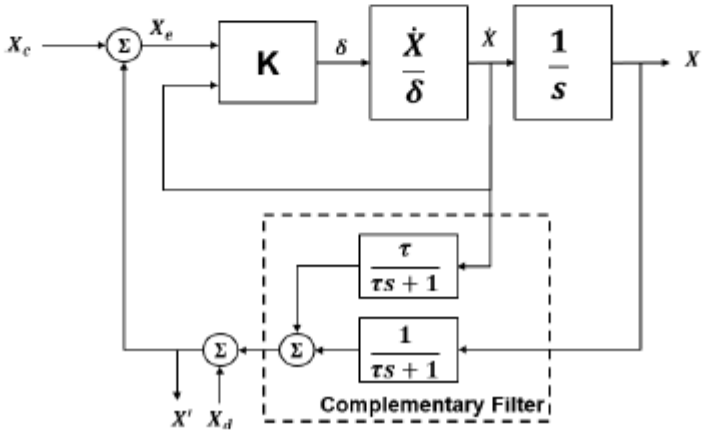
Alternatively:

$$\left|\frac{X'}{X_d}\right|_{reconstructed} = 1 - \left|\frac{X}{X_c}\right| = 1 + \left|\frac{X}{X_d}\right| \qquad (X = \phi, \theta, \psi, u, v, w, x, y, \text{or } z)$$

a) Disturbance rejection block diagram



b) Disturbance rejection block diagram with multiple feedbacks



c) Disturbance rejection block diagram with multiple feedbacks and hold variable complementary filter

FIGURE 31. Disturbance rejection block diagrams

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

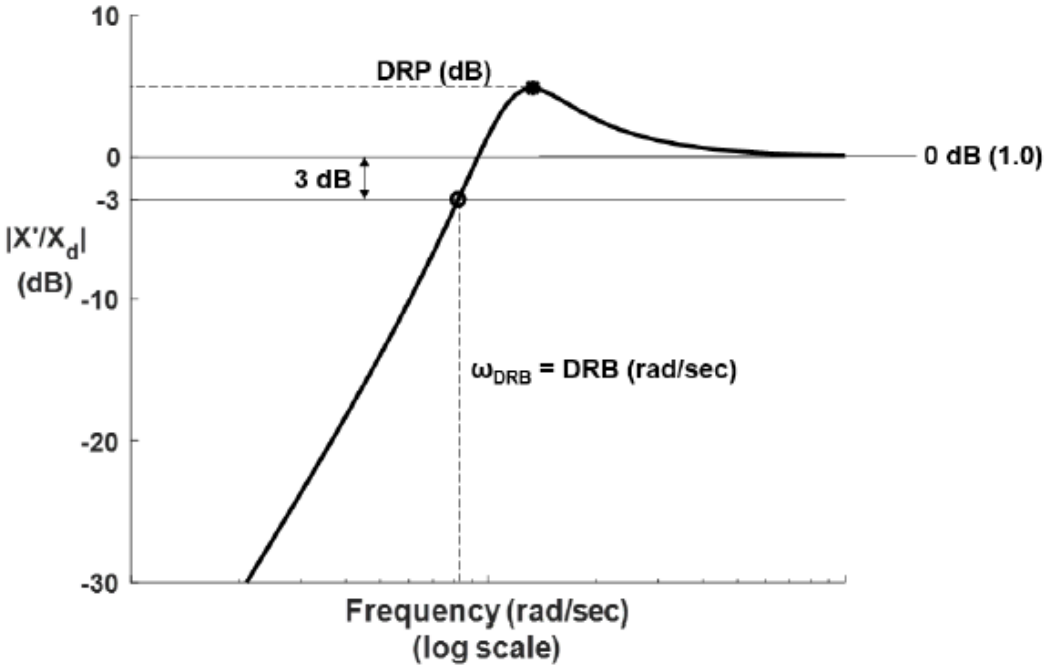
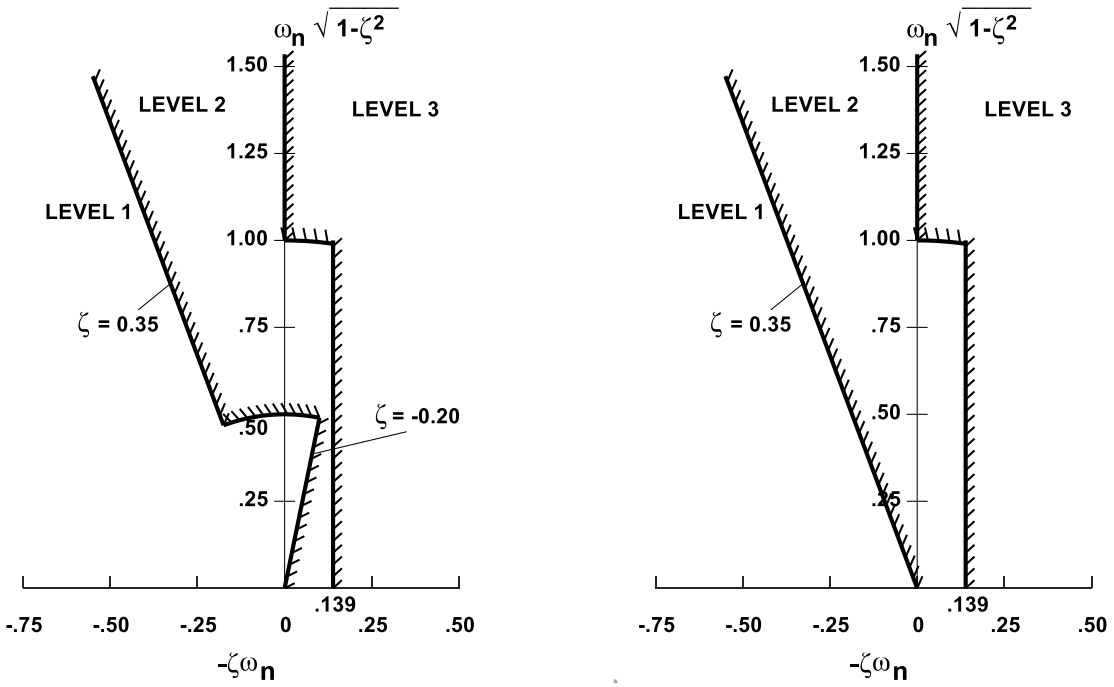


FIGURE 32. Definition of disturbance rejection bandwidth (DRB) and peak (DRP)



a) *Fully Attended operations* b) *Divided Attention operations*

FIGURE 33. Limits on pitch (roll) oscillations - hover and low speed

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

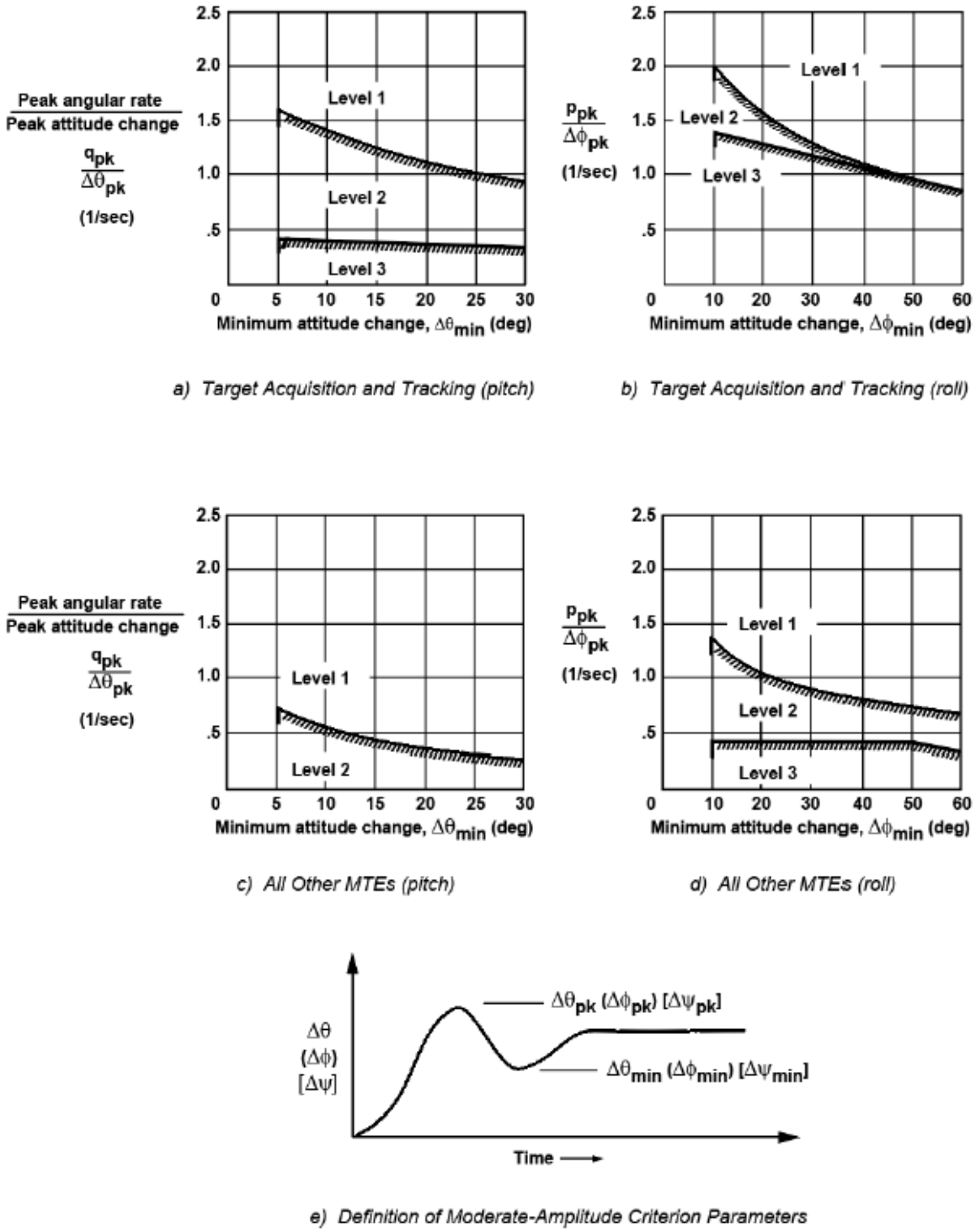
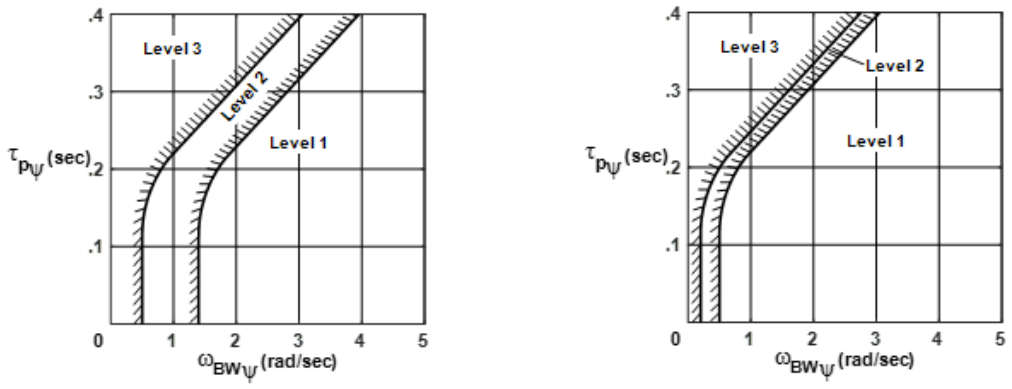


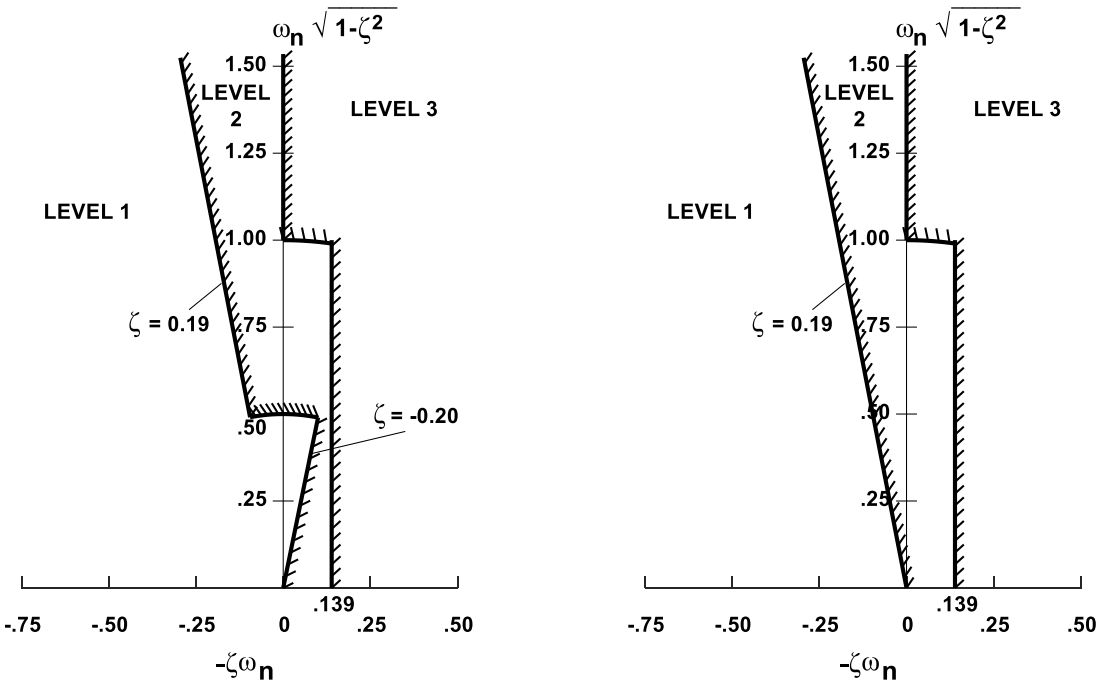
FIGURE 34. Requirements for moderate-amplitude pitch (roll) attitude changes – hover and low speed

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



a) Target acquisition and tracking *b) All other MTEs*

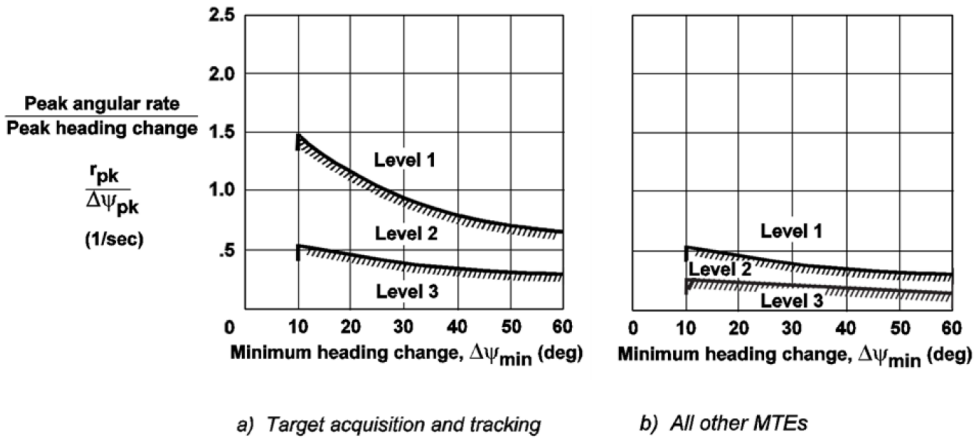
FIGURE 35. Requirements for small-amplitude heading changes – hover and low speed



a) Fully Attended operations *b) Divided Attention operations*

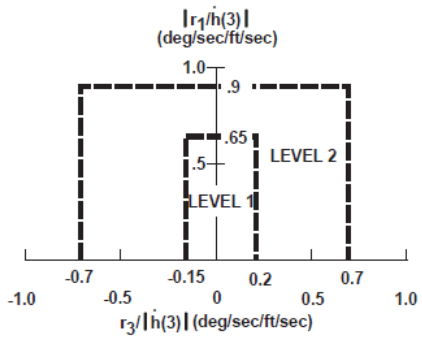
FIGURE 36. Limits on yaw oscillations - hover and low speed

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



a) Target acquisition and tracking *b) All other MTEs*

FIGURE 37. Requirements for moderate-amplitude heading changes – hover and low speed



Where:

r_1 = first peak (before 3 seconds) or r(1) if no peak occurs before 3 seconds

r_3 = r(3) - r_1 for $r_1 > 0$, or r_1 - r(3) for $r_1 < 0$

r(1) and r(3) are yaw rate responses measured at t = 1 and 3 seconds, respectively, following a step collective input at t = 0

In the unlikely event of more than one peak before 3 seconds, the largest peak (by magnitude) should be designated as r_1

FIGURE 38. Yaw-due-to-collective coupling requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

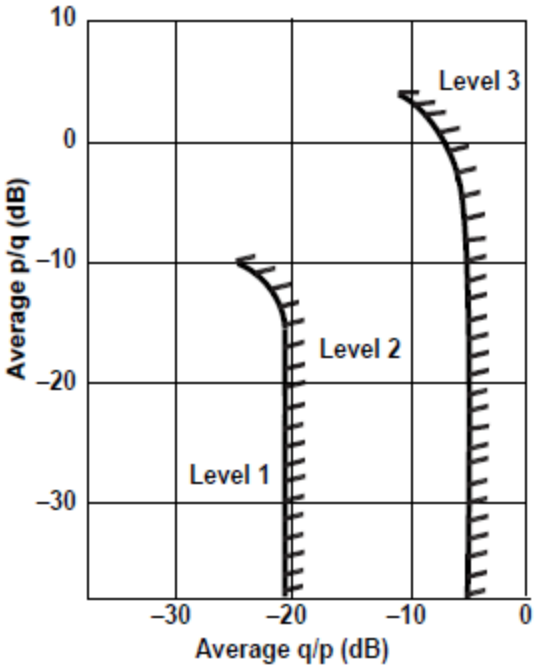


FIGURE 39. Requirements for pitch due to roll and roll due to pitch coupling for target acquisition and tracking

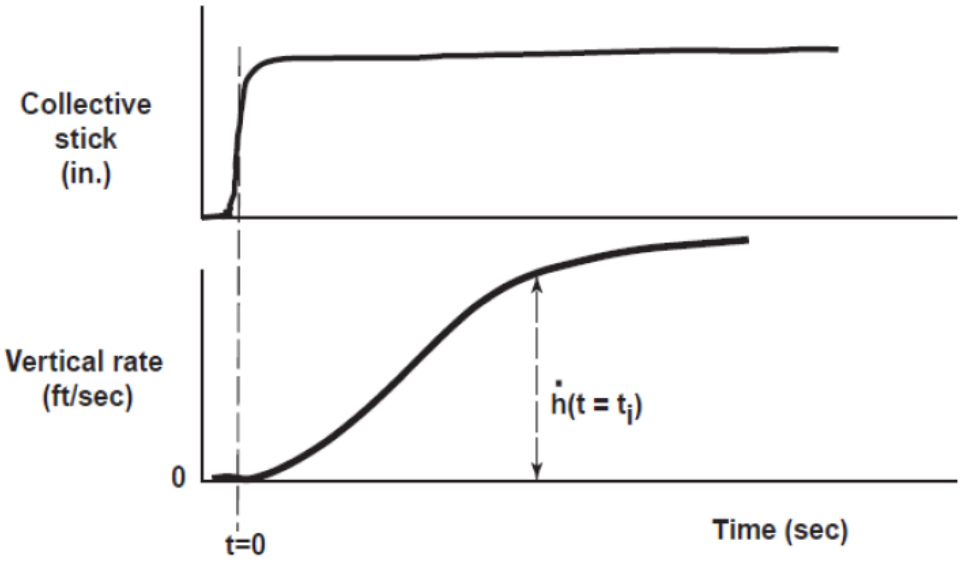
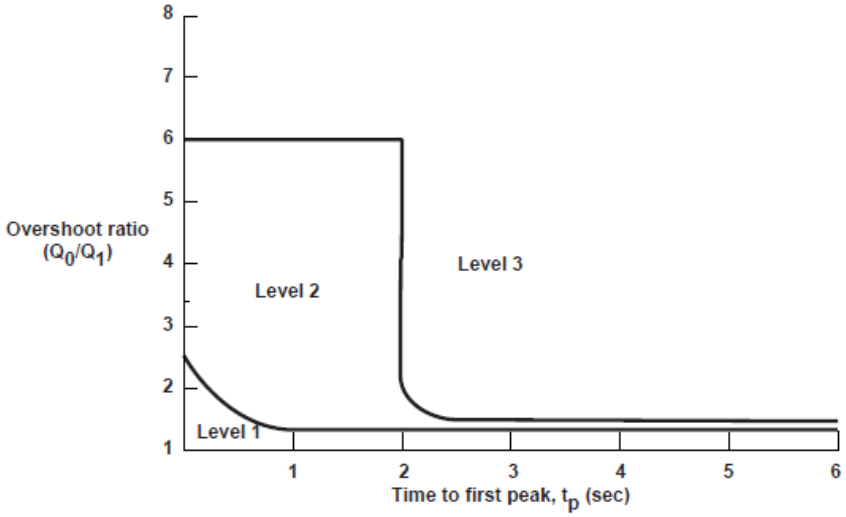
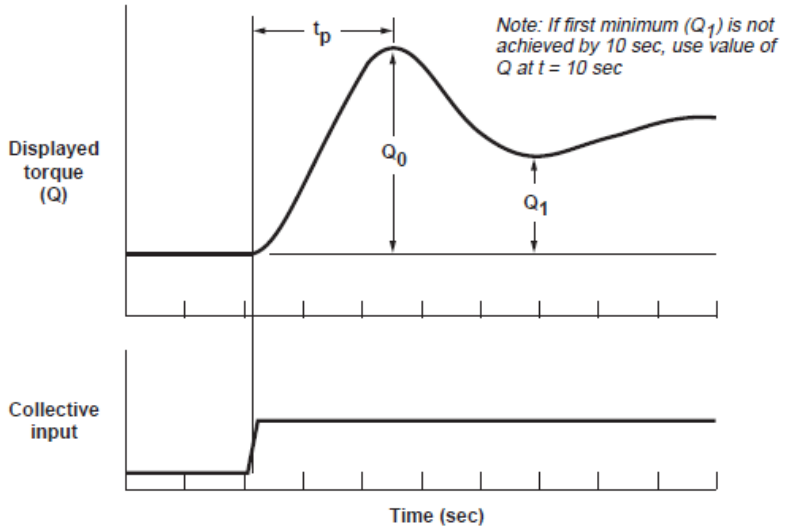


FIGURE 40. Procedure for obtaining equivalent time domain parameters for height response to collective controller

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



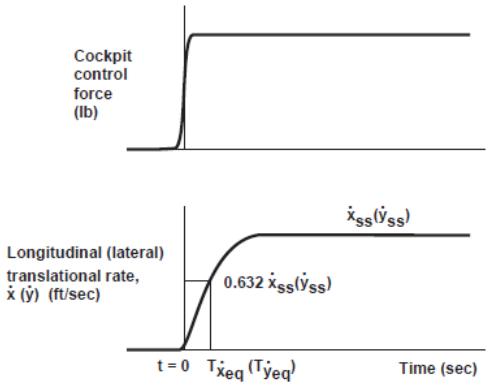
a) Requirement on dynamics of displayed torque based on step collective change



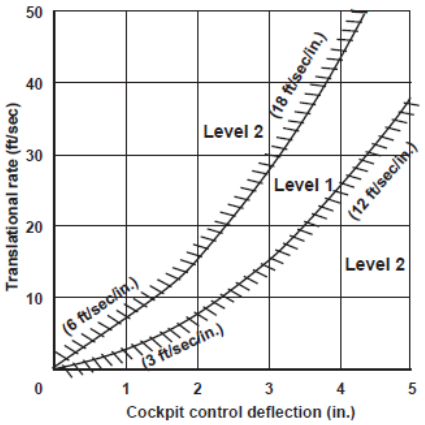
b) Definition of Q_0/Q_1 and t_p for displayed torque requirement

FIGURE 41. Displayed torque response requirement

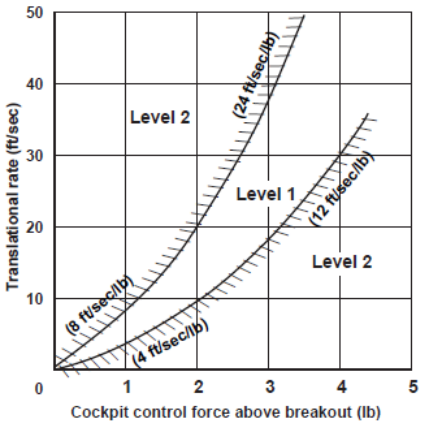
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



a) Definition of equivalent rise time, $T_{\dot{x}_{eq}}$ ($T_{\dot{y}_{eq}}$)



b) Control/response requirement for centerstick controllers



c) Control/response requirement for sidestick controllers

FIGURE 42. Requirements for longitudinal (lateral) translational rate response-types – hover and low speed

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

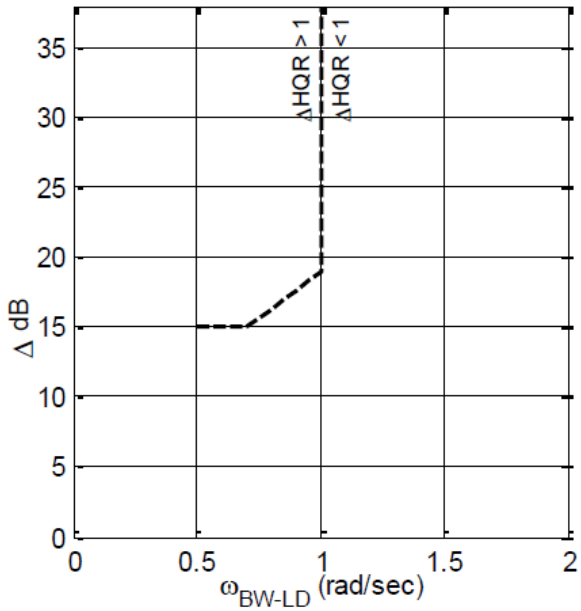


FIGURE 43. ΔHQR from baseline for externally slung loads

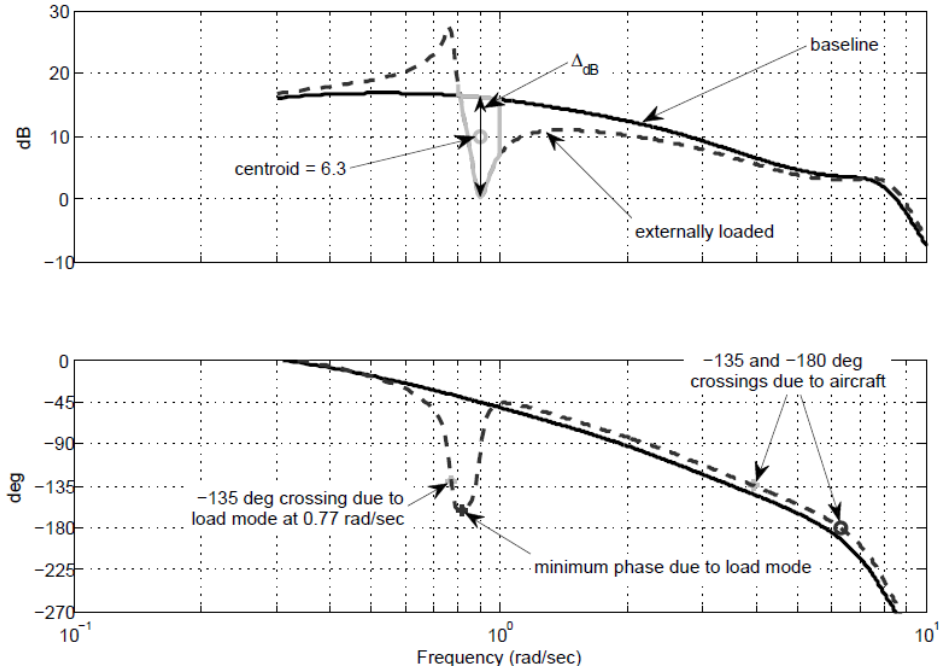


FIGURE 44. Definitions of slung load parameters

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

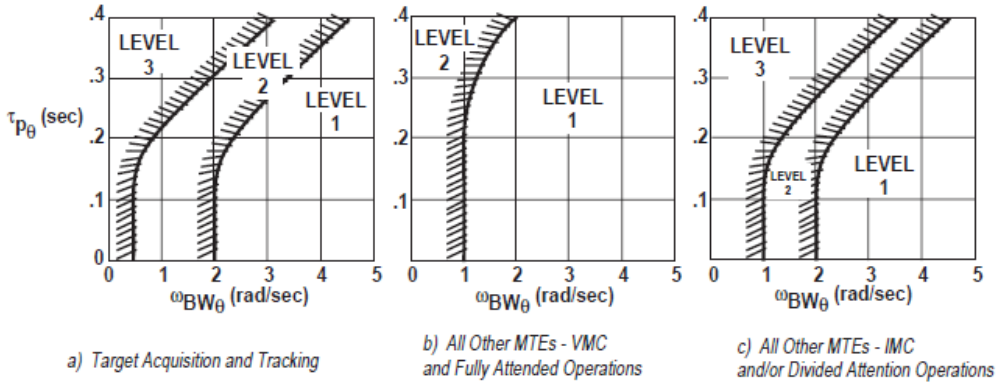


FIGURE 45. Requirements for small-amplitude pitch attitude changes – forward flight

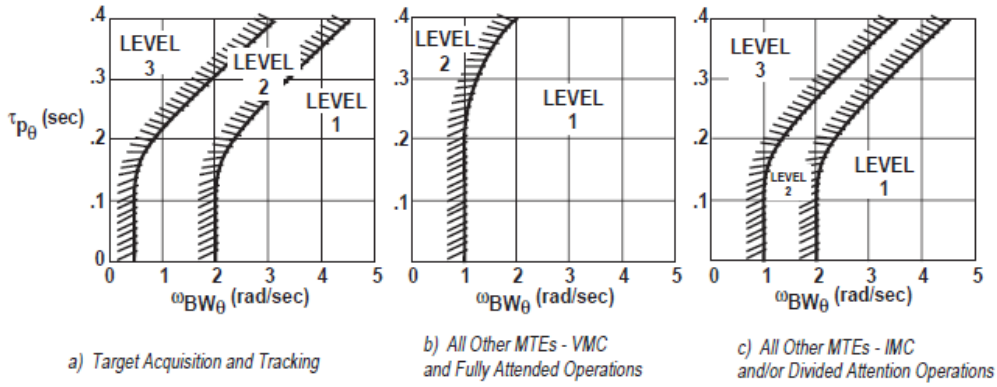


FIGURE 46. Requirements for small-amplitude roll attitude changes - forward flight

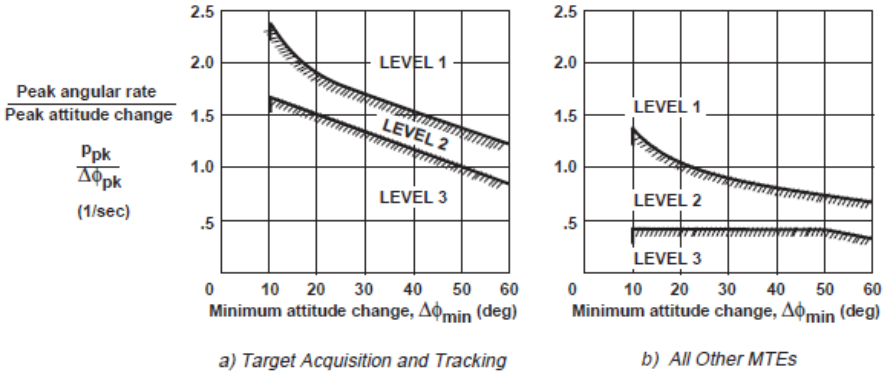
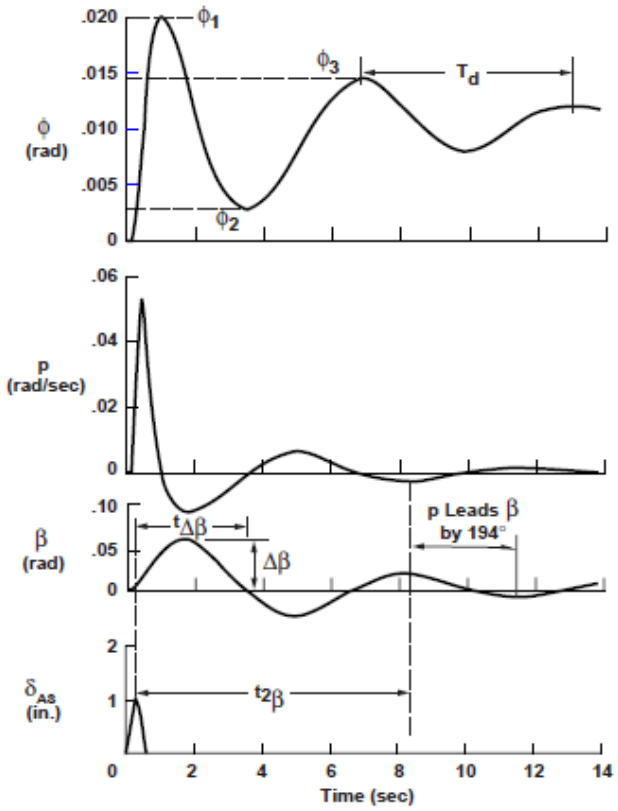
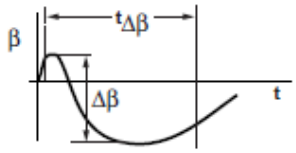


FIGURE 47. Requirements for moderate-amplitude roll attitude changes – forward flight

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



In this example, $t_{\Delta\beta} = T_d/2$ because $T_d/2 < 6$ sec



Example to illustrate definition of $\Delta\beta$ in a response with sideslip reversal at $t < t_{\Delta\beta}$

$$T_d = \frac{2\pi}{\omega_n \sqrt{1-\zeta^2}}$$

ζ, ω_n from paragraph 3.4.9.1

$$\frac{\phi_{osc}}{\phi_{av}} = \frac{\phi_1 + \phi_3 - 2\phi_2}{\phi_1 + \phi_3 + 2\phi_2} \quad (\zeta \le 0.2) \qquad = \frac{\phi_1 - \phi_2}{\phi_1 + \phi_2} \quad (\zeta > 0.2)$$

ϕ, β, δ_{As} change in roll attitude, sideslip, and lateral control position from trim.

$\Delta\beta$ the maximum change in sideslip following an abrupt roll control pulse command within time $t_{\Delta\beta}$

$t_{\Delta\beta}$ the lesser of 6 sec or $T_d/2$.

$t_{n\beta}$ time for the lateral-directional oscillations in the sideslip response to reach the nth local maximum for a right command.

Ψ_β phase angle expressed as a lag for a cosine representation of the lateral-directional oscillation in sideslip, where:

$$\Psi_\beta = -360\, t_{n\beta} / T_d + (n-1)\, 360 \quad \text{(degrees)}$$ with n as in $t_{n\beta}$ above

$|\phi/\beta|_d$ at any instant, the ratio of amplitudes of the bank angle and sideslip angle envelopes in the lateral-directional oscillatory mode.

FIGURE 48. Roll-sideslip coupling parameters

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

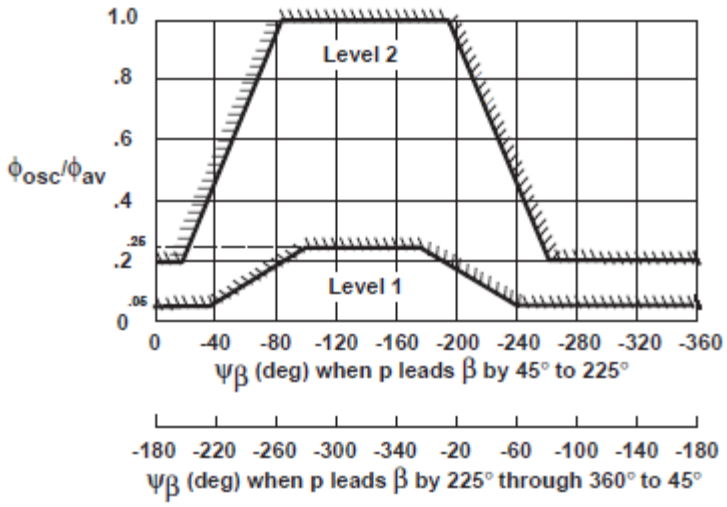


FIGURE 49. Bank angle oscillation limitations

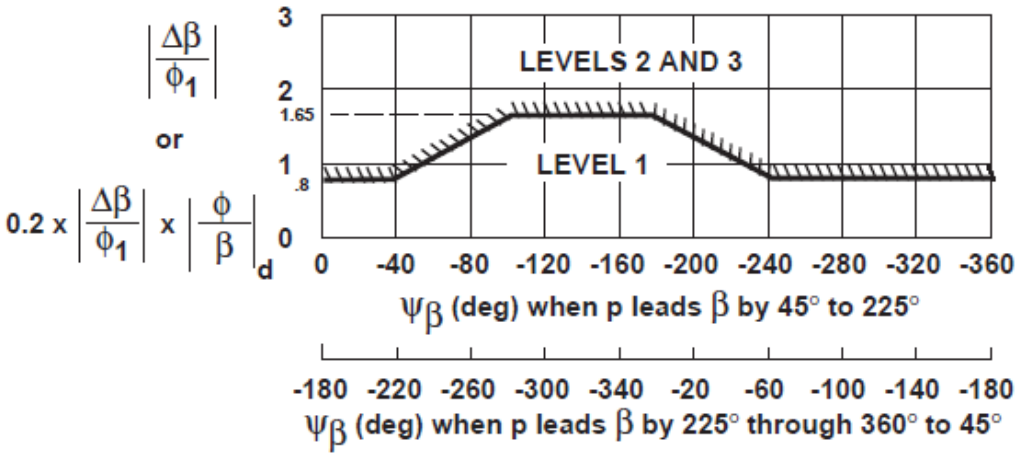
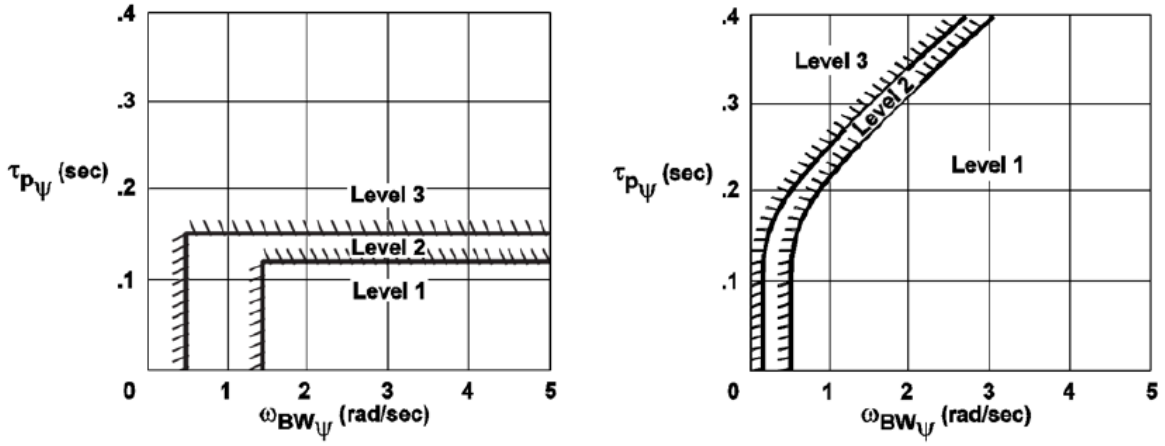


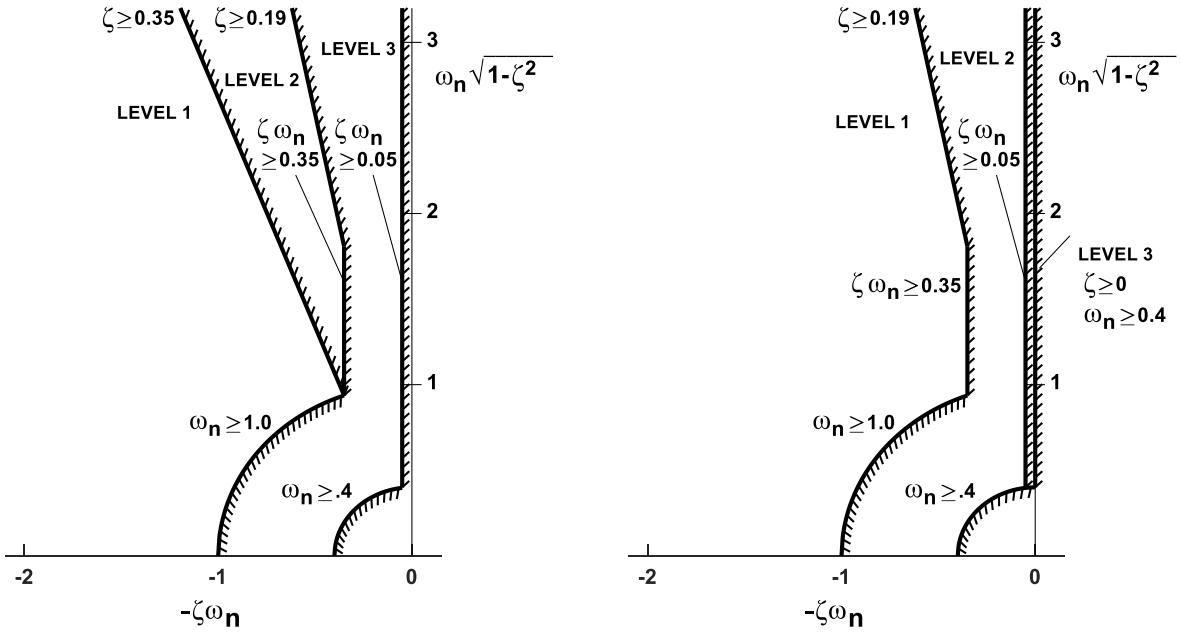
FIGURE 50. Sideslip excursion limitations

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.



a) Target acquisition and tracking *b) All other MTEs*

FIGURE 51. Requirement for small-amplitude yaw response - forward flight



a) Target acquisition and tracking *b) All other MTEs*

FIGURE 52. Lateral-directional oscillatory requirements

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

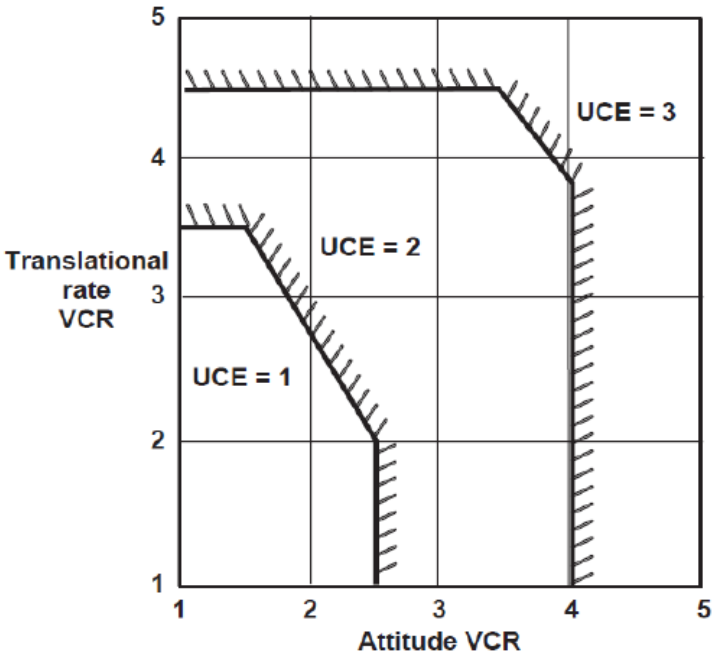


FIGURE 53. Usable cue environments for visual cue ratings

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

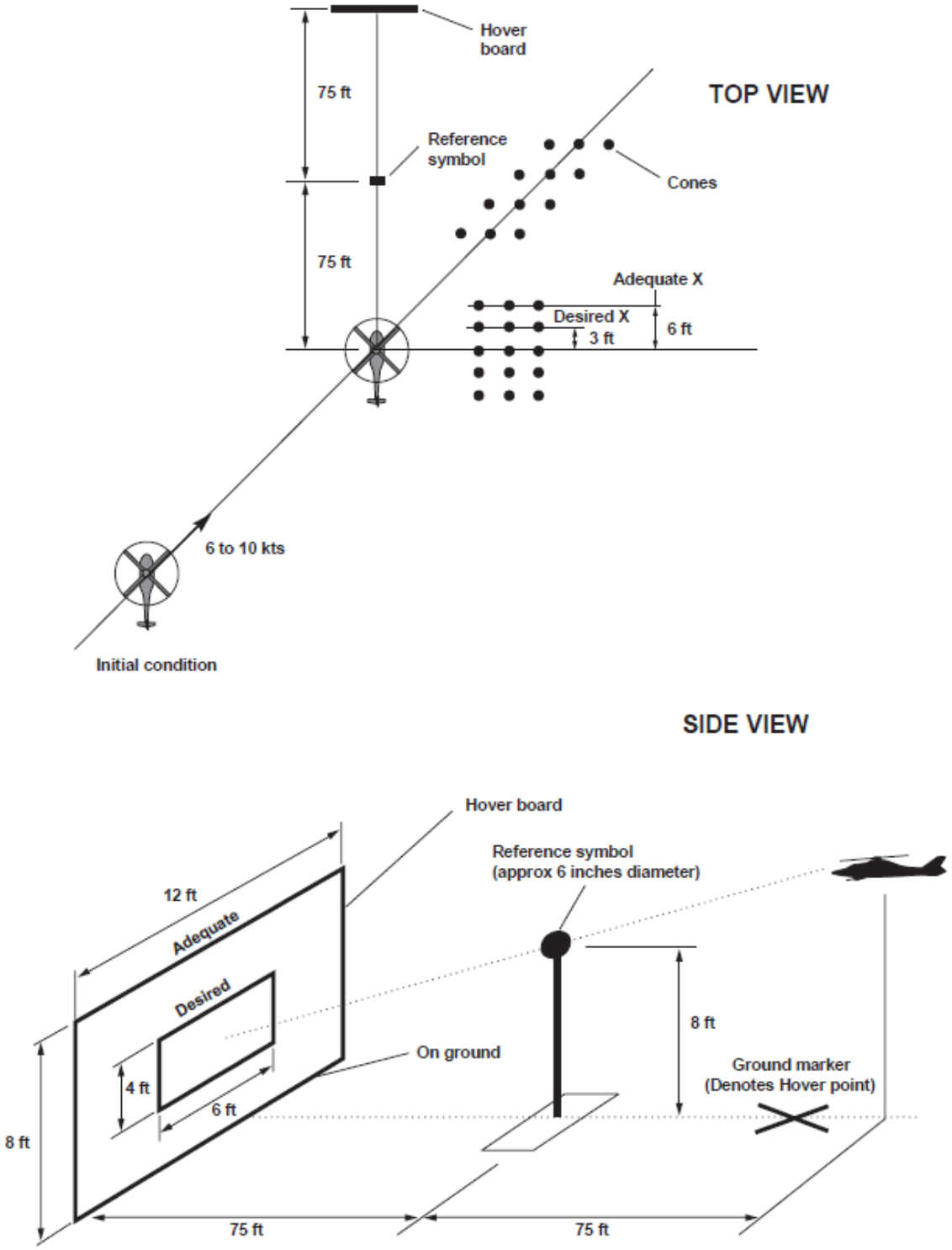


FIGURE 54. Suggested course for hover maneuver

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

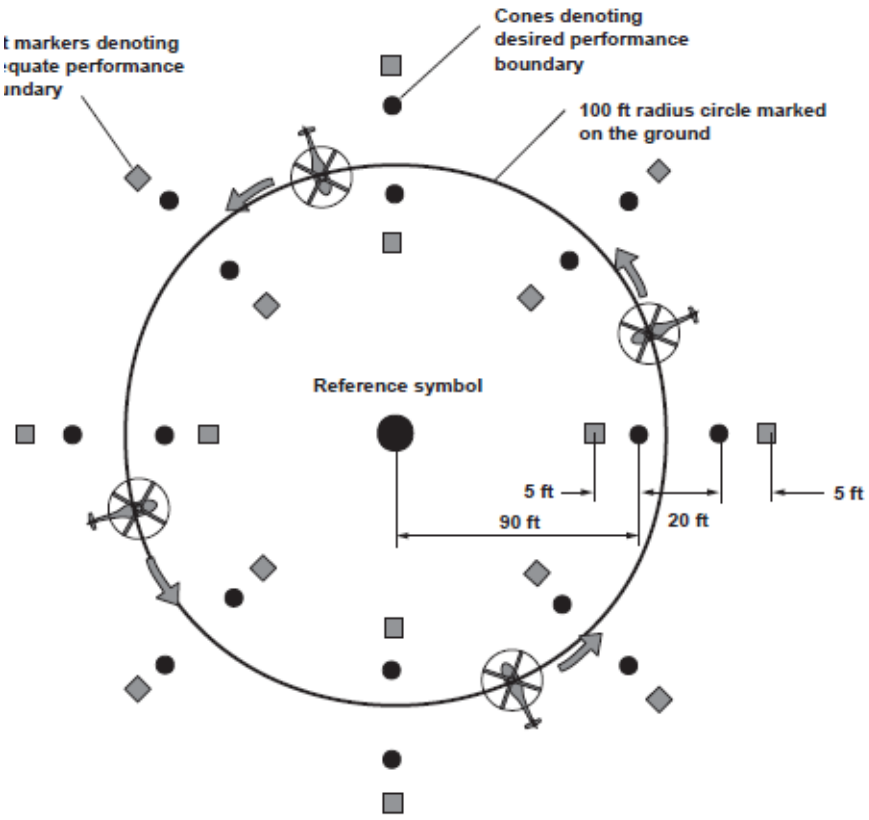


FIGURE 55. Suggested course for pirouette maneuver

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

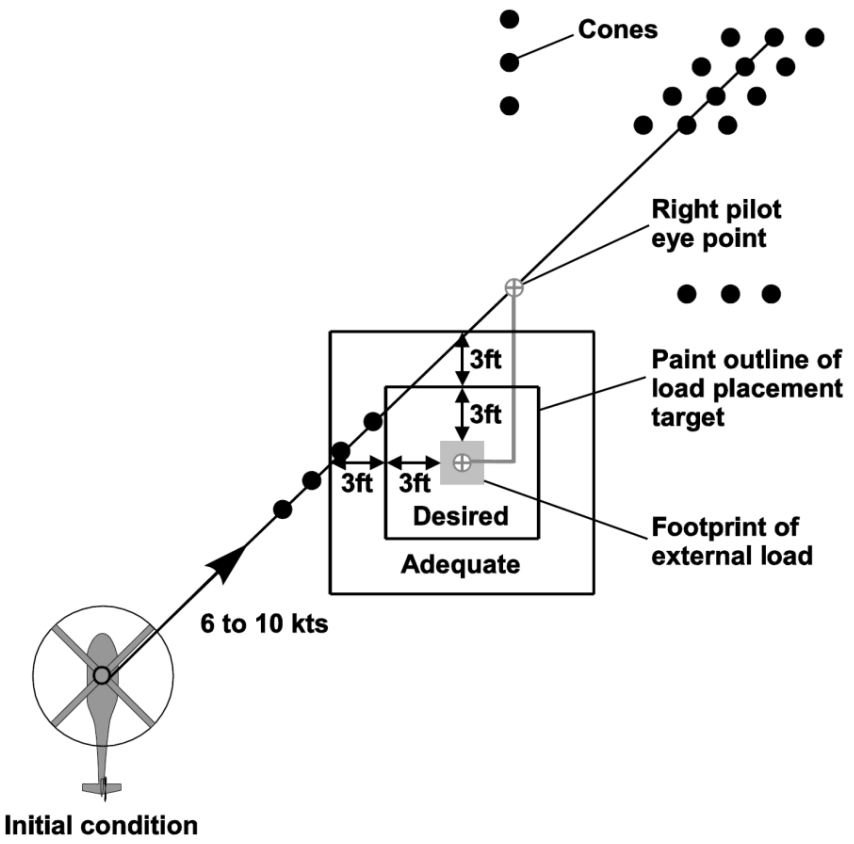


FIGURE 56. Load placement Mission Task Element (MTE)

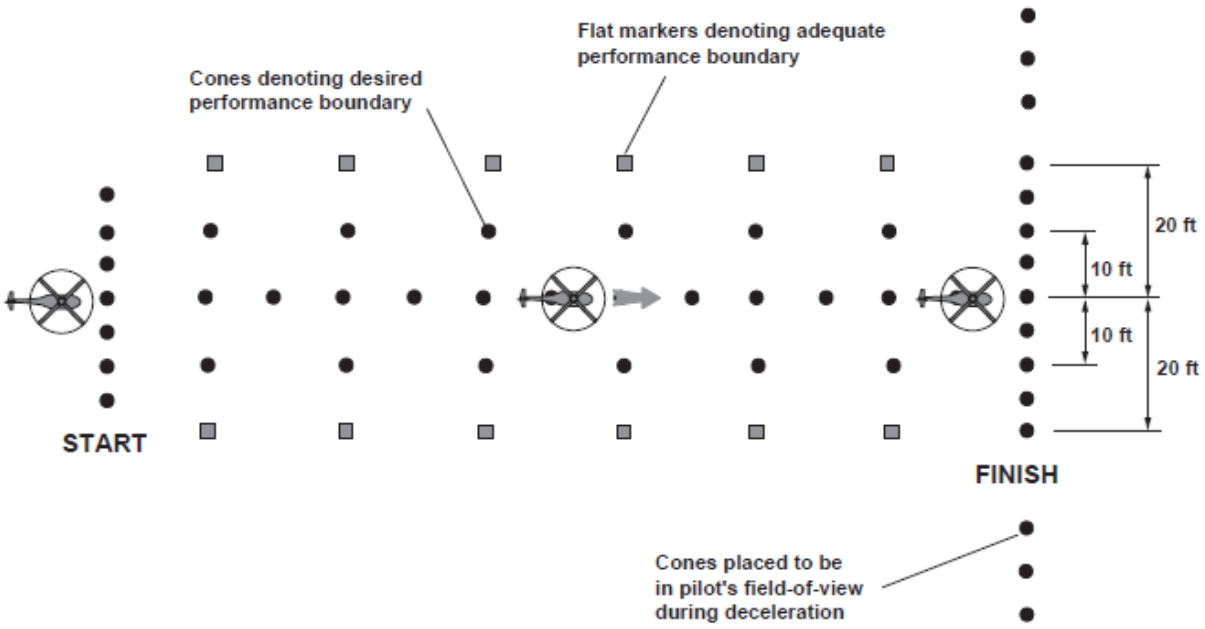


FIGURE 57. Suggested course for acceleration-deceleration maneuver

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

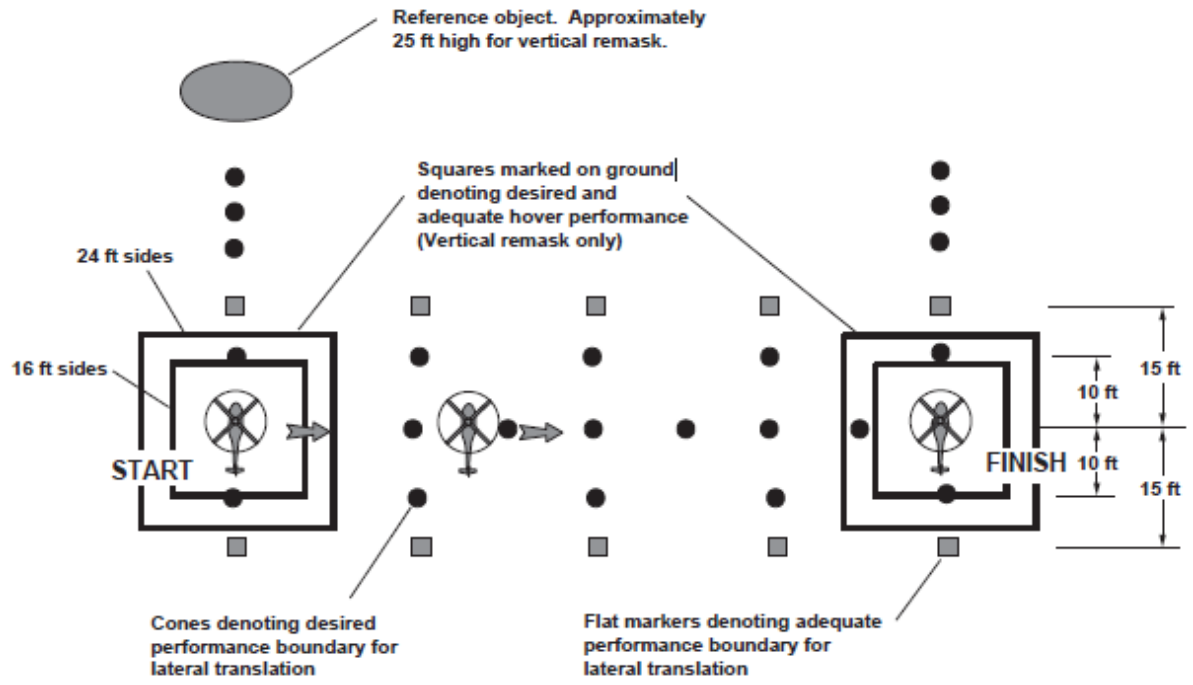


FIGURE 58. Suggested course for sidestep, lateral reposition, and vertical remask maneuvers

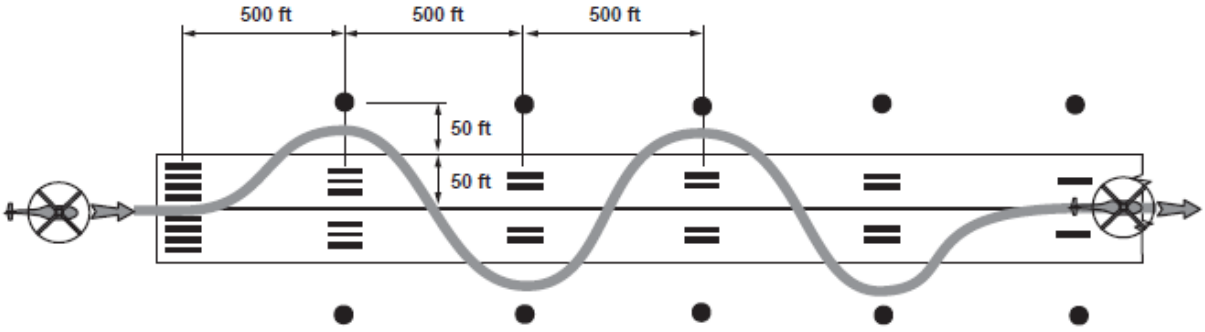


FIGURE 59. Suggested course for slalom maneuver

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – AR

Preparing activity:
Army – AR
(Project 1520-2023-003)

Review activities:
Army – MI

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T07:41Z
Check the source to verify that this is the current version before use.